

Essence

The Problem–Solution Canvas



Ivan Aaen

Essence: The Problem–Solution Canvas is a design methodology for complex and uncertain problems. Under such conditions, the situation we try to improve, the means at our disposal, and the ends we pursue might all change. Design essentially emerges as a solution to problems encountered during project planning and implementation; new opportunities arise as we explore ways to address these problems. The philosophical basis for Essence is Deweyan Pragmatism, which assumes that we experience and learn through working on problems. This basis is symbolized on the cover. The photo's misty trail symbolizes the uncertainty of the future, and the four colored circles suggest the need to apply different viewpoints while acting under such conditions. These viewpoints aim to understand the situation we work with, our contributions, how these contributions might improve the problem, and how to determine if we have succeeded.

IVAN AAEN is an associate professor of computer science at Aalborg University. He works with Problem-Based Learning (PBL), Software Engineering, Software Innovation, and Software Entrepreneurship.

Essence—The Problem—Solution Canvas

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Ivan Aaen

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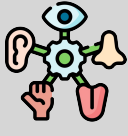
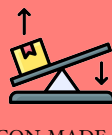
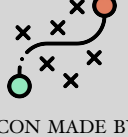
The author is an associate professor at the Department of Computer Science, Aalborg University, Aalborg.

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In writing I've often had the experience, when I get toward the close, of finding that what I said there involves considerable change in what I said in the early part of my discussion

John Dewey (1859-1952)

Chapter 1

Introduction

*To everything (turn, turn, turn)
There is a season (turn, turn, turn)
And a time for every purpose, under heaven*

*A time to be born, a time to die
A time to plant, a time to reap
A time to kill, a time to heal
A time to laugh, a time to weep*

*To everything (turn, turn, turn)
There is a season (turn, turn, turn)
And a time for every purpose, under heaven*

*A time to build up, a time to break down
A time to dance, a time to mourn
A time to cast away stones, a time to gather stones together*

*Pete Seeger
(1919-2014)*

In 1959, during the Cold War, Pete Seeger wrote the song "Turn! Turn! Turn!" The passage quoted here is based on the first eight verses of *Ecclesiastes 3*, where King Solomon reflects on the inevitability of change in life. Since the expulsion of Adam and Eve from

Eden, we have all become mortals. Therefore, King Solomon views change as another constant in a world created by his god.

Donald Schön made a similar point in his book *Beyond the Stable State* (Schön, 1971). In it, he portrayed organizations and societies as being in a permanent state of flux and transformation.

This is more true today than ever, and our hopes of ever returning to a stable state are gone. Change is occurring at unprecedented speed. We face constant demands for innovation, problem-solving, learning, and adapting. Institutions, organizations, and individuals continually face new challenges and develop innovative ways to address them.

This book is for designers engaged in digital innovation amid volatility, uncertainty, complexity, and ambiguity. In these scenarios, we need to go beyond conventional development methodologies.

Essence is designed with adaptability in mind. Grounded in Deweyan pragmatism, it perceives design as an ongoing dialogue between problems and solutions, where each influences the other (von Hippel and von Krogh, 2016).

In this chapter, we will discuss the loss of the stable state and how this affects innovation (Section 1.1), what we define as digital innovation (Section 1.1.6), and why we need to change how we design (Section 1.2.1).

1.1. Factors of Change

We will start by making four observations about why design occurs under fundamentally unstable conditions:

- 1 We live in the Anthropocene epoch: a man-made world.
- 2 We face Knightean uncertainties: fundamental unpredictabilities.
- 3 We work in a hypercomplex field where rational decisions may not always be practical or feasible.
- 4 We perceive and approach problems by imagining solutions, creating complex interdependencies between the two.

1.1.1. The Anthropocene

Firstly, we live in the Anthropocene epoch, a term geologists use to describe the profound impact of human activity on Earth's geology and ecosystems. This transformation presents unprecedented challenges to designers.

Metaphorically, the Anthropocene signifies that innovation projects are not isolated events but interconnected with past, present, and future human efforts, all occurring within interactive ecologies. We are designing and innovating for a world that is actively being constructed beneath our feet rather than merely existing.

We no longer create from scratch. Instead, we select, combine, and transform existing elements based on assumptions about their specific characteristics and under general uncertainties regarding the present and future. In other words, our designs may include dependencies and vulnerabilities related to the elements we use. Still, they may not be fully understood, and the future we are designing is uncertain.

1.1.2. Knightian Uncertainties

Secondly, we face *Knightian uncertainties*—i.e., uncertainties and opportunities that cannot be measured quantitatively. A century ago, American economist Frank H. Knight argued that risk and uncertainty are distinct categories. Risks refer to events we can evaluate based on their likelihood and potential impact. In contrast, uncertainties are inherently unpredictable and cannot be quantified similarly (Knight, 1921, pp. 19-20).

Knightian uncertainty typically connotes something unwanted, often with an aura of bad luck. Some recent examples of Knightian uncertainties include:

- The stock market crash in 2008.
- The *COVID-19* pandemic from 2019 on.
- The Brexit implementation in 2021.
- The blockage of the Suez Canal by a large ship—the *Ever Given*—in 2021.
- The disasters of flooding and drought caused by climate change.
- Russia's war on Ukraine in 2022 and on.
- The energy crisis in Europe following the outbreak of the war in Ukraine.
- The economic turmoil sparked by the energy crisis.
- The public release of *ChatGPT* in late 2022 and the rapid spread of generative AI.
- The Houthi attacks on shipping in the Red Sea in 2023 and on.
- The launch of *DeepSeek*, a Chinese generative AI newcomer, in January 2025. DeepSeek could become a significant challenge to the World dominance of US Tech companies.
- The trade wars initiated by the Trump administration in 2025.
- The altered defense landscape in Europe and other parts of the world in 2025 and beyond.

Such events cause chaos in nations, markets, companies, and individuals, while also creating new opportunities. New businesses replace those that fail, and new demands give rise to emerging markets. During the COVID-19 pandemic, widespread lockdowns sparked a boom in platforms like Zoom, Microsoft Teams, Discord, and others that enable remote work, education, socializing, and more. Digitalization suddenly became a top priority for everyone, from the King to the ordinary person, and future historians might view COVID-19 as a key driver in accelerating the Fourth Industrial Revolution. Similarly, Europe's reliance on Russian energy amid the Ukrainian War prompted significant investments in clean energy and accelerated the energy transition. Additionally, security challenges could cause Europe to reduce its dependence on Big Tech and invest in new defense technologies and suppliers.

Knightian uncertainties and opportunities are potential game-changers: they can create new markets and destroy existing ones. They can prompt radical changes in demand and accelerate the development of new technologies in response. Similarly, technology can drive radical change, disrupt markets, and transform our lifestyles. Examples include the steam engine, electricity, the production line, the computer, natural language processing, generative artificial intelligence, and more.

1.1.3. Hypercomplexity

Thirdly, projects—not least software-intensive projects—are becoming *hypercomplex*. Bohman defines hypercomplexity as a level of complexity that hinders or prevents rational decision-making within a reasonable time frame (Bohman, 1995).

In situations of hypercomplexity, it is impossible to determine the ideal starting setup for platforms, components, and data. Designers begin with a practical configuration for their initial efforts. Afterwards, they try to optimize each iteration. Every later step in the design process builds on previous experiences and structures.

1.1.4. Problems and solutions are interdependent

Fourthly, problems and solutions are interdependent (von Hippel and von Krogh, 2016): The problems we set out to solve are selected, perceived, and scoped by the options we see for solutions. Working in the Anthropocene under Knightian uncertainty and hypercomplexity deprives us of hope for a proper innovation method. Methods build on knowledge distilled from previous solutions to similar problems, but we cannot presume that similarities hold under these conditions.

Projects might seem solid: a problem and its solution might seem understood; cost and time estimates might be convincing; and staff and funds might have been allocated reasonably. However, situations can change suddenly. Despite our best efforts, the problem may turn out to be more complex than initially thought, and new ways of solving problems or unexpected opportunities may emerge.

Such developments make it increasingly complex to answer two critical questions in design:

- What are we building? What is our contribution?
- What are we building this for? Where will our contribution be used, and why is it needed?

We refer to what we are building as the *inner environment* (Simon, 1996). The inner environment denotes the substance and organization of our contributions. Likewise, we refer to what we are building this for as the *outer environment* (Simon, 1996). The outer environment relates to the surroundings in which our contributions will reside and operate. Both environments are significant sources of complexity and uncertainty.

The outer environment is problematic:

- The problem could be intricate and unclear, and we may not have the necessary knowledge to comprehend it.
- We may not have adequately scoped the problem, considering what might be feasible.
- Although third-party services may be available, we may not be aware of them.
- We may also be unaware of all available external sensors, equipment, and data sources.

Similarly, the inner environment is a significant source of complexity and uncertainty. We see radical increases in the means available to build substantial contributions:

- New technologies, infrastructures, standards, and knowledge.
- New components, sensors, data, platforms, and more allow diverse designs.

Collectively, both environments add even more complexity and uncertainty:

- Infrastructure with open interfaces is growing in number and strength, increasing the options available to designers.
- The rate of change implies dynamic working conditions for design and implementation. Conditions could change significantly between the start and the end.

1.1.5. Summing up the Challenges

The loss of a stable state alters our approach to problem-solving. Increasingly, our work is challenged by what is commonly referred to as VUCA (Bennis and Nanus, 1986):

- *Volatility*—i.e., conditions that change rapidly or unpredictably.
- *Uncertainty*—i.e., needs and solutions that are unsettled.
- *Complexity*—i.e., external or internal parts that form intricate meshes.
- *Ambiguity*—i.e., causes and effects that are mixed up.

Challenges like these require an experimental, flexible work process where solutions grow organically rather than being planned.

The photo on the cover of this book metaphorically represents the challenges we confront in entirely unfamiliar circumstances. Where will this path end? Will the path split into separate directions at some point? What directions lead to where we want to go? *Essence* is designed for design projects that operate under similar conditions. The circles on the front page represent the four core activities that *Essence* uses to help navigate the route.

Essence is designed to help teams manage unexpected events. No method can guide work on the unprecedented. Therefore, *Essence* provides tools for reflection, abstraction, and reasoning.

1.1.6. Digital Innovation—A Definition

So far, we have discussed digital innovation without providing a precise definition. Now is the time to delve into it more precisely.

Economists have long seen a strong link between *science and technology* and *economic and societal change* (Di Stefano et al., 2012; Karayiannis, 1998; Rosenberg, 1974; van den Ende and Dolfsma, 2005). We can divide these factors of change into *technology push* and *market pull*. The technology push highlights the role of science and technology in driving innovation and change. At the same time, the demand pull focuses on market features, users, overall economic developments, and other factors as drivers of innovation and change (Di Stefano et al., 2012).

Technology push refers to changes driven by the emergence and spread of new technologies. Digitalization, digitization, Big Data, natural language processing, generative AI, the Internet of Things, Industry 4.0, intelligent sensors, 5G, and many other technologies are key factors of change.

Demand pull happens in response to problems or needs. The UN's Sustainable Development Goals (SDGs), climate change, pandemics, cold wars, hot wars, trade wars, arms races, protection of critical infrastructure, obstruction of major waterways, and more are global factors that generate new demands and lead to significant changes in solutions, markets, and business models. These changes, in turn, generate new opportunities for digital innovation.

There are several definitions of digital innovation in the research literature, and they differ significantly. Arguably, Joseph Schumpeter is considered the father of innovation research within economics. Based on Schumpeter's 1934 landmark book, *The Theory of Economic Development*, Yoo et al. defined digital innovation as the implementation of new combinations of digital and physical components to create novel products (Yoo et al., 2010, p. 725). Yoo et al. primarily focus on product innovation, thereby clearly diverging from the earlier emphasis on process in IT innovation research. Their shift away from previous IT innovation studies is notable: Much of the earlier work concentrated on *internal innovation* in software development to enhance software processes. Yoo et al. instead focus on external innovation and the benefits innovative digital products can bring to customers, users, or markets. Although this definition marks a significant move from internal to external innovation, focusing solely on products introduces a notable limitation.

In 2014, Fichman et al. extended the term by defining digital innovation broadly as *a product, process, or business model that is perceived as new, requires some significant changes on the part of adopters, and is embodied in or enabled by IT* (Fichman et al., 2014, p. 330). This definition projects digital technology upon general innovation theory by outlining several types of innovation.

Three years later, Nambisan et al. (2017) specifically focus on value creation and conceptualize digital innovation as *the creation of (and consequent change in) market offerings, business processes, or models that result from the use of digital technology* (Nambisan et al., 2017, p. 224).

Based on similar discussions, Frey et al. combine innovation types and value creation. They define digital innovation as *all products, processes, or business models emerging within or between organizations that are based on technology and trigger or enable value* (Frey et al., 2020, p. 814).

In summary, the definitions from 2010 to 2020 link digital innovation to both physical and digital components, value creation, and potential changes on multiple levels, both externally and internally.

We largely agree with Frey et al.'s definition, but we choose to explicitly mention services and digital technology. In Essence, we, therefore, define digital innovation as

Products, services, processes, or business models that emerge within or between business units and where digital technology is a key factor in triggering or enabling value.

In this definition, the focus is exclusively on external innovation. Thus, process innovation refers solely to processes on the user side.

1.1.7. Types of Digital Innovation

We will now expand on the types of innovation discussed above. In their book *Managing Innovation*, Tidd and Bessant (Tidd and Bessant, 2009) see innovation as a diverse concept with four general types: *product, process, position, and paradigm*.

Following their example, we identify four types of digital innovation (Table 1.1). The four software innovation variants listed here are not necessarily mutually exclusive. For example, a telemedicine system could be a product for patients, but it could also support new processes within the hospital. Indeed, the combination of massive amounts of data from large numbers of patients will likely lead to paradigm shifts in healthcare regarding assessments and interventions. Finally, solutions from one project might later be adapted for other areas of healthcare and even for neighboring areas such as sports and fitness.

<i>Main definition</i>		Products, services, processes, or business models that emerge within or between business units and where digital technology is a key factor in triggering or enabling value
<i>Variants</i>	<i>Product</i>	New or radically changed products or services
	<i>Process</i>	New or radically improved ways to produce products or services
	<i>Position</i>	Products, services, processes, or business models that emerge by fitting earlier solutions into new uses
	<i>Paradigm</i>	Products, services, processes, or business models that emerge from radical changes in the mental models of what a business is; who the users are; or what the market is

Table 1.1: Four Types of Digital Innovation

Digital innovation encompasses a wide range of digital products and services, novel business processes that leverage digital technologies, innovative applications of existing digital solutions in new contexts, and paradigmatic shifts in specific domains or sectors of society enabled by digital technologies.

Our focus is on innovation for users or customers: innovations that offer something new to existing users, something familiar to new users, or something new to both.

1.2. Main Characteristics of Digital Innovation

What, then, are some main characteristics of *digital* innovation? Some key observations might help answer that question:

- *Digital innovation is open.* Digital innovation spans across various organizational entities and activities.
- *Digital innovation is software-intensive.* Digital innovation involves transforming practices and enterprises through the use of information technology.
- *Digital innovation is an extension of software innovation.* Software innovation enables digital innovation, but digital innovation has a broader scope.

The *first* observation—that digital innovation is open—is based on two schools of innovation theory: user-driven and open innovation.

User-driven innovation denotes innovation driven by lead users, communities of users, or user companies (Bogers et al., 2010; von Hippel, 1986). In this type of innovation, problem-domain knowledge and technological competence combine to create more effective digital solutions. User-producer interactions across knowledge areas drive digital innovation.

Open innovation is based on knowledge sharing and collaboration within or across organizational boundaries (Chesbrough, 2003). A basic tenet is that enterprises benefit from using external ideas alongside their internal ones. Open innovation thus supports collective creativity by sharing knowledge and solutions. Digitization typically spans organizational boundaries, including those between companies, branches, departments, and other entities. Under such circumstances, knowledge sharing and collaboration are crucial.

The *second* observation — that *digital innovation is software-intensive* — portrays software as the key enabling technology. By definition, *software-intensive systems* (Eckroth et al., 2007) are systems where software plays a key role in providing functionality and value. They are systems in which software is essential to their design, construction, or implementation.

The *third* observation — that *digital innovation is an extension of software innovation* — enables us to link it to existing literature in software development. Software innovation is not a well-established term; however, we will loosely define it as innovation enabled primarily through software. Contributions to this field are scattered over organizational levels and project stages. Some contributions are generic with little focus on either the managerial level or particular stages (Rose, 2010); some focus on the company level

(Pikkarainen et al., 2011); some on picking and improving promising ideas (Gorschek et al., 2010); some on ideation in the requirements stage (Eito-Brun and Sicilia, 2016; Mahaux and Maiden, 2008; Maiden et al., 2007); and some on innovation as part of an ongoing project (Aaen, 2008; Aaen, 2010; Aaen, 2013; Aaen and Jensen, 2014; Aaen, 2018; Campos, 2012; Conboy and Morgan, 2011).

On the other hand, digital innovation involves more than software. We can not meaningfully isolate software from digitization and digitalization. Our understanding of digital innovation must consider both software teams and external stakeholders. These concerns are like communicating vessels, where changes in one view affect the other.

1.2.1. A New Paradigm

As argued in Section 1.1, we are now beyond any stable state. This change in conditions necessitates a new design approach. Essence stems from a software engineering tradition. Within that tradition, there are currently two dominant paradigms: plan-driven and agile development.

The *plan-driven paradigm*—dating back to 1968 (Naur and Randell, 1969)—assumes explicit, known, and stable requirements. This paradigm advocates a document-oriented approach centered around requirements specifications. These requirements enable top-down planning and project management, facilitating the resolution of any disputes that may arise between parties.

The plan-driven paradigm generally overlooks creativity and innovation. Problems are defined from the beginning and set within familiar contexts. The search space for solutions is provided, although it can be pretty complex. To manage this, the plan-driven paradigm offers advanced tools and techniques for gathering requirements, ensuring requirement consistency, estimating project resources, planning projects, and verifying and validating compliance with requirements.

The *agile paradigm*—dating back to the mid-1990s and expressed in The Agile Manifesto (Beck et al., 2001)—assumes some requirements are *tacit* and complex to elicit up front. This paradigm advocates an incremental and user-oriented approach. Requirements are described, for example, as user stories and are subject to change as customers try out partial solutions. Still, the agile paradigm assumes that problems are stable and known in general.

Both paradigms generally involve building systems from the ground up in an orderly manner by applying generic methods. Both aim for quality and efficiency, but differ in how they achieve them. The plan-driven paradigm relies on process standardization and

documentation, while the agile paradigm relies on competent developers and customers, iterative processes, and sophisticated tools.

Neither of these two paradigms adequately addresses an inherently complex, uncertain, and unstable world. A third paradigm is needed.

The pragmatic paradigm is problem-oriented. We call it pragmatic to reflect the focus on understanding and solving problems in practical situations.

In this paradigm, systems are interconnected. Increasingly, systems become part of digital ecologies that may not even be fully understood at the time of design. New technologies, components, sensors, data, platforms, and more enter these ecosystems, and mobility constantly changes the operational context. Designers face complexities and uncertainties in their designs and the external environments they must adapt to.

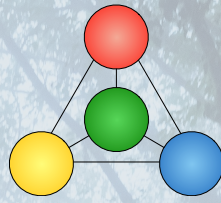
1.3. How to Use This Book

This chapter discussed some challenges designers face in an increasingly hypercomplex world. We outlined previous lines of thought—paradigms—for designing and developing software-intensive systems and defined key terms related to digital innovation.

Following this introduction, Part 1 present Essence's primary tool—the *Problem–Solution Canvas*—designed to explore solutions in a hypercomplex world.

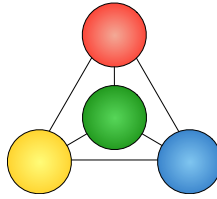
The remainder of the book focuses on using the Problem–Solution Canvas (PSC), but is organized so that consulting the chapters in any particular sequence is not required. You can visit those chapters in any order you wish. Writing a book that allows for reading parts on demand, however, requires that some chapters repeat and overlap with content found in other chapters.

The PSC functions as a visual inquiry tool rather than a strict method. I encourage you to approach it with curiosity, focusing on your problem rather than any idea of ‘proper’ PSC use. The best way to use the PSC is to customize it to support your investigation into the problem and help you find solutions. The canvas serves as a tool for inquiry and exploration more than for documentation.



Part 1

The Problem— Solution Canvas



Chapter 2

Essence Roles and Core Activities

*One who is too insistent on his own views,
finds few to agree with him*

*Lao Tzu
(6th century BC–5th century BC)*

How can we organize a team to work effectively under conditions of hypercomplexity? How can a team deal with needs, opportunities, and threats that emerge or change during the project? How do we utilize what is known beforehand and what is learned during action by individual team members? How do we integrate individual fields of expertise?

Any team is *heterogeneous* in one way or another. Even when all members have a similar educational background, there will still be differences in their knowledge, areas of expertise, and areas of focus.

Teams connect across organizational boundaries. Team members collaborate with customers, users, and individuals in their professional networks, and through

these connections, they access knowledge areas relevant to developing superior solutions. Sometimes, teams are indeed designed to be inter-organizational—for example, when developers and customers or users form teams together.

Heterogeneity may help a team work effectively and efficiently under conditions of hypercomplexity. Such conditions will lead team members to constantly reflect on whether the project makes good sense:

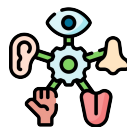
- Is the core problem and the context in which it appears stable and well understood?
- Are the planned interventions well chosen and applied?
- Is the scope of the problem meaningful given our resources? Have we selected the most effective means to amplify our contribution?
- Can we make reasonable decisions about whether to stay the course or take a new direction?

These questions are diverse and require various insights. One way to employ such insights might be to assign individual responsibilities. Therefore, Essence's strategy is to allocate a role to every team member. Roles serve to maximize existing differences among team members or to create differences even in the most homogeneous team.

Essence roles span different domains. This usually requires sharing insights and experiences, and one principal way to do so is through interaction among people. More than just listing requirements for a solution, we want to understand the needs and, at the same time, appreciate options for contributions that can address them. Roles are designed to facilitate learning, focus on what matters most, and ensure a balanced view of what needs to be addressed and what contributions would help meet these needs.

2.1. Four Core Activities

The four roles reflect four core activities in Essence projects (Figure 2.1):



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Perceive—understanding the problem and its context. This activity develops a mental model of the problem, including how it manifests, the environment where users are located, and the surroundings in which the design will operate. The icon represents how perception involves sensing, reflection, and communication.

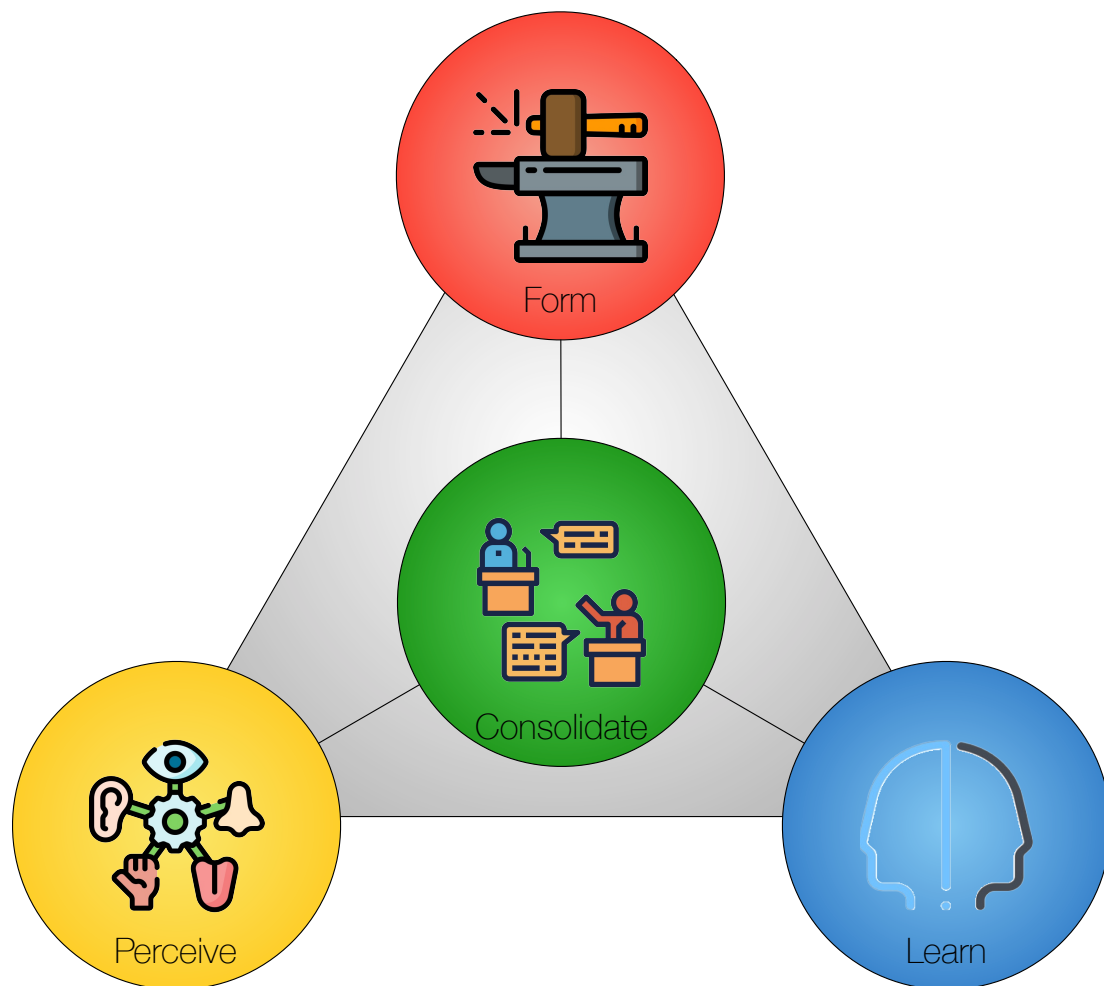


Figure 2.1: The Four Core Activities



Form—designing the constructive parts of a solution. This activity explains how components of a solution can be shaped or configured, what capabilities these components offer, and which leverage points to focus on. The icon illustrates how forming includes shaping, prototyping, and implementing.



Consolidate—integrating the design contributions into a clear whole that

addresses the situation. Consolidation focuses on the project's purpose and main idea, its strengths and weaknesses, and its overall value to the stakeholders. The icon shows how consolidating involves critical review, discussion, and decision-making.



- *Learn*—appraising the design. Learning centers on assessing whether we are on the right track: Are we meeting the mission? Is the solution strong and flexible? Will we effectively leverage our current skills while also developing new ones for the future? The icon represents how learning involves taking in and abstracting new knowledge, discarding outdated practices, and connecting knowledge to new needs.

We will break down these four activities into different levels of abstraction to simplify our approach and manage complexity by focusing on details at one level while ignoring others. Therefore, we will discuss these activities at the following three levels of abstraction:



- *Rationale*—the logical basis that makes actions meaningful and helps reason about strategy and tactics for solving the problem. At the rationale level, we identify the problem, consider what leverage we might use to solve it effectively, explore potential solutions, and analyze how this project creates strengths and opportunities. This icon represents how sound projects integrate well-understood problems, strong solutions, and economic factors.



- *Strategy*—the overall plan for addressing the problem, including its scope and context, the essential parts needed to build the solution, and any limitations or constraints that might influence its feasibility. At the strategy level, we also evaluate the potential for future growth. This icon shows how the project's overall goal makes each step meaningful.



Tactics—the manifestations of the problem we aim to address, the capabilities we offer for managing these manifestations, the value our solution provides, and how we will evaluate if we have achieved the project's mission. The icon indicates how each step in a project contributes to or modifies the overall goal based on the experiences gained.

The four activities are inspired by the classic elements: Earth, Water, Fire, and Air. Empedocles of Acragas (c. 495–435 BCE) argued in his *Tetrasomia*, or *Doctrine of the Four Elements*, that all matter is made up of these four elements. The name Essence comes from *Quintessence*, the cosmic fifth element suggested by Aristotle to supplement the four earthly elements.

The classical elements were linked to both objects and individuals. All matter—substances and materials—and the temperaments—states of mind—were believed to originate from these four elements.

The colors in Figure 2.2 reflect these four elements and the classical psychological temperament associated with each of them:

- Earth is associated with a melancholic temperament, which emphasizes foundations, resources, firmness, and balance. The *challenger* role in Essence is derived from this temperament.
- Water is associated with the phlegmatic, focusing on what is up and down, the facts, and the status of situations. The *anchor* role in Essence is linked to this temperament.
- Fire is associated with the choleric temperament, which is characterized as transformative, active, deliberate, and driving. The *responder* role in Essence is connected to this temperament.
- Air is associated with the phlegmatic temperament, which is characterized by changeability, dynamism, ease of movement, and a laid-back nature. This temperament inspired the *child* role in Essence.

These temperaments reflect the conflicting perspectives of ambitious, moderate, stable, and changeable moods in design projects.

2.2. Essence Roles

Essence emphasizes structures, tools, and techniques over process. These qualities encourage skilled individuals to work efficiently and meaningfully without being hindered by process rules.

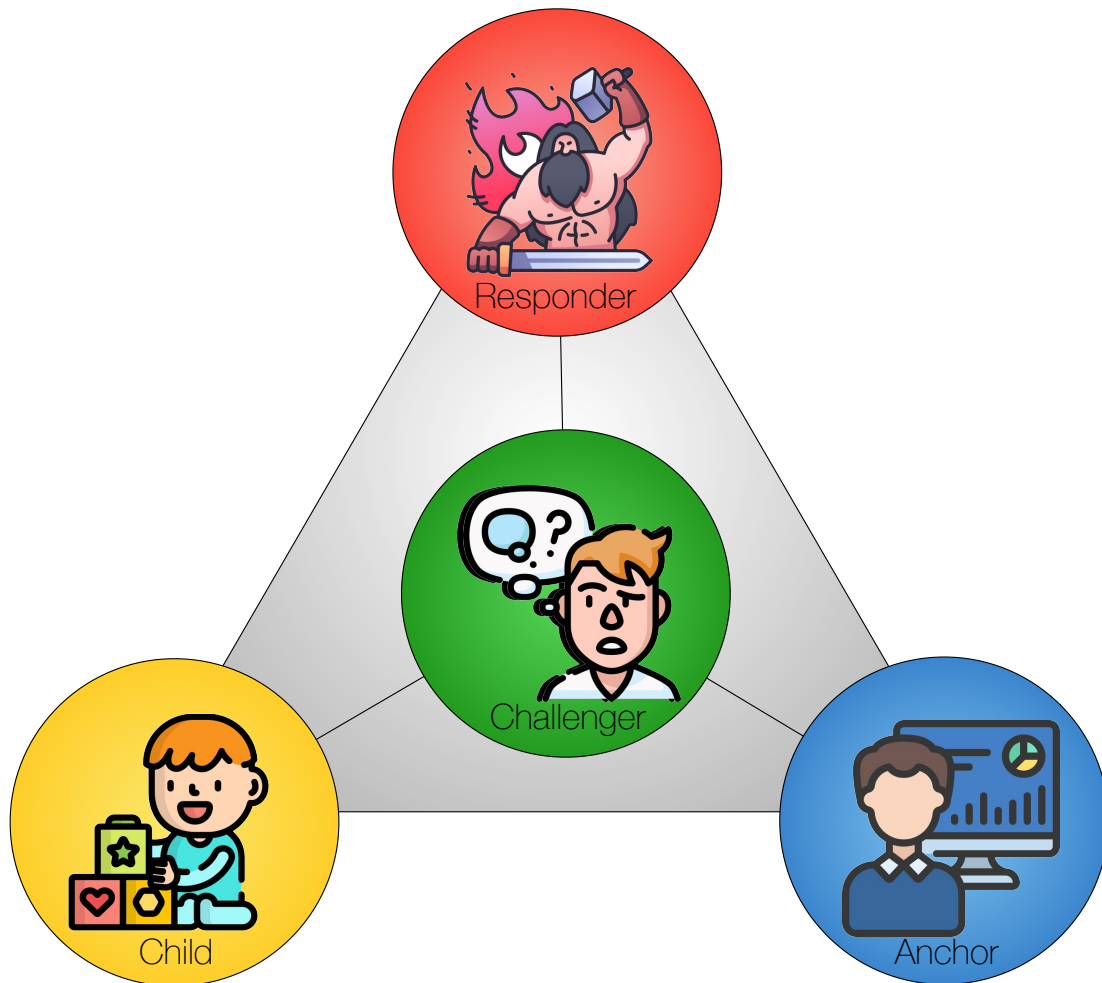


Figure 2.2: The Four Roles

Roles are structural concepts in Essence. They invite team members to take on diverse responsibilities in innovation projects. Just like in role-play, roles—by their characterization—inspire a line of thinking, a range of concerns, and particular viewpoints that may encourage a team member’s engagement in the creative process. The roles support diverse thinking by allowing individuals to view a problem or a prospective solution from different angles. The roles should also provide structure for personal work.

This chapter will present the four roles in Essence: responder, challenger, anchor, and child (see Table 2.1).

The *responder* is a designer and developer who applies a technological viewpoint, has functional ideas, and suggests ways to build and implement contributions. The *challenger* sees the project from a customer perspective and prioritizes and approves ideas for the contribution if they align with the problem. The *anchor* is the team's captain, responsible for liaising with management and outside stakeholders while ensuring that valuation is reasonable and appropriate for decision-

making. Finally, the *child* serves as the leading idea developer—not constrained by rules or laws—and explores alternative ways to perceive the situation and contribute.

	<i>Child</i>	Any team member exploring the <i>situation</i> . Unorthodox and creative. Offers ideas and proposals without responsibility. Questions conventional views on uses, needs, users, and the application of technology. This Role is fleeting.
	<i>Responder</i>	Any developer working on the <i>contribution</i> . Offers technological responses to challenges—always with a view to technological options and affordances that come up during the design process.
	<i>Challenger</i>	The team member who provides the reasoning and politics underlying the <i>solution</i> . Aiming for a solution with a good balance between problem and contribution in the specific situation. Open to new ideas and good arguments.
	<i>Anchor</i>	The team member in charge of <i>valuation</i> . A team captain (although not a team manager) who facilitates credible evaluations and provides a reasonable basis for decisions on whether to pivot or persevere.

Table 2.1: Role definitions

The child role is the only fleeting Essence role; all other roles are permanent, as a responder, a challenger, or an anchor, and they are held at all times except when assuming the child role.

Thomas Nagel wrote a famous paper: *What is it like to be a Bat?* (Nagel, 1974). In this paper, Nagel argues that being in another’s consciousness is impossible. Nagel argues against reductionism and contends that body and mind are inseparable. For that very reason, Nagel argues, it is fundamentally impossible to know what it would be like to be someone else. We can imagine what it could be like, but we can never know.

This limitation highlights fundamental problems in developing solutions for others, as developers often have a limited understanding of the user’s needs. Rephrasing Nagel, we could ask: *What is it like to be a user?* What are the genuine concerns of a user? Which solutions would be helpful? Such problems have been inherent in digital design from the beginning. Still, in innovation projects, we aim to develop contributions to situations that may not yet be fully understood or appreciated by anyone. We, therefore, need strategies to mitigate these problems. The Essence roles are instrumental in facilitating the experience and sharing of concerns among team members and stakeholders.

Roles enable technologists and customers, or their surrogates (when real customers and users are unavailable), to simulate potential solutions as realistically as possible. This way, roles also help technologists and customers or users better understand each other. Nagel’s problem can not be solved, but roles might help.

The role concept is no news to agile developers. Extreme Programming (Beck, 2000) introduced four roles: *programmer*, *customer*, *coach*, and *tracker*. Similarly,

Scrum (Schwaber and Beedle, 2002) presented three roles: *Scrum Master*, *Product Owner*, and *Scrum Team Member*. The Essence roles are related to these agile roles and should be accessible to agile developers.

Roles are also frequently used in management literature. Edward de Bono applies Six Thinking Hats (de Bono, 2000) to support diversity in analyzing a problem as part of creative work. Meredith Belbin recommends nine roles for successful management teams (Belbin, 1981), while Tom Kelley focuses on roles such as the anthropologist, experimenter, and storyteller in innovation projects. Kelley gives detailed accounts of 10 roles (Kelley and Littman, 2005). The characteristics of these roles can easily be aligned with the four Essence roles.

Essence roles are designed to match several concerns:

- Roles should be derived from existing team responsibilities. Essence is not redefining design; it is merely orienting it towards innovation.
- Roles (except child) are persistent. Each team member has a specific role to ensure that this perspective is fully absorbed. A role is not designed for episodic use but for the project's duration. The child's role is the only fleeting role.
- Roles should encourage any team member to utilize their expertise in dialogue with others. The role should fit well with a person's overall work in the team.

Roles may not be quickly learned. Some people experience difficulties identifying with the roles. There can be several reasons for this. One is when Essence is introduced outside of a project, i.e., when no real team is working on an actual project. In such situations, a mismatch may occur between the role and the required knowledge. A challenger must, by definition, be well-versed in the problem, just as any responder must master relevant technologies. Another problem is a mismatch between the role and the personality. For example, a timid person might not feel comfortable with the challenger role.

Two things may partly alleviate problems with roles: Firstly, it might be a good idea to think of the roles as points of departure for improvisation. The roles are designed to introduce diverse viewpoints within a team, based on the project's existing functional roles. Secondly, choosing a role that is most likely to match your career or career plans would probably be prudent. This should make it more likely that roles, personalities, and competencies align.

The following sections will introduce the four Essence roles.

2.3. Child

A child can be any team member exploring the problem. This role is perceptive, unorthodox, and creative. A child offers ideas and proposals without taking on responsibility and questions conventional views on use, need, users, and the application of technology. The role is fleeting.

The child is typically a significant contributor to ideas developed during a dialogue between customers and developers. This role involves exploring and identifying potential contact points and interactions between the designed contribution and relevant users, existing systems, and other ecosystem components. Therefore, the child inquires into the elements of the situation, how a contribution might meet its needs, and how new opportunities might emerge and be exploited.

Traditional customer relations are typically contractual and potentially antagonistic, as contracts outline the obligations and responsibilities of the parties involved. Agile customer relations are generally collaborative and based on partnering, but primarily on the assumption that needs are known and should be complied with. Innovative customer relations must allow inquiry into the project's core to develop interesting contributions. For this purpose, we need a role shared among customers and developers—the child role.

The child role is the only fleeting Essence role. It is used whenever one of the persistent roles—challenger, responder, or anchor—inquires into the situation.

Anyone engaged in idea development and exploration of the situation assumes the child role. Guests and external stakeholders will assume this role whenever they participate in ideation activities. Anyone offering new insights or perspectives on the project takes the child role.

The child engages in four types of activities:

- *Ideation*: Engaging in creative activities to open up alternative ideas or notions of context and needs.
- *Detailing*: Developing or challenging the manifestations or scope and contents of the project using, e.g., wireframes, simulations, or improvisations.
- *Evaluation*: Suggesting or questioning procedures for evaluating ideas or work products, offering alternative assessments, or suggesting other evaluation criteria.
- *Situation or contribution maturation*: Challenging, widening, or narrowing core assumptions of the project.

These activities occur at various times throughout a project, such as when discussing ideas and proposals, working on prototypes or interfaces, or when any team member needs to question the project's goals and objectives. The child role also serves as a safety valve when a team member feels the project may have lost focus or become too narrowly scoped.

Dialogue and role-play are dominant work forms. They use explicit tools and techniques or simple whiteboard sketches. Sketches may be in a free-form or more formal style, such as rich pictures, state-transition diagrams, or sequence diagrams. This way, options and solutions are visualized, discussed, and tested. The child thus helps ensure that ideas are thoroughly thought through.

One final feature of the child role is to reduce fixation problems (Zahner et al., 2010). Ideas produced early in a project often frame later ideas. The human mind frequently thinks along similar lines, and previous discussions and conversations can influence one's ideas. The child role encourages a change in perspective, and this, in turn, could stimulate thinking.

Children should express their thoughts, but they do not necessarily have the right to be heard. Decisions are made and implemented by the permanent roles in Essence; yet, the child significantly influences the ideas and discussions that precede decisions.

The child is optimistic, not given to worry, and believes that difficulties can be overcome. He or she is liberal, peaceable, flexible, informal, not averse to change, and focused on exploring and learning via collaboration.

The child is irresponsible and, therefore, able to continually raise ideas or issues, even when they contradict earlier decisions. He or she will produce ideas for improvements or new functionalities without concern for whether these ideas are feasible or even desirable.

This role has no direct parallel to mainstream traditional or agile development. These two approaches usually assume that contact points and interactions are known a priori. Interfaces are generally treated as given and invariant rather than as something to be negotiated or rediscovered during the project. However, negotiating and rediscovering interfaces and interactions are core to software innovation, as introducing sensors and actuators, big data, IoT, and more may transform the interfaces with humans and components within the situation's ecology.

The child role should help move the project from compliance to reflection. This means that the role should lead to discovering options and constraints relative to the situation as ideas mature in the team. The child role is shared among all team members and thus requires any team member to contribute with particular insights and engage in reflection and idea development. The child's working style is often playful. Playful here suggests discovery through playing out scenarios to build a shared understanding of needs, constraints, possibilities, and options. This could lead to paradigmatic changes, as paradigms are thought patterns,

and the child role can be instrumental in shifting from conventional thought patterns to new ways of perceiving needs, constraints, possibilities, and options.

2.4. Challenger

The challenger is the team member who offers reasoning and engages in politics to consolidate the solution. This role focuses on finding a balanced solution that considers both problems and contributions in the specific situation, while remaining open to new ideas and strong arguments.

The responsibilities of the challenger include both short-term and long-term duties. Short-term responsibilities focus on the current project, while long-term responsibilities relate to the team's commercial future. Will we be able to develop commercial value in light of future opportunities and challenges?

An Essence team comprises individuals with diverse qualifications and backgrounds. Still, most of them will be technology specialists, and the team will typically enjoy relative autonomy. The team members represent diverse viewpoints, and the challenger ensures a customer's perspective and insight into the project. The challenger may incorporate knowledge from future users or business analysts, focusing on the system level and across subsystems.

The challenger is the team's primary source of knowledge in the problem domain. The challenger shares knowledge about the problem and its context, thereby contributing to prioritization, informed decision-making, and the acceptance of solutions.

The challenger acts on behalf of the project's beneficiaries. The challenger links the team to external information sources as a boundary spanner. The challenger provides situational knowledge and identifies constraints from the application area. Still, the challenger is, first and foremost, the project champion and the one in charge of translating a problematic situation into a viable project that arrives at a meaningful solution.

The challenger is responsible for overall project planning, but planning must be open-ended and allow for experimentation. Whereas planning is usually based on resources and known assignments in the form of tasks, Essence's planning focuses on refining problem manifestations and realizing solutions for these manifestations to ensure they are viable, valuable, and contribute to the overall contribution. This is a key responsibility for the challenger.

The challenger is committed to being available to developers whenever they need information. The challenger reviews progress and re-evaluates priorities to optimize the contribution's value and ensure alignment with known user and customer needs.

The challenger thus has several functions, as he or she must:

- Be responsible for identifying the main manifestations of the problem.
- Engage in idea development.
- Prioritize features and formulate value propositions that cater to the customer's needs.
- Accept or reject contributions.
- State the overall reasoning behind the solution offered by the project.

A good challenger is sensitive, intuitive, and empathetic. They seek to be creative and self-thinking. As a minority on the team and with leadership responsibilities, a challenger must be able to work independently alongside other team members without being overly influenced by them.

The challenger role can be very demanding and requires qualifications that are difficult to find. Often, these qualifications are not readily available, and workarounds must be devised for compensation.

Having a challenger who is a real customer is desirable, as it fosters consistency between the ideas generated and the expected outcome of the finished design. Sometimes a real customer is unavailable, and the project may use a surrogate challenger, such as a sales representative or a product line manager with expert knowledge of the situation. This corresponds to a Surrogate Customer in Extreme Programming (Cao et al., 2004).

In traditional software development, this role overlaps with the customer and the project manager, as both roles prioritize features and plan tasks. In agile development, the role corresponds to the customer representative, known as the On-site Customer or Product Owner, across various approaches. The crucial difference is that innovation projects, by nature, are open-ended, as they address a problem rather than an assignment. This requires the challenger to perform their functions in a way that allows for reflection and experimentation.

In summary, the challenger role requires the project to address the situation rather than simply completing a list of assignments, and helps generate and mature the contribution.

2.5. Responder

A responder is any developer who forms solutions to the problem. The responder offers technological responses to challenges, always considering technological options and affordances during the design process.

In Essence, responders actively support the challenger in discovering the project's value propositions and complement the challenger by serving as the primary point of access to technological options and constraints. Responders are the

ones who create the project's tangible results. They build prototypes and solutions based on the team's combined knowledge and ideas. They provide an overview of relevant technologies and help the challenger choose among proposals.

This role is central to translating the problem into a clear end goal for the team. Ultimately, ideas must imply doable solutions with significant value. The exchanges between experts—personified in responders and challengers, respectively—produce input to this translation process.

The responder thus has several responsibilities as he or she must:

- Point out alternatives and options when building contributions.
- Help evaluate strengths, weaknesses, opportunities, and threats.
- Engage in idea development from a technological perspective.
- Help assess the feasibility and potential of wanted capabilities.

These responsibilities aim to enhance what technology can offer to foster innovation. Software innovation projects, by nature, are exploratory and experimental, which puts a special responsibility on the responders to seek out the most promising technological alternatives for the project.

The responders are responsible for creating a design that supports the desired capabilities and is conducive to additional capabilities as the project unfolds. This design must be flexible to accommodate new ideas, for example, by utilizing classic design strategies such as *information hiding* (Parnas, 1972) and *near-decomposability* (Simon, 1996) to minimize the number of components affected by any design change. Such strategies minimize path dependencies by reducing costs, risks, and the time required to implement design changes.

Typically, a team consists of several responders, and in projects that combine multiple technologies, responders may specialize in different areas.

Responders initiate or engage in improvisations, during which various sources of leverage and design options are tested—such as doing spikes as known from Extreme Programming—to ensure that the solutions being considered are feasible.

Responders are masters; they offer technical alternatives and imagination, point to technological opportunities and threats, evaluate and experiment with the challenger to aid in prioritizing and selecting options, and ensure that the contribution is based on technologies with the greatest potential.

The responder is an ambitious and generous developer who offers an overview of what is possible and valuable, helps decide on alternatives, and can implement the chosen solution.

A responder is dynamic, confident, and impulsive. They combine creativity with the ability to collaborate with other responders. This requires the responder to work independently while also being a team player. They are persistent—but not insistent—in promoting their ideas and arguing their value.

The responder strives, on the one hand, to identify technological potential relevant to a valuable solution and, on the other hand, to ensure that such potential is chosen in consideration of the available resources. In other words, the responder ensures that technologies are selected to maximize the value proposition within time, money, and competency constraints. The responders collaborate with the challenger to establish a clear and shared goal for the team. While the challenger acts as a boundary spanner, integrating contributions within the project environment, the responders verify that these contributions are realistic and consistently engaging for the team.

In traditional software development, this role corresponds to the *developer*, who usually contributes knowledge in areas such as analysis, design, programming, testing, or documentation. In agile development, the responder role corresponds to roles such as the *Programmer* in Extreme Programming (Beck, 2000) and the *Scrum Team Member* in Scrum (Schwaber and Beedle, 2002).

In summary, the responder should ensure that the situation is addressed with effective technological solutions rather than merely meeting requirements. Therefore, the responder engages in feature development, focusing on discovering promising combinations of technological options and user needs. The responder is responsible for identifying technologies that hold promise for the project without jeopardizing its implementation. The role of the responder in feature development is to keep doors open to technical alternatives, find ways to deliver desired features, and ensure strong designs. Thus, the responder takes charge of building technologically innovative yet sound products. The responder's working style is ambitious, generous, and persistent under the auspices of the challenger, who has the final say in disputes.

2.6. Anchor

The anchor is the team member responsible for learning. The anchor is a team captain (though not a team manager) who facilitates credible evaluations and provides a reasonable basis for deciding whether to pivot or persevere.

The responsibilities of the anchor include both short-term and long-term duties. Short-term responsibilities focus on evaluating the current project, while long-term responsibilities relate to the team's technical future. Will we be able to develop technical strengths to prepare for future opportunities and challenges?

The anchor is the team's master of ceremonies—the person who understands the work process, organizes and stages special events such as reviews and sprint planning, and ensures the project runs smoothly and the team learns from experience.

The anchor is the project's primary facilitator. They are responsible for administration, liaising with external stakeholders and management, and managing the team. The anchor supports the team's working processes and provides tools and techniques as needed. The anchor ensures the team is functional, productive, and sheltered from disturbances.

This role is central to maintaining the *flow* (Csikszentmihalyi, 1996) of the project. A good project is in constant flux as it iterates toward answering the situation. What has been achieved and settled, along with the ideas for what to do next, must be evaluated and put into perspective. The anchor facilitates this work.

In the Essence team, the anchor is responsible for providing stability when the team engages in chaotic situations as part of their creative work. The anchor offers stability and confidence in uncertain situations, represents calmness and moderation within the team, and facilitates experimentation and evaluation. They usually serve as responders with additional responsibilities for the project's evaluation and process aspects.

The anchor offers methods and tools for the team whenever needed. Methods and tools should facilitate work and be used with discretion. They are aids and must never take center stage in the project.

More than being a master of ceremonies, providing methods and tools, the anchor has essential social responsibilities. He should focus on the team's social dynamics, ensure the other roles function as intended, and remove any social impediments. The anchor ensures that the team stays focused on the challenge and makes valuable contributions.

The anchor must

- Ensure that the team is functional and productive.
- Suggest and adapt methods, techniques, and tools to support work.
- Ensure fairness in evaluations via appropriate methods and criteria.
- Take care of 'foreign affairs' by representing the project to stakeholders and shielding the team from interference during sprints.

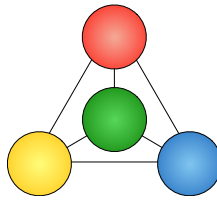
The anchor is particularly responsible for ensuring team reflection about solutions and work processes. Therefore, the anchor is also key whenever visions, ideas, and solutions are discussed and evaluated.

A great anchor is deliberative, reliable, relatively unaffected by pressure, and somewhat reserved. He or she remains calm and views things from a neutral perspective, without bias.

In traditional software development, project managers perform the anchor functions. The project manager is responsible for planning projects, allocating resources, supervising tasks, reporting progress, conducting reviews, and evaluating outcomes. In agile development, the project manager's role is rarely detailed, as agility relies on self-organizing teams. Still, the above-mentioned functions of the

anchor roughly translate to agile roles such as *Coach* and *Tracker* in Extreme Programming (Beck, 2000) and *Scrum Master* in Scrum (Schwaber and Beedle, 2002).

In summary, the anchor role ensures the team can work in a friendly, productive, focused, and creative manner, always aiming to produce results as valuable as possible under the given circumstances. This means that the anchor strives to maximize the potential of creative processes and ensure that evaluations focus on creating value propositions that matter to customers and users. The anchor's working style is pragmatic, deliberative, disinterested, and stable.



Chapter 3

The Canvas Building Blocks

First, the taking in of scattered particulars under one Idea, so that everyone understands what is being talked about ...

Second, the separation of the Idea into parts, by dividing it at the joints, as nature directs, not breaking any limb in half as a bad carver might

*Christopher Alexander
(1936–2022)*

The quote is based on Plato's Phaedrus 265D and E.

A system's design should convey its purpose, plan, and intent. At the same time, it should highlight the key components used to develop the system's primary features. This allows designers to define how the system interacts with the external environment and outline its internal architecture.

The *Problem–Solution Canvas (PSC)* helps designers develop the overall concept of the design and its interconnected parts. Thus, the canvas illustrates the overall design and breaks it down into 12 building blocks for more effective analysis. Each block concerns a key question about the problem and its solution.

The PSC is a *visual inquiry tool* designed for use by designers. It is intended to be used similarly to how designers and architects utilize prototypes, except that here, the prototype is a more abstract representation of a design than traditional prototypes. Visual inquiry tools support collaborative problem-solving by helping to explore options, generate ideas, and apply design thinking techniques (Avdiji et al., 2020). The canvas is designed to be useful throughout development, suitable for increasingly complex problems, and helpful when unforeseen challenges arise.

Dividing a canvas into 12 building blocks helps us understand the overall idea while also recognizing it as composed of individual parts. This way, we can discuss how changes to one part affect the others. If we encounter surprises or challenges in our project, the canvas lets us explore different ways to address them and discuss options.

In his landmark book, *The Sciences of the Artificial*, Herbert Simon argues that everyone *designs who devises courses of action aimed at changing existing situations into preferred ones* (Simon, 1996, p. 111). Simon thereby portrays design as a problem-solving activity in which goals are achieved under constraints.

Viewing design as a problem-solving activity doesn't necessarily mean starting with a problem and then designing a solution. This sequential view of design may not always be the most effective approach.

As von Hippel and von Krogh propose, a need and a solution are often discovered together and tested for viability as a *need-solution pair* (von Hippel and von Krogh, 2016). Nambisan et al. rephrase this when they talk about *dynamic Problem-Solution design pairing* (Nambisan et al., 2017, p. 226).

Essence is not based on assumptions about causality, sequence, or process. Solutions and problems may arise at random, but a thoughtful design will eventually provide a cohesive view of them. The Problem-Solution Canvas aims to offer a unified perspective and serves as a visual inquiry tool to facilitate this.

The Problem-Solution Canvas consists of 12 building blocks as defined in Table 3.1. Each building block is colored to indicate its relation to the core activities in Figure 2.1. The block names are printed in green, blue, or red letters to indicate their relation to rationale, strategy, or tactics, respectively (Section 2.1).

The following sub-sections present the 12 building blocks in the sequence indicated in Table 3.1, and demonstrate their practical application through a case study—*ClusterDetect*. The Problem-Solution Canvas, which combines all 12 building blocks, is shown in Table 4.1 on page 64.

ClusterDetect is a small design project focused on safety in crowded areas, such as concert halls, festivals, disaster zones, and similar locations. Where will we find the most congested areas in a place? How do we estimate the density of a cluster? We may need some indication of density to reduce the risk of super-spreading events during pandemics, prevent crowd crush disasters, and coordinate rescue operations, among other purposes. Our market strategy suggests that the

system hardware should be based on off-the-shelf components to target a diverse and cost-conscious market.

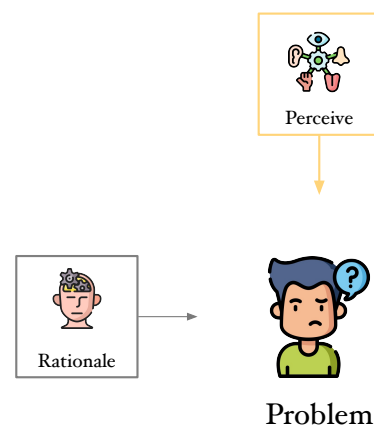
Problem  We see a problem as a situation to be handled, where problems appear as manifestations.	Outer Environment  The problem is found in the outer environment. The design is adapted to this domain.	Manifestations  Manifestations are objects, actions, or events that reflect or embody the problem.	Capabilities  Capabilities are the features that are designed to address manifestations.	Inner Environment  The inner environment is what provides the capabilities.	Leverage  Leverage points allow for the most valuable capabilities.
Solution  Why the problem is resolved, why it is important, and why the result is appealing.	Evolvability  The capacity to adapt to outer environments or to add new leverage points to the inner environment.	Merit  The value offered by the capabilities. The limitations of the solution and why these limitations are acceptable.	Mission  The mission presents the project's tactical objective: to solve the problem.	Potential  The potential defines the project's strategic objective: to create strengths and opportunities.	Horizon  Where might the outcomes of this project lead us? How do they align with the future we aim to be a part of?

Table 3.1: The 12 Building Blocks

ClusterDetect could use a camera, such as a standard 4K optical video camera, mounted fairly high above the ground to provide an overview of the relevant area. Such a system would only require simple on-site calibration for camera height and angle.

3.1. Problem

Conceptually, *problem* is a delicate term. A problem may seem simple, easy to understand, describe, and analyze. However, when we try to identify a problem, explain what it is, define its boundaries, or state what we expect from a solution, things tend to become complicated.



The icon above explains why the term problem is puzzling. A problem isn't a physical object; it has no body, taste, or sound on its own. It's a challenge rooted in something you observe or encounter. A problem cannot be separated from a person: an individual, a group of stakeholders, a scientific community, or other social entities see the problem, and they perceive it within a specific context, such as a place, time, or condition — and quite often, these perceptions differ.

The problem provides the rationale—the foundation—for a meaningful project, and defining a problem is a highly creative, iterative process. This process is closely linked to securing resources, defining scope and focus, setting priorities and goals, and motivating team members.

The *child role* discussed in Section 2.3 is intended for exploring the problem domain, as this role should combine empathy and creativity. Empathy is essential for understanding others and what matters to them, while creativity enables us to see problems from fresh perspectives, such as those arising from emerging technologies.

The problem building block defines the difficulty or challenge that must be overcome in our design. This serves as the foundation of the design project: what needs to be addressed and what defines the relevant scope?

We will use the ClusterDetect system as an example. This system tackles a common challenge and could, for instance, be described like this:

In crowded areas, safety could be compromised.

The problem is a combination of issues that, when viewed together, are seen as one. A problem is complicated but understood as a single thing. This problem is the primary focus of the project.

The problem provides a rationale for investigation and the development of solutions. Recognizing a problem involves perceiving the situation in which it occurs and exploring possible ways to solve it.

Essence defines *problem* this way:

A problem reflects an understanding of a situation and is used as an analytical instrument. It is formulated as part of inquiring into a problematic situation. Problems, therefore, have no objective status.

We view a problem as a situation to be handled, where problems manifest as specific issues. A problem is a situation or condition we aim to solve—something difficult to accomplish. When we examine a problem, we notice how closely it is connected to where we are, who we are, what we know, and more.

Problems reflect our worldviews. What we see as a problem is influenced by our knowledge, experience, and circumstances. A teacher might notice different issues in a school than a student, a janitor, a headmaster, or a parent. Who we are shapes the problems we perceive and the solutions we consider. Our knowledge, norms, and resources shape how we perceive a problem and what solutions we consider feasible.

In Essence, problems serve as tools for our investigation—we define problems as part of our inquiry into them. Problems represent our understanding of a situation; therefore, they lack objective existence.

Whenever a problem is expressed in words, this expression serves as an analytical instrument: We formulate it as part of investigating a situation. How we conceive the problem is crucially important since it determines (Dewey, 1938, p. 108):

- What mental representations are used to understand, reason, and address the situation?
- What hypotheses are conceived, valued, or considered?
- Which ideas are accepted and which are dismissed?
- What data is perceived, selected, or even discarded?

If we mistake the problem, we will not inquire with relevance. If we have no problem, our efforts will be *blind groping in the dark* (Dewey, 1938, p. 108).

Problems are contextual. When addressing a problem, we carefully consider the circumstances in which it arises. We strive to understand the needs with a

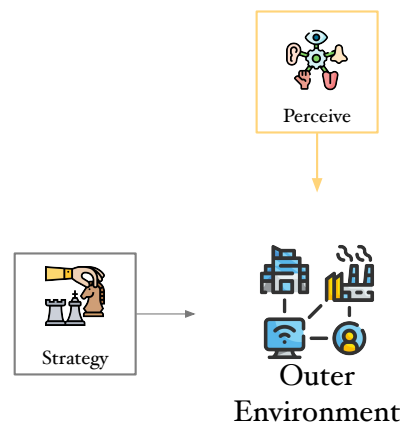
view to the resources available to us. We reflect on who is affected by the problem, when and where our contributions might be most useful, and the specific circumstances that need to be addressed. Expressing the problem in words helps us identify any misunderstandings and resolve any misalignments in our goals. This process encourages us to evaluate whether we can achieve all our goals simultaneously or if there are conflicts that need to be prioritized.

We usually become aware of a problem when it manifests in various ways. These manifestations are the visible or tangible parts of the problem we experience. Our understanding of the problem is primarily shaped by the signs we notice, which significantly influence how we perceive the problem and the methods we consider for solving it.

Problems are discussed in Chapter 6, where you will also find suggestions on how to get started on this building block.

3.2. Outer Environment

The *outer environment* building block concerns the system's surroundings. The name comes from Herbert Simon, who used the term to refer to the surroundings in which an artifact operates (*Simon, 1996, p. 6*).



As shown in the icon above, the outer environment is at the strategic level. What we decide to include in the outer environment reflects how we scope the project, and how we scope it in turn shows how we perceive the problem and what we believe we can do about it.

The outer environment contains the infrastructure on which we base our strategies. The outer environment encompasses the infrastructure that underpins our strategy and operations. It includes the objects we consider essential to our problem and the events related to it when it manifests.

The *child role* discussed in Section 2.3 is meant to explore the outer environment, as this role can perceive both the physical and social worlds with a fresh perspective, unbound by conventions, and with a lot of curiosity.

Usually, the outer environment is more a choice than a given. It can provide a variety of resources, such as data, sensors, services, utilities, and personnel, which may simplify or, in some cases, complicate the project. Therefore, selecting what to include or exclude is a strategic decision.

Essence defines the outer environment this way:

The surroundings in which our problem resides and where the contributions we design will operate. External services, implements, repositories, and people are examples of elements in the outer environment.

The problem is found in the outer environment. The design is adapted to this domain. Similar to leverage (Section 3.6), the outer environment is a complex concept that might include one or more of these categories:

- *External services*: Services that are used without modification and reduce what is left for us to deal with. For example, marketing information services, equipment tracking services, streaming services, digital infrastructures such as Starlink, cellular networks, and more.
- *External implements*: Equipment, sensors, or actuators available in the outer environment and accessed via trivial interfaces.
- *External repositories*: Repositories for collecting, storing, disseminating, and/or aggregating information via trivial interfaces.
- *External people*: Generally available people, professionals, stakeholders, authorities, networks, organizations, etc.

Trivial interfaces indicate that items of these categories are used as is, and anyone else can use them just as well as we can. Streaming services, weather forecasts, government databases, public webcams, municipal services, public information desks, and more exemplify items we can use just like anyone else. Choosing which items to include in the outer environment can either decrease or increase the project workload and may also affect our ability to modify or reuse our design for different situations.

In Essence, we assume the world to be hypercomplex, ambiguous, and uncertain, and in such a world, we cannot presume the outer environment to be given or known. Our world is constantly and dynamically shaped by the assumptions and interactions we encounter while designing our contributions: *Designing is learning*. A settled outer environment reflects our understanding of the problem domain at a particular point in time. In Essence, we assume that the conditions for goal attainment may change as we gain experience through designing and trying out.

In the *ClusterDetect* example, we might use one or more outer environment categories:

- *External service*: Access control systems—ClusterDetect could be designed to control access points such as gates and doors in critical situations.
- *External implements*: Venue information system—ClusterDetect could use such systems to inform the audience via screens or loudspeakers.
- *External repository*: Traffic information repositories might provide traffic count data or information about open or closed routes during emergencies.
- *External people*: Informants—perhaps a visitor module in the system can utilize in-app reporting to provide a better understanding of the situation. However, this solution could encounter problems with potentially misleading information.

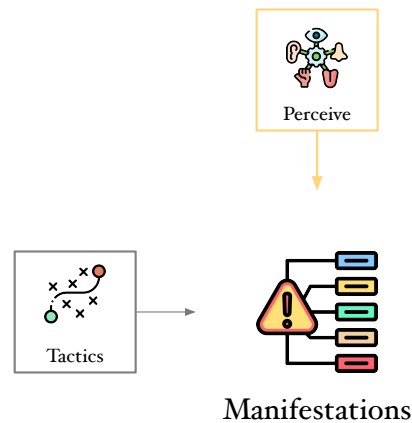
Deciding what to include in the outer environment depends on many factors. Therefore, we might keep a list of options we have not yet chosen.

The outer environment is discussed in Chapter 11, where you will also find suggestions for getting started on this building block.

3.3. Manifestations

Problems are not monolithic, indivisible, or uniform; they are complex and take many forms. We view a problem as a whole, but when addressing it, we must consider its numerous manifestations, which become clear or visible in specific situations.

Manifestations represent the problem at the lowest level of abstraction: The tactical level. We face manifestations, and the design project is fundamentally about developing capabilities (Section 3.4) to address them.



The icon above illustrates how the manifestations of a problem are perceptions shaped by tactical choices. People notice these manifestations, and their importance varies depending on them and their preferences. Even if we can see, feel, and measure a manifestation of a problem, it ultimately depends on people to decide whether to take action and address it in the project.

The *child role* discussed in Section 2.3 is meant to suggest and explore manifestations unbound by conventions, and with a focus on options. Should this manifestation be included in the project? Should it be addressed more thoroughly than usual in similar projects? Or less? Are there manifestations that haven't been included in similar projects before?

Essence defines manifestations this way:

An object, event, or action that reflects or gives a tangible or visible form to the problem. Capabilities might match such objects, events, or actions as part of the project's contribution.

Manifestations are objects, actions, or events that reflect or embody the problem.

The word manifestation refers to making something perceptible to the senses. In Essence, a manifestation reflects or gives a tangible or visible form to parts of the problem. Observations of manifestations help identify and understand specific aspects of the problem and potential ways to address them.

Returning to the *ClusterDetect* system, we may wonder about the underlying motivation for solving the problem. This interest is part of the situation and helps us select and scope contributions. For ClusterDetect, we aim to prioritize safety in crowded locations where obstacles to movement may delay emergency responses, lead to crowd crushing, or create other safety risks.

Based on this, some immediate manifestations could be:

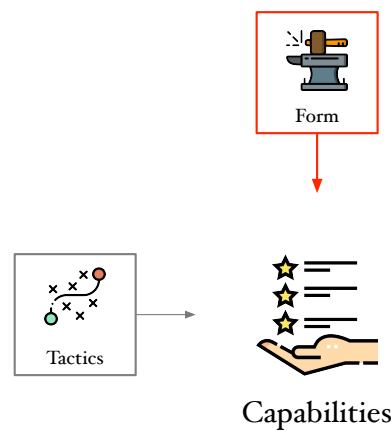
- Potential crowding—locations where population density might rise.
- Places of congestion—locations where freedom of movement is impeded.
- Potential starts of panic—outbreaks of collective anxiety or agitation.
- Potentially disruptive behavior—accidents, fights, or other events that might agitate a crowd.

When we understand the underlying interests in solving a problem, we can clarify its manifestations and explore ways to address them. Manifestations help identify risky areas, hazardous events, or actions that might trigger such events. Once we consider how to handle manifestations, we start thinking about functional capabilities.

Manifestations are discussed in Chapter 16, where you will also find suggestions for getting started on this building block.

3.4. Capabilities

Capabilities and manifestations are deeply connected. Manifestations (Section 3.3) are objects, events, or actions where we sense the problem and conceive the need for contributions. *Capabilities* are specific contributions aimed at managing manifestations. Capabilities—like manifestations—are thus addressed at the tactical level.



The icon represents a set of capabilities that help address various manifestations of the problem. These capabilities are developed to be helpful and valuable to stakeholders.

At the tactical level, each capability serves as a partial step toward a comprehensive solution to the overall problem. This building block is thus notably constructive, with every added capability evaluating the effectiveness and efficiency of the overall contribution while also assessing whether the problem has been appropriately identified.

The *responder role* discussed in Section 2.5 is intended to suggest and explore capabilities based on their expertise and specialties. Since the team may include several responders with different or complementary strengths, there may be opportunities to consider competing ideas before selecting a specific implementation of a capability.

Essence defines capability this way:

An ability or facility devised to handle manifestations and provide value as part of a solution to a problem.

Capabilities are features designed to address manifestations, and, in general, we want a correspondence between manifestations and capabilities. This means that every manifestation should be matched by one or more capabilities and vice versa. This way, we can discuss the distinctive contributions of each capability in a meaningful manner.

Conceptually, capabilities are high-level. A capability, for example, could be *customer profiling*, rather than a set of *Create*, *Read*, *Update*, or *Delete* operations, if we design a system for online sales. Conversely, if we want to optimize file operations or memory use, a read operation might be considered a capability.

A solution should provide capabilities to handle the manifestations effectively. Every capability offers *value* that will be evident to the stakeholders.

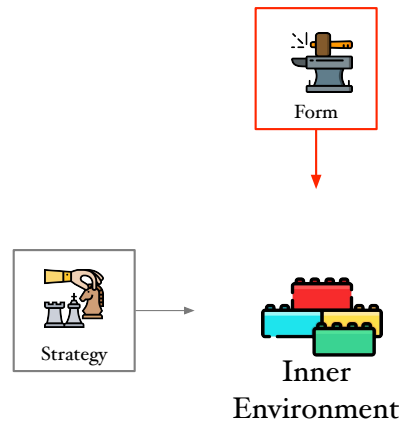
The *ClusterDetect* example illustrates what capabilities could be:

- Identify and manage crowds where population density exceeds a specified threshold.
- Identify and handle congestion—areas where crowds slow down.
- Identify and handle panic—indications that suggest crowd panic or agitation is occurring.
- Identify and handle disruption—actions that might agitate a crowd.

Capabilities are discussed in Chapter 17, where you will also find suggestions on how to get started on this building block.

3.5. Inner Environment

The inner environment building block describes the architecture and main components used to exploit leverage points (Section 3.6) and develop capabilities. Inner environment is a term introduced by Herbert Simon. He uses this term to denote the *substance and organization of the artifact itself* (Simon, 1996, p. 6).



The icon depicts the inner environment as composed of the fundamental architectural elements that contribute to it. This building block is situated at the strategy level, indicating that strategic decisions are made with regard to both short- and long-term considerations, flexibility, preparedness for uncertainties and risks, among other factors.

The *responder role* discussed in Section 2.5 has the primary responsibility for exploring the inner environment, as this role reflects the expertise and specializations of developers within the team.

The inner environment is what provides the capabilities. Conceptually, Essence defines the inner environment this way:

The substance and organization of what we design and implement.

Simon defines design as concerned with the inner and outer Environments. To Simon, design is about *attaining goals by adapting the former to the latter* (Simon, 1996, p. 113). This adaptation is bi-directional. As Simon says:

If the inner environment is appropriate to the outer environment, or vice versa, the artifact will serve its intended purpose. (Simon, 1996, p. 6)

Building this relationship between the two environments is dynamic and iterative as we learn more through implementation and reflection.

To be appropriate, the inner environment must be reasonable in three ways:

- The boundary between the inner and outer environment must balance the *scope of work* with the *scope of effect*. The contribution must be relevant and worthwhile.
- The leverage points must be exploited in the inner environment to reap potential benefits.
- The relationship between the inner environment and capabilities must be tight. The inner environment must reflect design values such as simplicity, clarity, flexibility, effectiveness, and verifiability. This means designing with a clear focus on current and future needs, without over-engineering or 'gold-plating'.

In the ClusterDetect example, we might consider these internal environment modules (square brackets indicate options pending final decision):

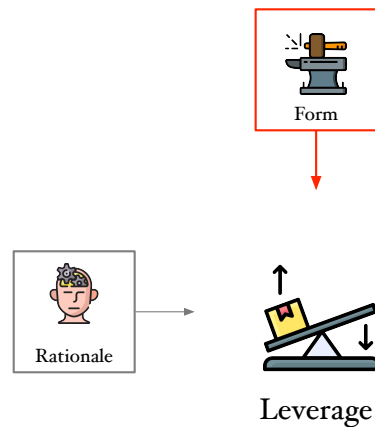
- Density monitor [Thermal?]. This module would utilize state-of-the-art AI-based video analytics technologies, possibly in conjunction with a thermal camera. It will support crowd location, congestion, panic, and disruption capabilities.
- Panic and disruption locator. This module would utilize recent or new AI-based video analytics technologies and support capabilities designed to mitigate panic and disruption.
- General modules for on-the-ground staff and visitors. These modules will support staff and visitor communication capabilities.
- [Customized module for on-the-ground staff]. If built, this module would provide capabilities to support specialized staff.

The inner environment is covered in Chapter 12, where you can also find suggestions on how to get started with this building block.

3.6. Leverage

Leverage is short for leverage points—the keystones of a great solution. A leverage point presents an opportunity for a solution with enhanced features, lower costs, greater flexibility, increased robustness, or other qualities that distinguish it.

Leverage points are crucial for the current project and for preparing the team for future endeavors. Using a new leverage point or expanding existing experience in a leverage point within an ongoing project can help build future strengths. Therefore, selecting leverage points is about both the current project and its potential to generate future leverage.



The icon above depicts leverage as a lever on a fulcrum. A lever amplifies an input force to produce a greater force on the output side. Fundamentally, the choice of leverage point affects both how the current problem is solved and the ability to address future challenges. Leverage is a key part of the project's rationale and forms the backbone of the inner environment (Section 3.5).

The *responder role* discussed in Section 2.5 is meant to identify and explore leverage points based on their expertise and specialties. Since the team may include responders with varying or complementary strengths, there may be opportunities to explore alternative leverage points before choosing a specific direction. Leverage points allow for the most valuable capabilities.

Essence defines leverage this way:

Keystones of the inner environment that help create a great contribution. Technologies, components, information, and human resources are examples of keystones.

This definition of leverage demonstrates that leverage is a complex concept that may involve one or more of these categories:

- *Technology*: Scientific knowledge, machinery, and equipment on which a design might be based.
- *Components*: Rare equipment, sensors, or actuators that will be accessed via crafted interfaces to build a solution.
- *Information*: Used for collecting, storing, disseminating, and/or aggregating information via crafted interfaces.
- *Human resources*: Specific users, categories of users, experts, authorities, social or professional networks, organizations, or others with particular functions in the resulting solution. Cyberhuman systems are prominent examples of systems where internal people are leverage points.

Crafted interfaces suggest that specific categories of items require particular skills or expertise to handle effectively. If we use a standard online service or a regular camera, we won't have any advantage over others, since anyone can use these as effectively as we can. Leverage should give us an edge over others.

In the *ClusterDetect* example, we might use one or more leverage categories:

- *Technology*: AI-based video analytics. Several technologies are available, but a new approach to crowd detection—DFNet—was proposed in 2023 and may be considered. DFNet is a deep feature network with multi-scale fusion (Yan et al., 2023). This approach could offer improvements compared to current state-of-the-art methods. The approach is currently in development and may entail some risks.
- *Components*: Thermal surveillance cameras could be used in locations with low visibility or lighting. Such cameras might make the system more useful in various places, but they will also increase installation and development costs. It is uncertain if a larger market will be enough to cover the additional expenses.
- *Information*: Problem spots—maybe the system can use historical data to predict crowded areas and congestion and inform on-the-ground staff to take precautionary measures.
- *Human resources*: If the system targets a more specialized market, such as concert venues, festivals, or theme parks, it could include modules tailored for on-site staff at each venue. Implementing modules that support or coordinate staff can significantly enhance the system's effectiveness. This would increase module development costs; however, the system could still benefit from economies of scale by utilizing generic components across similar installations.

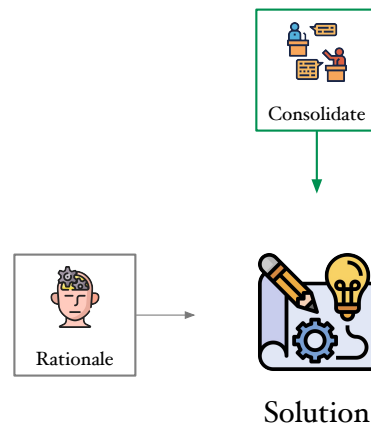
Every potential leverage point has strengths and weaknesses. Deciding on them requires research and, likely, experimentation. Such decisions will have implica-

tions for both the inner environment and the capabilities. To start, we might list options in brackets to indicate that they have not been decided on yet.

Leverage is discussed in Chapter 7, where you will also find suggestions on how to get started on this building block.

3.7. Solution

The *solution* building block explains why our design solves the problem, why it is important, and why it is appealing. In other words, it highlights the alignment between the problem and our proposed solution. This block is assertive—it consists of statements that affirm the solution will be meaningful.



The icon represents our developed solution idea, along with the reasoning behind it. At the rationale level, consolidation emphasizes the purpose and main idea, ensuring that the solution's benefits outweigh its drawbacks.

The *challenger role* discussed in Section 2.4 emphasizes the reasoning and politics of the project. The project should be reasonable in terms of focus, scope, costs, benefits, and relevance both now and in the future. The challenger plays a key role in identifying and responding to changes in both the problem and solution domains, helping the project address these challenges as meaningfully as possible.

The solution explains why the problem is resolved, why it is essential, and why the result is appealing. Essence defines a solution this way:

Why the problem might be resolved by what is to be built, why this is important, and why the result is attractive? Solution combines prospect, warrant, and backing.

From this definition follows that the solution building block is complex and consists of three parts:

- *Prospect*—What we believe to be a solution if used properly. A prospect reflects our understanding of the problem and how to solve it.

- *Warrant*—Why solving the problem is essential. Why is a solution likely to spread to other users and prevail in regular use?
- *Backing*—Why the proposed contribution provides an attractive solution to the problem. Why will the benefits exceed the costs?

We will use the *ClusterDetect* example to illustrate. The prospect names the solution and suggests key contributions; the warrant argues for the need for a solution and the backing reasons why this design is attractive.

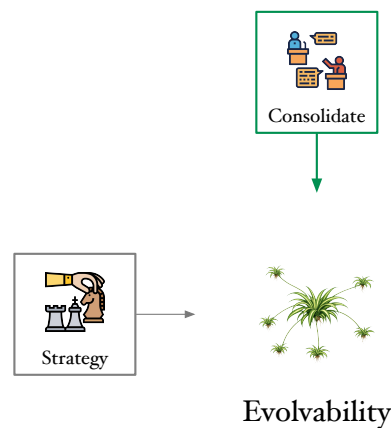
- *Prospect*: ClusterDetect reduces congestion and disruption.
- *Warrant*: Crowded areas are widespread and dangerous.
- *Backing*: The contribution is effective and inexpensive.

Solutions are usually imperfect and come with limitations. The question is how severe these limitations are and whether we can live with them. The *qualifications* building block (Section 3.8) is concerned with these questions.

The solution is discussed in Chapter 8, where you will also find suggestions for getting started with this building block.

3.8. Evolvability

The term *evolvability* refers to the ability to access other markets or contexts. Since digital technologies handle information and data, which exist in any system, we expect designs based on these technologies to have potential across various contexts. In times of hypercomplexity, we need to care about how to spread and adapt a product or idea to address challenges and explore opportunities.



The icon represents our pursuit of fertile ground for our products or ideas. The spider plant sends out runners and spiderettes to find suitable places to grow. Likewise, we should use new projects to face challenges and develop strengths and expertise. It prompts two strategic questions: should we explore new markets? Should we invest in new leverage points?

The *challenger role* discussed in Section 2.4 is instrumental in such discussions by combining problem-domain knowledge with an understanding of available resources. Deciding whether to improve on current solutions in existing markets or to modify and explore new markets depends on priorities, decision-making, and consolidation. The challenger acts as a mediator, attempting to balance the conflicting interests of stakeholders. Central to this mediation is the exchange of information and preferences among stakeholders and team members.

Evolvability helps determine whether we can pursue future growth in current or new environments based on existing and future strengths. Essence defines evolvability this way:

Evolvability is the capacity to adapt to outer environments or to add new leverage points to the inner environment.

Building on the spider plant metaphor, we will focus on the spiderettes. The runners move them to other environments, but once there, the spiderettes must fit with the new surroundings.

If the runners spread to new soil, the spiderettes must be able to settle there; if they reach soil similar to their original environment, they need to evolve to better capitalize on the opportunities available than their competitors.

To make evolvability practical, we introduce two auxiliary terms: diffusibility and adoptability.

Diffusibility is the ability of an innovation to spread into new environments or to reinvent existing ones.

Adoptability is the capacity of an innovation to be adopted.

Diffusibility involves either introducing an innovation to new environments or upgrading or renewing it within current ones. Adoptability describes how easy or difficult it is to implement the innovation in this environment—whether it will require changes in technology, organization, skills, or other areas.

Again, we will use the ClusterDetect example to illustrate. In our example, the system's initial design is based on off-the-shelf 4K optical video cameras mounted above locations of interest. Let us now consider two scenarios:

- Changes to the *outer* environment: we want to enter a new market: security efforts in trains, buses, and metros. The idea is to use surveillance cameras on public transportation to help identify and address threats, crime, and anti-social behavior.
- Changes to the *inner* environment: we want to offer stronger support to the security staff by introducing generative AI to enable early identification and handling of emerging problems.

How would such changes affect diffusibility and adoptability?

The proposed changes to the outer environment are likely to be difficult in both terms of diffusibility and adoptability. Entering the security market for public transportation will require significant updates to technological and organizational systems and will face major compatibility challenges. This implies that diffusibility and adoptability might not support this scenario.

The proposed changes to the inner environment will likely be less difficult if we assume the design is highly decomposable. Adding generative AI to the system could significantly increase the value of the services, thereby boosting the system's competitiveness. The system won't need major organizational changes. This indicates that diffusibility and adoptability might better support this venture;

however, the market remains mostly unchanged, and growth prospects seem limited.

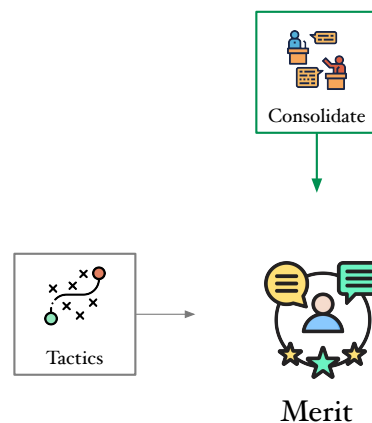
Based on these observations, we might summarize:

- *Diffusibility*—ClusterDetect is highly specialized and has low diffusibility towards new markets.
- *Adoptability*—ClusterDetect is limited to the current market, but it can address new needs and thereby strengthen existing market positions.

Diffusibility is discussed in Chapter 13, where you will also find suggestions on how to decide on strategies.

3.9. Merit

Within law, merit refers to the inherent rights and wrongs of something. In Essence, we use this term to refer to the combination of the value and the qualifications of the design produced by the current project.



The icon reflects our assessment of both the positive and negative aspects of our solution. It raises an important question: Should we accept a solution even if it doesn't meet all our needs? Should we work on making improvements?

Merit is a tactical concern related to consolidating this particular project. It is tactical and consolidating because it highlights the *value* of the capabilities and their limitations. Ensuring that every capability adds value is a vital part of consolidating the project, but so is *qualifying* the overall design for its limitations.

The overall value of a design derives from its ability to solve the problem. We might not be able to total this value as the sum of individual contributions from each capability, but each capability should add some identifiable value. At the same time, we should also qualify the solution by considering its limitations, their significance, and whether it remains acceptable despite them.

The term qualification is ambiguous in everyday language. It can refer to a quality that makes something suitable for a specific purpose or to a limitation that reduces the certainty of a claim, making it less absolute. We use the word in both senses here: qualifications describe limitations in the design and explain why these limitations are acceptable despite any shortcomings or imperfections. We will call limitations *reservations*, and the reasons why these reservations are acceptable will be referred to as *rebuttals*.

The challenger role discussed in Section 2.4 has in-depth knowledge of the manifestations and is therefore capable of ensuring a close alignment between manifestations and capabilities. Hence, this role is crucial in scoping the manifestations and capabilities to optimize the cost/benefit ratios.

The challenger plays a key role in qualifying the solution, given their knowledge of the problem area and understanding of the resources available to deliver a meaningful result.

The challenger acts as a mediator, attempting to balance the conflicting interests of stakeholders. Central to this mediation is the exchange of information and preferences among stakeholders and team members. In this sense, the challenger is a boundary spanner, providing situational knowledge to the team and constraints to the stakeholders.

Essence defines merit this way:

The qualified value of the contributions produced by the current project.

The customer or user perceives the value of this solution based on the importance of its manifestations and the high value they place on its capabilities. This appreciation, in turn, relates to the needs of the customers or other stakeholders.

The merit building block combines value and qualifications. It is complex and consists of three parts:

- *Value*—the overall value of the capabilities.
- *Reservations*—Limitations in the solution (why the design might not support a complete solution to the problem).
- *Rebuttals*—Why limitations to the solution would be acceptable anyway.

Essence does not offer any metrics to evaluate value. If metrics are needed, the challenger must, working with key stakeholders, develop them. The overall value of the design depends on how well its capabilities meet the needs of customers and stakeholders, and the solution shows how effectively it addresses those needs. Together, the reservations and rebuttals qualify the solution.

Again, we will use the ClusterDetect example to illustrate. In our example, the system's initial design is based on off-the-shelf 4K optical video cameras mounted above locations of interest.

- *Value*: Finding crowds, congestion, panic, and disruption, and helping to intervene in risky situations. Locate impediments that reduce freedom of movement.
- *Reservation*—Requires calibration and clear sight.
- *Rebuttal*—Many users will accept these limitations in a low-cost system.

Merit is discussed in Chapter 18. In this chapter, you will also find suggestions for getting started on this building block.

3.10. Mission

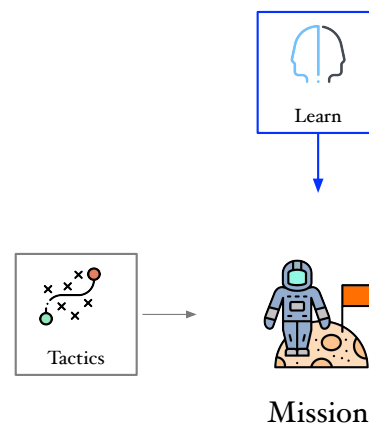
The mission building block aims to validate our design at the tactical level: Will the design solve the problem? Is our mission accomplished?

By mission, we mean the project's primary objective: to develop a system centered on a problem–solution pair that delivers value to users and customers. The mission block indicates whether the assignment has been achieved or completed.

It is crucial to understand what to look for when validating a design. Usually, criteria are used to check if we meet the requirements. However, developing such criteria requires a clear understanding of what matters most. We refer to these ideas as hypotheses, and these hypotheses will vary across abstraction levels.

Hypotheses have a key feature: they express an assumption about the solution that, at a later stage, might prove to be wrong. A failed hypothesis indicates either a flawed design or a revised understanding of the problem. Defining and testing hypotheses is an integral part of a learning-based design approach.

On the tactical level, we focus on the immediate value our solution delivers to the customer or user and question whether it creates enough value for them. A positive answer to this question increases the likelihood of attracting and retaining users and customers.



The icon illustrates the mission as something that can be validated at the tactical level as part of learning about the specific problem and the solution. Italian economist Mariana Mazzucato developed the mission concept, based on the Apollo moon project, as an example of a challenge that drives innovation when pursued (Mazzucato, 2017).

The *anchor role* discussed in Section 2.6 has the primary responsibility for learning. At the tactical level, this learning emphasizes completing the mission—the team learns by doing. Solving the problem includes understanding the effects

of our design. We also develop skills by using tools and techniques in the project. Finally, we learn by applying known or new leverage points (Section 3.6) for the benefit of this and future projects.

The anchor serves as the master of ceremonies, facilitating methods and techniques for the team and conducting evaluations. Facilitating evaluations at the tactical level involves developing hypotheses to determine whether the mission is accomplished, defining criteria for each hypothesis, proposing review methods, and overseeing their practical implementation within the team.

The mission presents the project's tactical objective: to solve the problem. In Essence, the term is defined this way:

The mission defines the project's tactical objective: to develop a solution that attracts and retains users and customers.

As mentioned earlier, we use hypotheses to determine if we achieve our expected goals. A hypothesis is a proposition that serves as a basis for further investigation. It serves as a starting point for evaluating whether the contribution yields a valuable problem–solution pair. Once we establish our mission hypotheses, we need to set review criteria to consider them.

The mission focuses on viability in terms of value for the user or customer. The mission hypotheses concentrate on three other canvas cells: *manifestations*, *capabilities*, and *value*. These three cells describe the circumstances in which the problem occurs, the capabilities designed for those situations, and the value users or customers gain from using these capabilities. These three cells should support the following broad hypotheses to attract and retain users and customers.

- *Manifestations*: The manifestations are essential to the primary users or customers.
- *Capabilities*: The capabilities are efficient and effective.
- *Merit*: The cost/benefit ratio is attractive to the primary users or customers.

The criteria used to validate these generic hypotheses in practice will reflect the specifics of the project in question. In the *ClusterDetect* example, we might develop criteria for topics like these:

- Do we address the essential manifestations for specific users and customers?
- Does the solution deliver the essential capabilities they need? Are these features effective and efficient?
- Do we offer good value to these users and customers? Can we retain them? Can we attract more?

The mission concept is discussed in Chapter 19. In this chapter, you will also find suggestions for getting started on this building block.

3.11. Potential

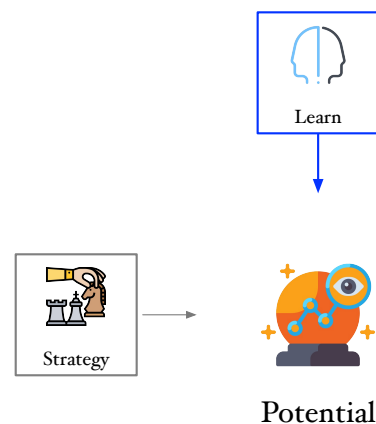
The potential building block emphasizes how the mission's outcome can provide future leverage. While the mission building block emphasizes the immediate value delivered to the customer or user, this block explores the strengths and opportunities for future projects that may stem from the current project.

Potentials are valuable attributes, qualities, and skills that create opportunities for future growth and development. This building block focuses on attracting and retaining new users and customers in the current market or in future markets.

At the strategic level, we focus on growth potential—on the project's longer-term outcomes. A highly specialized solution will likely only be valuable for very similar applications. To ensure growth potential, a solution must be usable by new users within an organization or by customers in a new market.

For that reason, potentials may possess qualities such as:

- *Flexibility* to easily modify the solution by changing or replacing existing components or adding new ones later.
- *Scalability* to ensure the design can handle changes in workloads and user traffic without compromising performance.
- *Robustness* to withstand or overcome adverse conditions such as errors and disruptions.



The icon symbolizes potential as the act of creating something desirable at a strategic level—something rooted in the present that offers hope for the future. This hope combines logic based on information with predictions or guesswork.

The *anchor role* (Section 2.6) is essential for learning. At the strategic level, this learning highlights the broader implications of the current project—what can we learn from this project to help the team prepare for future projects or markets? Can the knowledge gained lead to growth in future projects or markets?

While the challenger role emphasizes creating value for stakeholders and customers, the anchor takes on greater responsibility for the technical aspects of design. To unlock the potential of the current project, the anchor stresses the importance of generalizing and abstracting potential so that it can be applied in broader contexts. It also aims to minimize dependencies and boost robustness, making it easier to reuse parts of the solution in future projects.

Essence defines the potential term this way:

The potential defines the project's strategic objective: to develop strengths and opportunities for growth.

The potential hypotheses focus on three cells: *outer environment*, *inner environment*, and *diffusibility*. These cells describe the possible interfaces to the outside world and the internal trade-offs among flexibility, effectiveness, and efficiency. The cells are crucial for adoption, retention, and growth in an organization or market:

- *Outer environment*: the external interfaces are generalized and abstracted to support growth.
- *Inner environment*: the internal structure is flexible and supports growth through modularization.
- *Diffusibility*: the project's results can foster future growth.

In the *ClusterDetect* example, the specific hypotheses are identical to the generic growth hypotheses. To evaluate these hypotheses in the *ClusterDetect* example, we could consider criteria for topics like these:

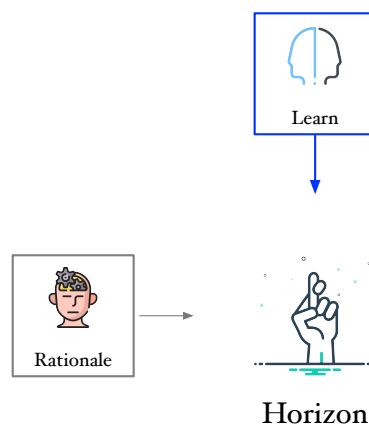
- Are the interfaces suited for this and similar application areas?
- Is the architecture flexible enough to support growth in other areas?
- Does the system support relevant interventions when needed?
- How will camera position and visibility affect accuracy and system setup?
- Will results from this project be diffusible and adoptable to other applications?

The potential concept is discussed in Chapter 14. In this chapter, you will also find suggestions for getting started on this building block.

3.12. Horizon

The horizon cell highlights the link between the mission, the potential, and the direction we want the developing organization to take. We use the horizon cell to ensure that the outcomes of our efforts in this and other projects align with the future we aim to be a part of.

As we complete missions, we build growth potential. The horizon cell links the immediate results with the overall direction we believe the team should pursue in preparation for the future.



The icon represents the rationale for the current project and the long-term strengths we plan to develop. When considering the rationale, both the present and future are taken into account. Fingers crossed symbolize our hope that our expectations will come true and that our chosen path will be meaningful.

The anchor role (Section 2.6) is crucial for learning. At the level of rationale, this learning emphasizes the bigger picture: are we addressing the overall problem? Are we solving it effectively? Is the problem and solution significant? Are we building future leverage points and moving toward future markets through our current activities? Have we identified a meaningful future to strive for?

While the challenger role mainly focuses on generating commercial value, the anchor bears greater responsibility for the technical future. To maximize the benefits of this and upcoming projects, the anchor observes trends and technological advancements that could be crucial to the team's future.

Essence defines the horizon term this way:

Where might the outcomes of this project lead us? How do they align with the future we aim to be a part of?

Thereby, the horizon integrates both short- and long-term perspectives.

Over the long term, we evaluate major trends in technology, legislation, culture, society, markets, the economy, industries, and other factors that could affect us. In the short term, we concentrate on the current project and aim for a meaningful alignment between the problem, the leverage, and the solution.

We need hypotheses for validation both in the short and long term. The horizon hypotheses focus on three cells: *Problem*, *Leverage*, and *Solution*. These three cells address the project's immediate results and the team's overall direction.

Essence proposes these generic hypotheses for the horizon cell:

- *Problem*: The problem covers the issues faced by key users or customers. The formulation should be generic and understandable to both insiders and outsiders, allowing for reuse in future projects.
- *Leverage*: Leverage is strategically vital for both current and forthcoming projects. Leverage should ensure the solution's long-term success and usefulness, promoting its adoption and diffusion over time.
- *Solution*: The solution answers the key parts of the problem in a meaningful way. By answer, we mean the prospect is a significant, essential, and valuable response to the situation, where essential refers to the warrant, and valuable to the backing.

The prospect hypothesized as a solution to the problem is based on what Ries calls a *leap of faith* (Ries, 2011, p. 81). This is the foundation for the design and is postulated without evidence. It must be tested throughout the project.

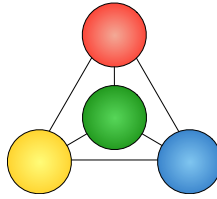
The *warrant* part of the hypothesis—that solving the problem is valued by stakeholders, customers, and users—is inspired by what Ries calls the *growth hypothesis* (Ries, 2011, p. 61). This part posits that the solution will likely spread and become widely adopted among potential adopters.

The *backing part*—that the prospect is an attractive and valuable solution—is inspired by what Ries calls the *value hypothesis* (Ries, 2011, p. 61). This hypothesis posits that the solution is appealing and the benefits exceed the costs.

In the *ClusterDetect* example, the specific hypotheses are identical to the generic growth hypotheses. To evaluate such hypotheses, we could consider criteria like these:

- *Problem*: Is the problem accepted by users or customers? Is it generic?
- *Leverage*: Are the leverage points the keystone for the current project? Are they strategically important for upcoming projects?
- *Prospect (leap of faith)*: Is the prospect meaningful?
- *Warrant (growth)*: Does it address a significant class of problems? Can it create a viable market?
- *Backing (value)*: Does it create enough value? Does it motivate adoption among users and customers?

The horizon concept is discussed in Chapter 9. In this chapter, you will also find suggestions for getting started on this building block.



Chapter 4

The Problem–Solution Canvas

I begin with an idea and then it becomes something else

Pablo Picasso (1881–1973)

The Problem–Solution Canvas is a Visual Inquiry tool ((Avdiji et al., 2020)). The canvas is designed to help explore the problem and various options for solving it using available resources. It helps generate and assess ideas in teams, as well as for communicating with key stakeholders. The canvas shows a current status—the canvas and its building blocks—at a specific point in time. When unexpected challenges arise, this status provides a basis for identifying potential responses and finding effective ways to address them.

The Problem–Solution Canvas is shown in Table 4.1. The table combines the 12 building blocks described in Chapter 3 in four rows—one for each core activity—and three rows—one for each abstraction level as described in Section 2.1.

As mentioned in Chapter 3, two of the building blocks are composites:

- *Solution*: This cell contains three elements: *prospect*, *warrant*, and *backing*.
- *Qualification*: This cell contains two elements: *reservation* and *rebuttal*.

	 PERCEIVE	 FORM	 CONSOLIDATE	 LEARN
 RATIONALE	 PROBLEM	 LEVERAGE	 SOLUTION PROSPECT WARRANT BACKING	 HORIZON
 STRATEGY	 OUTER ENVIRONMENT	 INNER ENVIRONMENT	 EVOLVABILITY DIFFUSIBILITY ADOPTABILITY	 POTENTIAL
 TACTICS	 MANIFESTATIONS	 CAPABILITIES	 MERIT VALUE RESERVATION REBUTTAL	 MISSION

Table 4.1: The Problem–Solution Canvas

Table 4.2 presents an early Problem–Solution Canvas for the ClusterDetect system discussed in Section 3.1 through Section 3.12. Items in brackets indicate options that are yet to be decided.

The remainder of this chapter outlines four ways to get started on the canvas.

4.1. Fulcrums

The Problem–Solution Canvas consists of 12 cells, and getting started can be challenging. Where should we begin? Which parts of the canvas set the foundation for the rest? Which cells should be addressed first?

If Essence were a method, one approach could be to suggest a series of steps. This sequence might include defining the problem, pinpointing leverage points, identifying the solution to pursue, and completing the remaining building blocks in a top-down manner.

Essence, however, is not a method. Methods are used for problems that have been solved many times before. In general, a method is a summary of experiences gained from working on many similar challenges.

	PERCEIVE	FORM	CONSOLIDATE	LEARN
 RATIO-NALE	 PROBLEM In crowded areas, safety could be compromised	 LEVERAGE Technology: AI (DFNet) Components: [Thermal camera] Information: Problem spots Human resources: [On-the-ground staff]	 SOLUTION PROSPECT ClusterDetect reduces congestion and disruption WARRANT Densely populated areas are common and dangerous BACKING The Contribution is effective and inexpensive	 HORIZON The project creates strengths and opportunities - The Problem is understood and generic - The Leverage is future-problem - The Prospect solves the Problem - The Warrant suggests growth - The Backing suggests motivation to adopt
 STRATEGY	 OUTER ENVIRONMENT External service: [Access control] Implements: Venue information system Repository: Traffic information People: [Informants]	 INNER ENVIRONMENT Density monitor [Thermal?] Panic and disruption locator General modules for on-the-ground staff and visitors [Customized module for on-the-ground staff]	 EVOLVABILITY DIFFUSIBILITY ClusterDetect is highly specialized and has low diffusibility towards new markets ADOPTABILITY ClusterDetect is limited to the current market, but it can address new needs and thereby strengthen existing market positions	 POTENTIAL The design has strategic potential: - Outer Environment is generalized - Inner Environment is near-decomposable - The Qualifications are non-critical
 TACTICS	 MANIFESTATIONS Potential crowding Places of congestion Potential starts of panic Potentially disruptive actions	 CAPABILITIES Identify and handle crowds Identify and handle congestion Identify and handle panic Identify and handle disruption	 MERIT VALUE Locates impediments. Finds and handles crowds, congestion, panic, and disruption RESERVATION Requires calibration and clear sight REBUTTAL Many users will accept these limitations in a low-cost system	 MISSION ClusterDetect solves the Problem: - The Manifestations are essential - The Capabilities are efficient and effective - The Value offered ensures sustained use

Table 4.2: Early Problem–Solution Canvas for the ClusterDetect system

Innovation projects are inherently unique, so Essence cannot provide a one-size-fits-all or established process. However, Essence might suggest exploring what motivated the team to start their project and focusing on the core elements that define this starting point. For this purpose, Essence introduces four key fulcrums that could offer different ways to begin.

We use the word fulcrum with inspiration from Archimedes (287 BC – 212 BC):

Give me a lever long enough and a fulcrum on which to place it, and I shall move the world.

A fulcrum is an enabler of a lever, and it decides where we lift.

In Essence, we have four kinds of fulcrums, one for each core concern:

-  *Manifestation fulcrum*—with a focus on the *perceived* problem.
-  *Capability fulcrum*—with a focus on what will be *formed*.
-  *Merit fulcrum*—with a focus on *consolidation*.
-  *Mission fulcrum*—with a focus on *learning*.

4.2. Manifestation Fulcrum

Manifestation fulcrums address problems (needs, inconveniences, difficulties, and more) experienced by users, customers, or other stakeholders.

 PROBLEM	 LEVERAGE	 SOLUTION PROSPECT WARRANT BACKING	 HORIZON
 OUTER ENVIRONMENT	 INNER ENVIRONMENT	 EVOLVABILITY DIFFUSIBILITY ADOPTABILITY	 POTENTIAL
 MANIFESTATIONS	 CAPABILITIES	 MERIT VALUE RESERVATION REBUTAL	 MISSION

Table 4.3: Manifestation Fulcrum

The ClusterDetect case could have begun by outlining how the problem would manifest itself. The issue was ensuring people's safety in crowded areas, specifical-

ly a venue. Situated in this outer environment, the manifestations were perceived as follows:

- Places of congestion.
- Potential starts of panic.
- Potentially disruptive actions.

Based on this initial understanding of the situation, the next steps could be to outline capabilities, values, and mission.

Problem scenarios (Section 6.3) can be used to start projects based on a manifestation fulcrum.



4.3. Capability Fulcrum

Capability fulcrums rely on leverage. The design team possesses technologies, components, information, and staff that enable it to form contributions that few others can achieve.

 PROBLEM	 LEVERAGE	 SOLUTION PROSPECT WARRANT BACKING	 HORIZON
 OUTER ENVIRONMENT	 INNER ENVIRONMENT	 EVOLVABILITY DIFFUSIBILITY ADOPTABILITY	 POTENTIAL
 MANIFESTATIONS	 CAPABILITIES	 MERIT VALUE RESERVATION REBUTTAL	 MISSION

Table 4.4: Capability Fulcrum

The ClusterDetect case could have begun by identifying leverage points where the design team has comparative strengths, such as thermal cameras. These leverage points could then serve as keystone capabilities.

- Locate crowds and congestion.
- Locate panic and disruption.

From these first ideas for a contribution, the design team could suggest manifestations where such capabilities would offer significant value and mission hypotheses.

Leverage scenarios (Section 7.3) can be used to launch projects from a capability fulcrum.



4.4. Merit Fulcrum

Merit fulcrums are based on innovative ways to generate value through designing new or significantly improved products and services. The team will need to consolidate the project early by verifying if these improvements are feasible and commercially viable.

 PROBLEM	 LEVERAGE	 SOLUTION PROSPECT WARRANT BACKING	 HORIZON
 OUTER ENVIRONMENT	 INNER ENVIRONMENT	 EVOLVABILITY DIFFUSIBILITY ADOPTABILITY	 POTENTIAL
 MANIFESTATIONS	 CAPABILITIES	 MERIT VALUE RESERVATION REBUTTAL	 MISSION

Table 4.5: Value Fulcrum

One example could be enhancing the user experience by using AI-based user profiling to anticipate users' needs more accurately and earlier than competing systems.

The *ClusterDetect* case could have started with a team wondering if it would be possible to reduce congestion and disruption hazards in crowds via precautionary measures. The team could point to venues and the use of technologies like generative AI to create these values:

- Eliminate locations with a high probability of crowding.
- Avoid obstacles that might block people's escape.
- Identify early signs of panic or disruption.

The design team could suggest manifestations and capabilities from these initial solution ideas and outline a qualified value proposition.

Solution scenarios (Section 8.3) can be used to start projects based on a value fulcrum.



4.5. Mission Fulcrum

Mission fulcrums are based on quantitative or qualitative goals for a design. Can we create something fundamentally new or significantly improve speed, volume, evaluations, or similar aspects? Will what we learn from doing so help us align with a future we aim to be part of?

The ClusterDetect case might have begun with a team questioning whether it could outperform state-of-the-art solutions in crowd security. The team could do this by clarifying terms like ‘reliably,’ ‘timely,’ ‘identified,’ and ‘located’ in both quantitative and qualitative ways when defining criteria such as these.

- Are crowds and congestion reliably and promptly recognized?
- Are panic and disruption recognized reliably and promptly?
- Are impediments identified and located?

 PROBLEM	 LEVERAGE	 SOLUTION PROSPECT WARRANT BACKING	 HORIZON
 OUTER ENVIRONMENT	 INNER ENVIRONMENT	 EVOLVABILITY DIFFUSIBILITY ADOPTABILITY	 POTENTIAL
 MANIFESTATIONS	 CAPABILITIES	 MERIT VALUE RESERVATION REBUTTAL	 MISSION

Table 4.6: Mission Fulcrum

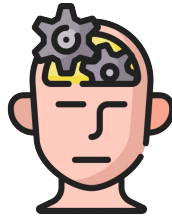
This clarification should lead to the development of new performance standards for similar systems, accompanied by a compelling mission statement. From there, the team could suggest manifestations and capabilities to make relevant value propositions stand out. Beyond value, the mission should also suggest potential growth opportunities.

Horizon hypotheses (Section 9.3) can be used to start projects based on a mission fulcrum.



Part 2

*Rationale:
What We Devise*



Chapter 5

Rationale: What We Devise

Everyone designs who devise courses of action aimed at changing existing situations into preferred ones

*Herbert A. Simon
(1916–2001)*

The rationale level concerns what we create beyond our specific projects. Our focus is on how we provide what we should while developing resources and capabilities to prepare for future challenges. What do we aim for as we take steps in this and other projects?

By *devise*, we mean what we plan or invent through careful thought, what combinations or applications of ideas we form in our minds, and what we intend to obtain or bring about over time.

Projects have specific goals and operate within constraints such as budget, duration, and staffing. Typically, projects are managed and evaluated in accordance with these goals and constraints.

At the same time, projects are assigned to organizations, teams, and staff based on merit, availability, and capability. The responsibilities and options we receive reflect our past and present. During times of unprecedented change,

we should try to add resources and capabilities that we consider strategically important.

The rationale level aims for a balanced, resilient innovation strategy while simultaneously advancing the current project. We should make the most of short-term projects to develop strategic capabilities and deploy them tactically whenever possible. For example, insights from previous and current projects can inspire new long-term development directions, while ongoing and new research and development efforts also benefit the current project.

We should keep stakeholders informed about the need to balance short- and long-term efforts and manage expectations regarding the potential value across different timelines, reinforcing how this potential supports future growth.

5.1. Limits to Rationality

A rationale is a logical foundation that makes actions meaningful and provides reasons supporting a specific action. In Essence, the rationale level represents the basis for our strategy and tactics in solving a problem. Unfortunately, despite our best efforts, this “rationale” may not be entirely rational.

The term rational refers to something based on sound reasoning and logical thinking. When working on a problem, we aim to understand it and find the best or acceptable solutions. Ideally, we strive to identify the problem, determine its root cause, explore potential solutions, and select the most effective one. After implementing a solution, we evaluate the results to see how effectively the problem was addressed.

In reality, problems, however, are often ambiguous and complex. Ambiguous because they are not tangible—they are perceived and cannot be measured, let alone scoped independently of our aspirations. Aspirations, in turn, originate from us, not the problem itself: We define who has the problem, where and when it occurs, and how much we are willing to spend on solving it. We scope the problem, select solutions, and do so collaboratively.

“Rational” or not, the rationale level in Essence expresses the project's logical foundation and helps us think about strategy and tactics. To work on a problem meaningfully, we must:

- State the problem as clearly as possible.
- Suggest ways we can utilize our abilities and resources.
- Explain how our efforts contribute to solving the problem.
- View our contributions as part of building resources for what we believe could be future challenges or opportunities.

5.2. Theory: Knowing and Doing

Essence is rooted in Deweyan pragmatism. From this perspective, we do not assume that we will know the actual problem or all possible solutions from the start, nor that the problem will remain the same throughout the project. Therefore, we cannot assume a linear problem-solving process. Problems and solutions can change significantly during, and often as a result of, the actions we take. This means that the reasoning behind our actions is limited by what we know and understand at a given moment.

John Dewey proposed a theory of knowledge based on inquiry (Dewey, 1938). Inquiry involves gaining knowledge through both theoretical and practical efforts to solve a problem. In this way, we slowly shift from understanding without solid facts towards justified beliefs.

In this section, we will examine Dewey's perspective on problems—what they are and how they are addressed—and introduce Deweyan inquiry as a process of reflective thinking. We will also include work by authors heavily influenced by Dewey, notably Donald Schön, who linked Dewey's ideas to professions and design (Schön, 1983; Schön, 1984; Schön, 1987; Schön and Wiggins, 1992; Schön, 1992a).

A key idea in Dewey's philosophy is that physical and mental constructs are part of the same creative process in problem-solving. We learn about a problem while we explore ways to resolve it. As we create theories, prototypes, and mock-ups, our understanding of the problem evolves. We explore the problem from different angles and adjust our focus accordingly. Donald Schön aptly describes this process as a reflective conversation with the materials of a situation (Schön, 1992b).

To both Dewey and Schön, practice is research-like: What we do is based on our knowledge about the situation and our expectations for what our actions will bring about. This outcome, in turn, confirms, expands, or rejects our knowledge. Knowing and doing, therefore, are inseparable (Schön, 1987, p. XI).

When we work on problems, we combine two kinds of knowledge (Dewey, 1938, p. 143):

- *Warranted assertions*—beliefs or facts backed by evidence.
- *Apprehensions*—understandings or leaps of faith—without factual grounds for asserting their existence.

When we *apprehend*, we understand an idea or notion without any evidence. Apprehension means seizing or grasping a meaning intellectually, without questioning (Dewey, 1938, p. 143). We are familiar with the concepts of an atom or a virus,

even if we do not collect evidence ourselves. We identify something as a computer if it resembles our past experiences with computers. We recognize objects at a glance as belonging to specific categories, and we interpret meanings in conversations or in a book based on our previous encounters with discussions and texts.

These two kinds of knowledge are easily confused, and we often consider apprehensions to possess properties similar to warranted assertions. Based on experiences from earlier situations, conversations, and media, we understand the meanings of new conversations, computers, or movies, and the principles for operating a new phone, TV, or computer. Such understandings are mediated through memories, habits, experiences, and conclusions drawn from them (Dewey, 1938, p. 143).

There is no guarantee that we will eventually move from apprehending something to having any final assertion about it. Apprehension is based on experience, and this familiarity may be obstructive, leading us to match observations with old routines rather than seeking data and suggestions elsewhere (Dewey, 1938, p. 144). In other words, what we think we know might stand in the way of gaining new insights.

Just like assertions and apprehensions differ, so do what we sense and perceive. *Sensation* and *perception* are distinct processes. Sensation is passive: Light, sound, smells, temperatures, etc., stimulate our sensory receptors. Perception is active: We choose, organize, interpret, and understand these sensations as they become meaningful. Physiologically, these processes might be sequential, but psychologically, they are inseparable: *The reality we perceive is selected and structured by our interests and by the standards which our interests generate* (Vickers, 1968, p. 108).

We perceive what matters to us and what makes sense, and our reality and value systems are essentially products of our experience, learning, and education. To Vickers, perception grows through life. Gradually, the child learns to understand what the eye ‘sees’ as their experience builds *schemata* to select and order impressions. Medical students build schemas as they learn to recognize signs of particular diagnoses by repeatedly scanning examples of chest X-ray images. Continually reviewing known examples builds up the schemata required to identify future instances (Vickers, 1968, p. 104).

Based on this line of thinking, Vickers goes on to describe how professionals depend on schemata in their work:

The lawyer learns to distinguish situations giving rise to legal liability from those which do not; to distinguish those that create only civil liability from those which also constitute crimes; to distinguish civil liability deriving from contract from that which is based on ‘tort’. His mental processes appear to be the same as those of the doctor to diagnose diseases, although, unlike the doctor, he relies not at all on visually manifest cues (Vickers, 1968, p. 105).

Schemata facilitate seeing; they help us select and group impressions, making us more attentive to what we are used to seeing. However, this has both positive and negative consequences: Schemata help us recognize familiar phenomena more quickly and accurately, but they also limit our ability to notice new things. Schemata *fixate* our way of seeing and hinder alternative perceptions. It is as hard for the teacher to ignore the schema that makes a pulmonary radiograph significant as it is for the student to learn it (Vickers, 1968, p. 105).

Schemata are indispensable for professionals to recognize and classify situations precisely and act accordingly. At the same time, schemata also limit our abilities to appreciate the unfamiliar and the unusual:

Anything so unfamiliar that no existing schema can be bent or stretched to contain it passes in blank incomprehension—until further experience evokes an ordering schema (Vickers, 1968, p. 105).

Apprehension thus serves as a basis for our inquiry by offering an understanding and a base for our conversations; at the same time, this apprehension may lead us to fall victim to conventional wisdom, prejudice, and unfounded assumptions.

Inspired by Geoffrey Vickers, Schön extends this discussion on apprehension to describe how practitioners of a profession rely on and share appreciative systems. These systems include values, preferences, and norms used to make sense of situations, set goals, and point to lines of action per professional conduct (Schön, 1987, p. 33).

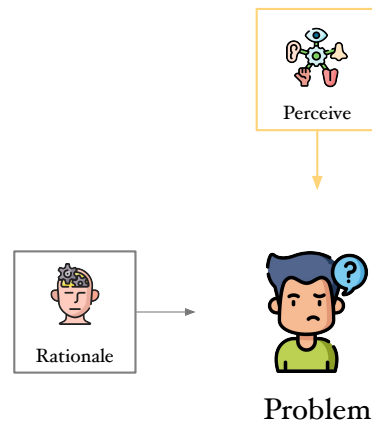
5.3. Chapters in this Part

In this chapter, we discussed rationality and why solving problems in a sequential, orderly manner is often impossible. We argued that knowledge and action cannot be separated, and that our experiences and training shape our perceptions of problems and their solutions. In other words, our ability to address problems depends on the frameworks or schemata we have developed through experience.

The remaining four chapters in this part are:

- Chapter 6 (*Problem*) explores the problem concept and the situation—the context—in which the problem is found. This chapter discusses the relationship between the problem and the context. It presents four different types of contexts, ranging from simple to chaotic. Lastly, the chapter proposes using problem scenarios to explore a range of problems and contexts.
- Chapter 7 (*Leverage*) discusses the leverage concept and the experimentation and reasoning used to explore the contribution.

- Chapter 8 (*Solution*) develops the solution concept and discusses the use of means and ends to steer towards a resolution.
- Chapter 9 (*Horizon*) discusses the horizon concept and valuation based on hypotheses to ensure that the short- and long-term goals are met.



Chapter 6

Problem

For every complex problem there is an answer that is clear, simple, and wrong

*H. L. Mencken
(1880–1956)*

Deciding on a problem–solution pair is a learning process. Problems can never be seen in isolation and cannot be understood without reference to what the absence of the problem might look like. In that sense, problems are reflected in their possible solutions.

This chapter discusses the problem and the context in which it occurs, based on the conditions we understand in our efforts to solve it.

Conceptually, a problem isn't as simple as it sometimes seems. It exists within a changing situational context—physical, social, and more—that influences the qualities we want in a solution. Therefore, a problem is part of an ongoing dialogue between the designer and the context in which the problem is embedded.

In everyday language, a problem is a matter or condition we want to overcome—something difficult to accomplish. At first, we might think of it almost as if it were a thing—something that exists in its own right because of its particular qualities—but once we inquire into it, we realize it is connected with ourselves.

Problems reflect our worldviews and those of other stakeholders. Recognizing something as a problem reveals our identity and perspective. A teacher may notice different issues in a school than a student, janitor, headmaster, or parent would. Our identity shapes the problems we identify and the solutions we pursue. Our knowledge, values, resources, and experiences influence how we perceive a problem and its potential solutions.

Following Dewey, we use problems as analytical instruments—we formulate problems as part of investigating them. A problem reflects an *understanding* of a situation, and, as a result, it has no objective existence as such.

Dewey defines a problem this way:

A problem represents the partial transformation by inquiry of a problematic situation into a determinate situation. (Dewey, 1938, p. 108)

This definition contains three keywords—*situation*, *transformation*, and *inquiry*—that will be explored further in this and other chapters. For now, we will make a few observations about them:

- The *situation* reflects that problems are *contextual*—conceived and acted upon in a world of objects and events.
- *Transformation* suggests a fleeting quality—problems may change or be replaced by others as our understanding develops through our efforts to solve them.
- *Inquiry* suggests that problems are part of an intellectual effort to understand and resolve a problematic situation—problems are perceived, and our perception of the problem is instrumental in this inquiry.

The following two sections will discuss situations and problems in more detail.

6.1. Situation

In Essence, we are concerned with the *situation* since problems are perceived in a context we are somewhat acquainted with:

What is designated by the word 'situation' is not a single object or event or set of events for we never experience nor form judgments about objects and events in isolation, but only in connection with a contextual whole. This latter is what is called a 'situation' (Dewey, 1938, p. 66).

Dewey defines three types of situations and orders them sequentially:

- Indeterminate situation—before we begin inquiring into the problem.
- Problematic situation—when we try to solve the problem.
- Determinate situation—when we find the problem has been settled.

The *indeterminate situation* is the precursor to inquiry. This situation is *not only "open" to inquiry, but it is open in the sense that its constituents do not hang together* (Dewey, 1938, p. 105). The elements of the situation are not determined—let alone understood—and the scope of inquiry is not presumed yet.

Dewey characterizes indeterminate situations as *disturbed, troubled, ambiguous, confused, full of conflicting tendencies, obscure, etc.* (Dewey, 1938, p. 105), but also as being unique:

It is this unique quality that not only evokes the particular inquiry engaged in but that exercises control over its special procedures. Otherwise, one procedure in inquiry would be as likely to occur and to be effective as any other (Dewey, 1938, p. 105).

Simply put, the procedures must be designed to reflect the specific situation. There will be no universal method of inquiry. In many cases, the process applied will change over the course of the project as experience is gained and circumstances develop.

Dewey insists that indeterminate situations are existential and exist outside our minds. For that reason, we can not fundamentally change the situation solely by meditation, wishful thinking, or similar:

[S]ituations that are disturbed and troubled, confused or obscure, cannot be straightened out, cleared up and put in order, by manipulation of our personal states of mind (Dewey, 1938, pp. 105-106).

Indeterminate situations exist irrespective of our cognition, and Dewey describes them as precognitive. They are the necessary conditions of our cognitive operations, since they force us to consider the situation as one that requires inquiry. Problems are not things; climate change, traffic jams, coronavirus, etc., are not problems per se. However, once a problem is perceived, the constituents of its situation start to hang together. This marks the beginning of the problematic situation:

The indeterminate situation becomes problematic in the very process of being subjected to inquiry (Dewey, 1938, p. 107).

The *problematic situation* begins when our uncertainty and doubtfulness make us stop and reflect on how to settle it. *To see that a situation requires inquiry is the initial step in inquiry (Dewey, 1938, p. 107).* At that moment, we start considering objects and events relevant to the situation: Who plays a role? When? Where? How? What things matter in this situation? etc.

For illustration, we can return to the *ClusterDetect* case. We might, e.g., wonder about the possible risks involved in giving, attending, or arranging a concert. This wondering might identify some limitations in existing safety apps for concert venues: Such apps might not be helpful in enclosures such as concert halls. Depending on who the user is, the problem might be to advise a music lover to avoid risky parts of the venue, whereas, for example, an administrator of a concert hall might see a problem in how to prevent crowd formation.

Dewey describes perceptions and conceptions as logical divisions of labor for investigating problematic situations. Perceptions serve to locate the problem, while conceptions represent possible solution methods. Therefore, they are highly connected and checked by their capacity to work together during the inquiry (Dewey, 1938, p. 111).

For example, in the *ClusterDetect* example, who is the customer—who should we help? One option is private individuals, while another is officials or authorities, such as the concert hall manager or on-the-ground staff.

Having settled on the customer, we might ask where and how we want to help.

- Outdoors, we might seek to advise individual users to stay out of an area.
- Outdoors, we might advise the police before crowds grow too big so that they can divert the public.
- Indoors, we might intervene by blocking access to a crowded zone.
- Finally, we might help officials intervene indoors or outdoors before crowds are formed.

Subsequent intervention involves devising and testing contributions to deal with the problem. If we are successful, we arrive at a determinate situation.

The *determinate situation* is the outcome of the inquiry. This is a situation where our uncertainty and doubtfulness are resolved and replaced by a closed, finished, and unified state:

The determinate situation on the other hand, qua outcome of inquiry, is a closed and, as it were, finished situation or “universe of experience” (Dewey, 1938, p. 105)

We are done with the problem. The situation is no longer problematic for us.

6.2. Problems

Problems arise in physical, social, and situational contexts. The environment influences everything related to the problem and its solution. It determines the issues we notice, the ideas we generate and select, the evidence we rely on, the hypotheses we value, and the reasoning we apply to address a problematic situation.

In that context, a problem serves as a tool for understanding and addressing the situation. Recognizing a problem is crucial for understanding the circumstances in which it occurs and exploring potential solutions.

Dewey gives a simple example of how we explore problems and situations:

When an alarm of fire is sounded in a crowded assembly hall, there is much that is indeterminate as regards the activities that may produce a favorable issue. One may get out safely or one may be trampled and burned. The fire is characterized, however, by some settled traits. It is, for example, located somewhere. Then the aisles and exits are at fixed places. Since they are settled or determinate in existence, the first step in institution of a problem is to settle them in observation. There are other factors which, while they are not as temporally and spatially fixed, are yet observable constituents; for example, the behavior and movements of other members of the audience. All of these observed conditions taken together constitute "the facts of the case." They constitute the terms of the problem, because they are conditions that must be reckoned with or taken account of in any relevant solution that is proposed (Dewey, 1938, p. 109).

When we recognize a problem, we also perceive the situation. Our perceptions influence how we address the issue. For example, children will likely seek out their parents, as parents are typically the ones who resolve problems. A firefighter in the audience will notice things differently from most others in the hall and may react accordingly.

6.3. Problem Scenarios

How do we identify the problem? We cannot determine if we have solved it without a clear understanding of it. To explore our understanding, we might use problem scenarios. Problem scenarios can be an effective way to initiate projects centered around a manifestation fulcrum.

The simplest way is to write up the problem as briefly as possible, stating it as you see it. Gertrude Stein said that a rose is a rose is a rose; paraphrasing her, we might say that a problem is a problem is a problem. Things are what they are. We

believe we understand the problem well, so we strive to describe it in a manner that can be easily shared among stakeholders.

This approach might work well for simple contexts, but many are far from simple. For that reason, we might use a typology for contexts to help us classify the type of problem we are dealing with when investigating the problematic situation.

David J. Snowden (Snowden and Boone, 2007) defines four types of contexts in his Cynefin framework (Table 6.1).

CONTEXT	CHARACTERISTICS
<i>Simple</i>	The domain of <i>known knowns</i> . This is where there is a clear relationship between cause and effect. Where this relationship is evident to everyone, and where there is a right answer. In this domain we sense, <i>categorize</i> , and respond.
<i>Complicated</i>	The domain of <i>known unknowns</i> . This is where cause-and-effect relationships are discoverable but not obvious to everyone, and where there might be multiple right answers. In this domain we sense, <i>analyze</i> , and respond.
<i>Complex</i>	The domain of <i>unknown unknowns</i> . This is where we can't know a priori what will work, and where patterns emerge for handling this flux and unpredictability. In this domain, there are many competing ideas and no right answers. Here we <i>probe</i> , sense, and respond.
<i>Chaotic</i>	The domain of <i>unknowables</i> —the domain of Knightian uncertainties. This is where cause-and-effect relationships cannot be determined and where there is high turbulence, high tensions, and limited time. In this domain, we <i>act</i> , sense, and respond. We look for what works to transform from chaos to complexity and then identify emerging patterns.

Table 6.1: Four Context Types (adapted from Snowden and Boone, 2007)

The simple type is not what Essence is made for. Unless we run into surprises, this context is hardly problematic. Essence is made for the three other types, where the problem is discovered as part of solving it.

One analytic approach to start formulating the problem is to ask the questions summarized in Kipling's poem about the *Six Serving Men*:

*I keep six honest serving-men
(They taught me all I knew);
Their names are What and Why and When
And How and Where and Who.*

Sometimes, these five Ws and one H are supplemented by a *How much* to form the W_5H_2 approach. Addressing such questions helps deepen our understanding of the problem.

An even more analytical approach is to explore whether the problem is indeed a singleton or whether it would be helpful to consider variations of it. To examine this issue, we could use problem scenarios.

Problem scenarios contrast alternative problem formulations and contexts. They define four quadrants, allowing us to explore four problem variants. Each interpretation defines unique concerns and different requirements for handling them.

We will use one vertical and one horizontal axis for this. Each axis is dichotomous (e.g., positive or negative, day or night), and we will use the W5H2 questions for inspiration when forming the axes.

For the vertical axis, we will use the who or the why question as outlined in Table 6.2.

We will use the question that best stimulates our imagination and points to significant alternatives. There are no external criteria to help make that choice. The team must choose between the ‘who’ and the ‘why’ questions.

PROBLEM	VERTICAL AXIS
<i>Who</i>	Who is affected? Who are the stakeholders?
<i>Why</i>	Why is this a problem? Why do we do this project? Is it about money, health, convenience, safety, aesthetics, law, social responsibility, ecology, ...?

Table 6.2: Problem Scenarios—Vertical Axis

PROBLEM	VERTICAL AXIS
<i>Who</i>	Who has the problem? One option could be an event planner at the venue. Another option could be on-the-ground staff handling crowd building during an event.
<i>Why</i>	Why is this a problem? Why do we do this project? The project could aim to target a market by addressing a general problem. Alternatively, it could grapple with the challenges of a particular venue.

Table 6.3: ClusterDetect Problem Scenarios—Vertical Axis

Let us use the ClusterDetect case presented in Chapter 3 for illustration. This case concerns detecting crowds in venues such as festivals and concert halls. The initiating problem was that *in crowded areas, safety could be compromised*. In this case, our dichotomies for the questions in Table 6.2 could be as shown in Table 6.3.

PROBLEM	HORIZONTAL AXIS
<i>What</i>	What type of problem? Is it simple, complicated, complex, or chaotic? (see Table 6.1)
<i>Where</i>	Where do we meet the problem? Where is it located or experienced? Where are those affected by it located?
<i>When</i>	When does the problem occur? Is it constant, occasional, accidental, random, immediate, gradual, delayed, ...?
<i>How</i>	How does the problem cause trouble? Does it affect directly, indirectly, as a network effect, as a side-effect, ...?
<i>How much</i>	How much does the problem bother stakeholders? Can the problem be quantified by what is lost or what could be earned by solving it? Can it be quantified in terms of time, money, manpower, risk, ...? Is it about life, money, or comfort?

Table 6.4: Problem Scenarios—Horizontal Axis

For the horizontal axis, we will choose one question among the remaining five as outlined in Table 6.4. Once again, the team will select a question that stimulates the imagination and explores significant alternatives.

Returning to the *ClusterDetect* case, our dichotomies for the questions in Table 6.4 could be as outlined as shown in Table 6.5.

LEVERAGE	HORIZONTAL AXIS
<i>What</i>	What type of problem? This question could relate to the project's scope. One approach might be <i>complicated</i> and limit the scope to address only known unknowns. Another approach could be <i>complex</i> and recognize that unknown unknowns may emerge. This would allow for a broader scope and, hopefully, a more robust outcome solution, though it may come at the cost of increased complexity risk.
<i>Where</i>	Where do we encounter the problem? Once again, this question could impact the project's scope. One option is to focus on specific venues to narrow the scope, while another is to aim for any venue to broaden its utility.
<i>When</i>	When do we encounter the problem? The team could consider predictable situations, such as breaks or when an event concludes. Alternatively, it might consider accidental scenarios, like a fire outbreak or a heart event attack.
<i>How</i>	How does the problem cause trouble? One alternative could be direct threats, such as trampling or stampedes; another could be indirect threats, such as overcrowding that may impede paramedics from reaching a visitor experiencing a heart emergency attack.
<i>How much</i>	How much does the problem concern stakeholders? One consequence could be financial loss if the venue is closed due to safety issues; another could be loss of life in the event of an accident.

Table 6.5: *ClusterDetect Problem Scenarios—Horizontal Axis*

The team could, for instance, select the *why* cell—market versus a specific venue—as the vertical axis and the *when* cell—predictable versus accidental situations—as the horizontal axis. If so, we will have the coordination system outlined in Figure 6.1.

We now have four quadrants, and while we could have used other W5H2 questions and defined the chosen axes differently, we have arrived at four meaningful quadrants, each suggesting a scenario. These scenarios could, for example, be labelled as shown in Figure 6.1:

- I *Site-specific planning and support.*
- II *Site-specific analysis and optimization.*
- III *General analysis and optimization.*
- IV *General planning and support.*

Within every quadrant, we could describe many alternative scenarios. Even after deciding on the ‘why’ and ‘when’ cells, we still have several open questions. As an illustration, we can discuss some of the options for Quadrant I:

- *Why:* What does it mean to focus on a particular *venue*? We could, for example, map the infrastructure and equipment in this venue or collect data on the location or movement of people.

- *When*: What does a focus on an *accidental situation* imply? We could, for example, focus on contingency planning to avert some situations. Alternatively, we could focus on the need for support in such situations.

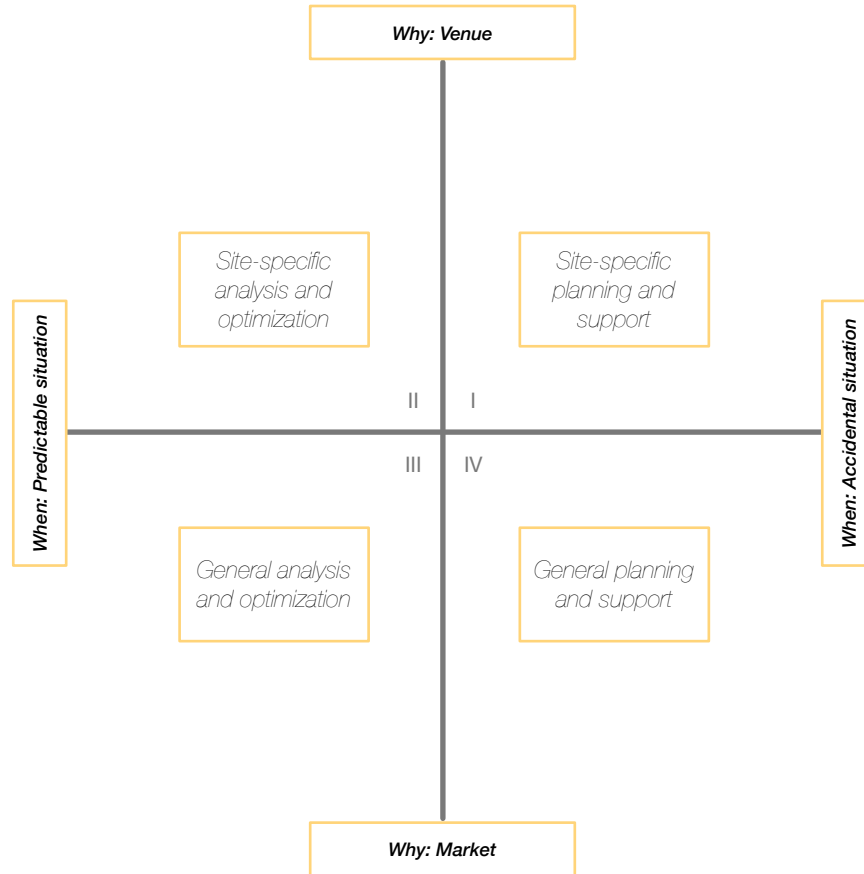
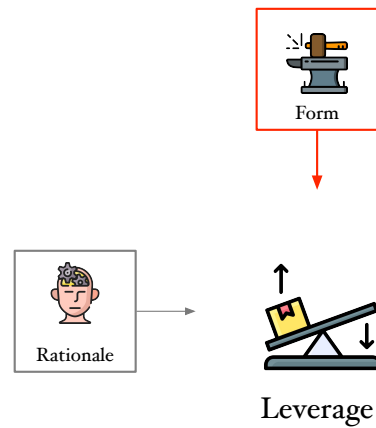


Figure 6.1: ClusterDetect Problem Scenarios

We started with this initiating problem: *Safety might be compromised in densely populated venues*. Using the scenarios in Figure 6.1, we might rephrase this problem for every quadrant. For quadrant I, the problem could be: *Can ClusterDetect help reduce safety hazards and aid in managing accidents at specific locations?*

Again, there could be many alternative formulations of the problem, and it is up to the team to formulate a meaningful statement for each of the four quadrants.

A problem is not merely a problem but a perspective shaped by circumstances and beliefs. Working through scenarios triggers alternative viewpoints before settling on a reasoned one. Problem scenarios stimulate idea generation about the problematic situation before arriving at a specific formulation. This is not a linear process in which we step by step reach the single best formulation. Instead, it is an exploratory process that encourages inquiry into alternative perspectives on the problem–solution pair.



Chapter 7

Leverage

Give me a lever long enough and a fulcrum on which to place it, and I shall move the world

*Archimedes
(287 BC – 212 BC)*

Deciding on a problem–solution pair involves simultaneously developing our understanding of the problem and ideas for addressing it. This chapter focuses on ideas that enhance our contributions. These ideas are the keystone of our contribution to this project. Still, we should also look beyond and carefully select ideas with the potential to provide leverage in future projects.

In many situations, we will choose a means we already know well. This approach has obvious limitations: As Maslow said, *it is tempting, if the only tool you have is a hammer, to treat everything as if it were a nail*. We assume that competent teams will have a range of available technologies, components, information, and human resources. Therefore, the team should evaluate potential means and assess their effectiveness in solving the problem.

We examine a problem through interaction, which depends on assumptions about the expected outcomes of our actions. When we create something, we anticipate how our constructs will interact with and change the situation. As a result, our experiences from this interaction will influence our perception of the problem and the effectiveness of the available tools.

At its core, our contribution consists of the constructs we create as part of a solution. Since the problem can evolve, the constructs we develop should also be adaptable, given their experimental and reflective characteristics. A construct is evaluated based on how well it helps to address the problem.

Every step we take is essentially experimental. Experiments—whether successful or not—help us better understand the problem because their results either support or challenge our assumptions about the situation and potential solutions. Additionally, any experiment might lead us to change our strategy. As a result, we may face a dilemma between sticking with what we already have and exploring new options that could lead to a better solution. We need to carefully select the option that offers the greatest leverage, considering various alternative approaches. We do this through real or hypothetical experiments with different candidates.

The following two sections will discuss contributions and leverage in more detail.

7.1. Contribution

Experimentation and reasoning are fundamental methods for exploring a contribution and understanding how our ideas relate to our real-life experiences from actual interactions.

To Dewey, ideas are possibilities—something that may happen (Dewey, 1938, p. 109). Developing and evaluating ideas is central to determining a problem and possible solutions. Therefore, ideas may be vague when we start working in a not-too-familiar context and become clearer as we gain insights. *Every idea originates as a suggestion, but not every suggestion is an idea* (Dewey, 1938, p. 110). Ideas begin as suggestions that occur to us. Suggestions have no logical status but may become ideas when examined for their capacity to resolve the situation (Dewey, 1938, p. 110).

Dewey defines ideas as forecasts—expected outcomes—of what may occur when specific operations are carried out under observed conditions. The facts we observe and our ideas evolve in correspondence with one another (Dewey, 1938, p. 109).

Ideas stimulate the activities we undertake to resolve a problem. Therefore, we must examine the idea's functional fitness and ability to contribute to a solution (Dewey, 1938, p. 110). The situation talks back to us as we try out ideas. This *back-talk*, as Schön calls it—the discoveries and surprises—forces us to reflect and reevaluate the situation and our judgment of ideas.

Some observed facts point to an idea [...] for a possible solution. This idea evokes more observations. Some of the newly observed facts link up with those previously observed and [...] rule out other observed things [...]. The new order of facts suggests a modified idea (or hypothesis) which occasions new observations whose result again determines a new order of facts, and so on until the existing order is both unified and complete (Dewey, 1938, pp. 113-114).

At the beginning of the *ClusterDetect* case, we may not know the accuracy of the data we collect or the requirements of the practical situation. This uncertainty might lead us to start modestly concerning the problem to solve and our choice of technology. We might wait and see what is learned from the first experiments.

7.1.1. Types of Means

Means are instruments to attain an *end*—something helpful in achieving a result. A database, a server, an app, etc., could be ends in and of themselves, but they might also be means for building a system to serve a higher end. Similarly, ideas such as to-do items, components to build, features to implement, etc., are means, even before they are built, by representing possible lines of action to resolve a problematic situation. Ideas, therefore, are means of inquiry.

In the *ClusterDetect* case, concepts and data related to ticket-holders, staff, managers, locations, regulations, events, interventions, and more are abstractions that might serve as means, enabling us to understand the situation. Like the components we build, the things we see as constituting the outer environment may change with our experiences and sense-making.

Not all means are things. Some are instrumental in facilitating the process through procedures and principles. Dewey, therefore, distinguishes between *material means* and *procedural means* to resolve the situation:

Both types of means help determine a resolved situation (Dewey, 1938, p. 136). The material means—including observed data, facts, and artifacts—are needed to solve the problem. In contrast, the procedural means determine how and whether such a resolved situation can be attained. Material means could be existing com-

ponents, platforms, objects, events, to-do items, etc. Examples of procedural ones could be work stages, sprints, plans, reviews, tests, evaluations, and coordination meetings, among others.

The methods used to address a situation are subject to critical reflection throughout the project. They are continuously compared with real-world conditions through design, testing, and execution, and are adjusted based on experience. Methods, therefore, serve both as a foundation for and a result of the inquiry.

There is continuity in inquiry. The conclusions reached in one inquiry become means, material and procedural, of carrying on further inquiries (Dewey, 1938, p. 140).

Our perceptions and actions are often shaped by convention: the objects we see, what matters to us, and ideas of professional conduct—such as what professional designers are expected to do—and more. This also applies to means.

A designer commonly employs patterns, abstractions, paradigms, methods, tools, techniques, prototypes, models, and components. All of these represent some of the many apprehensions available to the designer. Beyond this, any team member has a vast repertoire of understandings from daily life—e.g., concepts such as user, family, company, supermarket, sale, airport, flight, hotel, hospital, nurse, etc. —whose meanings are usually taken for granted. These examples illustrate how our appreciative systems shape what we see and what we do.

When discussing the *ClusterDetect* case, apprehensions are used extensively. We use concepts such as concert, ticket, concert hall, staff, ticket-holder, and regulations when discussing the problem. We also consider apprehensions such as technology and components when exploring possible solutions.

Apprehensions—or prestructures, as Schön calls them—represent a knowing built into our everyday routines of action that guide our seeing, thinking, and doing in familiar situations (Schön, 1992a, p. 129). Apprehensions —particularly those we know well from experience—help us save time and energy when starting a project. Our *a priori* knowledge of them helps reduce complexity and establish an overview, and since we know them from experience, we may not need to explore them explicitly.

[T]here are conceptual objects, and objects of perceptual experience, which have been so instituted and confirmed in the course of different inquiries, that it would be a waste of time and energy in further inquiries to make them objects of investigation before proceeding to take and use them (Dewey, 1938, p. 140).

One disadvantage of apprehension is that it can hinder our thinking if we don't look beyond what's familiar and blindly follow conventional wisdom, preconceptions, and groundless assumptions.

7.1.2. Transactions

Transactions refer to interactions between the inquirer and the materials at hand—the elements deemed part of the situation and the means available for resolving it. Donald Schön describes transactions as conversation-like, where the situation shapes an inquirer's actions and where these actions, in turn, shape it (Schön, 1992a, pp. 125–127).

To Schön, transactional inquiry means the same as *designing* in the design professions (Schön, 1992a, p. 127). Any professional design activity is, therefore, illustrative of transactions. In transactional inquiry, the inquirer is in interaction with the elements constituting the situation:

The inquirer does not stand outside the problematic situation like a spectator; he is in it and in transaction with it (Schön, 1992a, p. 122).

Seeing design as a conversation-like interaction implies that design and reflection are two sides of the same coin. Both serve to explore what the problem is and how it could be addressed. Schön calls this unity *reflection-in-action*.

The term conversation with the situation refers to a type of reflection-in-action understood from Dewey's transactional perspective. Here, an inquirer, in transaction with the materials of a situation, encounters a surprise in the form of “back-talk” that momentarily interrupts action, evoking uncertainty. The inquirer goes on to transform the situation in a way that resolves uncertainty, at least for the moment. [...] The inquirer is in the situation, influenced by his appreciation of it at the same time that he shapes it by his thinking and doing—in Dewey's words, “instituting new enviroining conditions that occasion new problems” (Schön, 1992a, p. 125).

7.2. Leverage

Leverage is short for leverage points—technologies, components, information, or human resources that could be key elements in forming an excellent solution. A leverage point might lead to a solution with more advanced features, lower cost, increased flexibility, robustness, or any other quality that makes a design stand out.

Proposed leverage points are ideas that forecast possible effects on the project or product.

Leverage is a complex concept. Table 7.1 suggests four categories of leverage points to inspire the generation of leverage candidates for consideration in a project. The crafted interfaces described in the table show that items from these

categories need to be integrated into the design in a non-trivial way. For example, simply using a standard online service or an off-the-shelf camera might not give us much leverage since anyone else can use them just as easily. Leverage should give us a competitive edge.

CATEGORIES	CHARACTERISTICS
<i>Technology</i>	Scientific knowledge, machinery, and equipment on which a design might be based.
<i>Components</i>	Rare equipment, sensors, or actuators used actively in building a solution and accessed via crafted interfaces.
<i>Information</i>	Used for collecting, storing, disseminating, and/or aggregating information to be accessed via crafted interfaces.
<i>Human resources</i>	Specific users, categories of users, experts, authorities, social or professional networks, organizations, or others with particular roles or functions designed to build a solution. Cyberhuman systems are prominent examples of systems where internal people are leverage points.

Table 7.1: Four Leverage Categories

Leverage closely relates to the team and its partners, reflecting our identity, knowledge, and connections. A leverage point provides a significant advantage in developing solutions. Therefore, standard tools, norms, libraries, and other resources are *not* considered leverage points because they are accessible to everyone and offer no comparative advantage.

CATEGORIES	CHARACTERISTICS
<i>Technology</i>	<i>Trilateration</i> using Bluetooth signal strength from beacons and smartphones to locate individuals and calculate group density.
<i>Components</i>	<i>Smart cameras</i> equipped with body recognition software could detect individuals, and by combining images from multiple angles, it would be possible to estimate group density in a particular location.
<i>Information</i>	Humans emit body heat, and this radiation can be used to generate <i>empirical heat maps</i> , which help identify current high-density groups using thermal cameras.
<i>Human resources</i>	<i>Trained staff</i> could monitor areas via visual inspection. Such staff could coordinate across enclosures and reduce population density in areas before crowding becomes a problem.

Table 7.2: Leverage Point Ideas for the ClusterDetect Case

In the *ClusterDetect* case, several alternative leverage points may exist. Table 7.2 lists four leverage point ideas. Each alternative has strengths and weaknesses. Choosing among them will affect the inner environment and the range of possible capabilities.

In most situations, we know which leverage points we want to use. We understand the leverage we can expect from them. However, we sometimes need to assess the attractiveness of a particular idea. And sometimes, we have too many or conflicting alternatives to make our choice obvious. For such occasions, Essence presents several tools for idea evaluation in Chapter 20. Here, we will demon-

strate a tool developed specifically for Essence named PCRT. The name of the tool is an acronym:

- *Power* (how powerful might this option be?).
- *Cost* (how costly might it be to exploit it?).
- *Risk* (how risky might it be?).
- *Time* (how might this option affect the time to complete?).

While cost, risk, and time are straightforward, the power factor is more intricate. What does power mean for a project or a team? In Essence, the power of a leverage point should be assessed in relation to the project and its significance for future initiatives. Thus, the power factor in PCRT should focus on both aspects. Inspired by Eric Ries, we will distinguish between the value and growth potential of a leverage point (Ries, 2011):

- The *value potential* refers to the value for current customers or other stakeholders. Will the solution become more valuable—more effective, easier to use, or more appreciated in different ways—if this leverage point is utilized?
- The *growth potential* refers to the project's future outcomes. Will the results lead to increasing benefits over time for current users or other stakeholders and potentially open new market segments if this leverage point is used?

PCRT has two variants: *qualitative*, which evaluates single ideas, and *quantitative*, which assesses multiple ideas. We will first illustrate the qualitative variant by analyzing a leverage point idea from the *ClusterDetect* case: Trilateration (Table 7.3). The qualitative version is not meant for comparisons and describes the evaluation with words rather than numbers.

Trilateration (Technology)	POWER		
	<i>Medium.</i> The technology is non-intrusive, and almost everybody wears equipment emitting Bluetooth signals. The accuracy in location equipment does, however, seem relatively low. Bluetooth range is relatively short, and this might limit where this technology is helpful. The potential for this approach might have reached its peak, and further developments in accuracy seem unlikely.		
	COST	RISK	TIME
	<i>Low</i> since this technology can use cheap and standard equipment. Development costs should also be low in teams with the relevant competencies.	<i>Moderate to high.</i> Bluetooth signal strength varies based on the brand, models, and other factors related to the equipment present in the area. Such variations significantly impact accuracy.	<i>Low</i> since the complexity of design and implementation will likely be low.

Table 7.3: *Qualitative PCRT Analysis of Trilateration (technology)*

We will use the same ClusterDetect case to illustrate the quantitative variant of the PCRT tool. This tool is designed to support the comparison of different ideas

using the same four factors. We assign a number to each leverage point to express its value to the project. Power is considered positive, while cost, risk, and time are negative. The scale for power consequently uses positive integers, while the other scales utilize negative values. We suggest using a 1–10 scale for power and a **5–1** scale for cost, risk, and time.

Table 7.4 assesses four leverage ideas for the ClusterDetect case (see Table 7.2). Trilateration is rated based on the reasoning in Table 7.3. It receives a medium score for power, low scores for cost and time, and medium to high scores for risk. A negative score of 1 indicates this technology might not be worth considering. The other three ideas in Table 7.2 were evaluated similarly. The evaluations in Table 7.4 show that smart cameras are the most promising, followed by trained staff.

LEVERAGE POINT IDEAS	ASSESSMENT						SUM	TOTAL
TRILATERATION (<i>Technology</i>)	POWER						5	1
	COST	1	RISK	4	TIME	1	6	
SMART CAMERAS (<i>Components</i>)	POWER						7	4
	COST	1	RISK	1	TIME	1	3	
THERMAL CAMERAS (<i>Components</i>)	Power						8	2
	COST	3	RISK	1	TIME	2	6	
TRAINED STAFF (<i>Human resources</i>)	POWER						3	6
	COST	6	RISK	1	TIME	2	9	

Table 7.4: Comparative PCRT analysis of the ClusterDetect case (Leverage ideas)

7.3. Leverage Scenarios

Having decided on leverage points, the next question is how to maximize their impact—how can they help design outstanding capabilities?

Leverage scenarios are designed to assist in this process. This tool is recommended when a team is unsure how to apply the chosen leverage point. Leverage scenarios can be an effective way to initiate projects centered around a capability fulcrum.

The scenarios are based on the classic W5H2 questions: *what, why, when, where, who, how*, and *how much*. These questions are applied in the Essence leverage scenarios, which aim to explore whether alternative solutions could offer different qualities.

Leverage scenarios define four quadrants to explore four solution variants, each with unique capabilities. We will use one vertical and one horizontal axis for this. Each axis is dichotomous (e.g., positive or negative, day or night), and we will use the W5H2 questions for inspiration.

For the vertical axis, we will use the ‘who’ or the ‘why’ question as outlined in Table 7.5.

We will use the question that best stimulates our imagination and points to significant alternatives. There are no external criteria to help make that choice. The team must choose between the two questions.

LEVERAGE	VERTICAL AXIS
<i>Who</i>	Who will be part of the contribution? Who are the operators?
<i>Why</i>	Why is this a contribution? Is it bespoke or generic? Is it contextual, universal, preventive, ameliorative, mitigating, compensating, active, passive, supportive, ...?

Table 7.5: Leverage Scenarios—Vertical Axis

LEVERAGE	VERTICAL AXIS
<i>Who</i>	Who will operate the system? One option could be safety assurance specialists working at the venue. Another option could be on-the-ground staff handling crowd building during an event.
<i>Why</i>	Why is this a contribution? Will the system help prevent dangerous situations, or will it help control disasters? Will it address a specific situation or a broader range of problems? Will it only be helpful concerning disasters, or will it have wider applications?

Table 7.6: ClusterDetect Leverage Scenarios—Vertical Axis

Let us use the ClusterDetect case presented in Chapter 3 for illustration. This case concerns detecting crowds in venues such as festivals and concert halls. The initiating problem was that *in crowded areas, safety could be compromised*. In this case, our dichotomies for the questions in Table 7.5 could be as shown in Table 7.6.

LEVERAGE	HORIZONTAL AXIS
<i>What</i>	What are the inner characteristics of the contribution? Is it flexible, decomposed, integrated, platform-based, open, closed, standardized, ...?
<i>Where</i>	Where will the contribution be used? Where will it have an effect? What part of the problem is targeted?
<i>When</i>	When will the contribution be used? When will the effects be seen (immediately, delayed, or gradually)? Will the contribution be used according to a plan, on demand, constantly, repeatedly, only once, ...?
<i>How</i>	How will it work? Will the contribution work directly, indirectly, or via network effects? Will it be active or passive, intrusive or non-intrusive, proactive or reactive, switched or always-on?
<i>How much</i>	How much will the contribution cost? Can the expenses be quantified in terms of investment and operations? Can the savings be quantified in terms of time, money, manpower, risk, ...?

Table 7.7: Leverage Scenarios—Horizontal Axis

For the horizontal axis, we will choose one question among the remaining five as outlined in Table 7.7. Once again, the team will select a question that stimulates the imagination and explores significant alternatives.

Returning to the ClusterDetect case, we assume our contribution will be based on smart cameras (Table 7.4). If so, the questions in Table 7.7 could be reformulated as shown in Table 7.8.

LEVERAGE	HORIZONTAL AXIS
<i>What</i>	What characterizes the use of smart cameras in this system? Is the system highly integrated for efficiency, or is it decomposed for flexibility and a longer lifespan?
<i>Where</i>	Where will the smart cameras have an effect? Will the system work directly with people, or will it work with the location's infrastructure?
<i>When</i>	When will the smart cameras be used? During planning and prevention work or in critical situations?
<i>How</i>	How will the smart cameras be used? Will they passively collect and present information, or will they be used actively to coordinate or control interventions?
<i>How much</i>	How much will a system based on smart cameras cost? How much will be saved? Will cost considerations mainly depend on investments or operations? Will savings come mainly from reducing manpower or from minimizing risks?

Table 7.8: ClusterDetect Leverage Scenarios—Horizontal Axis

For example, the team could choose the *who* cell—*safety specialist* vs. *on-the-ground staff*—as the vertical axis and the *how* cell—*passive* vs. *active* behavior—as the horizontal axis. This could give us the coordination system outlined in Figure 7.1.

We now have four quadrants, and while we could have used other W5H2 questions and defined the chosen axes differently, we have arrived at four meaningful quadrants, each suggesting a scenario. These scenarios could, for example, be labelled as shown in Figure 7.1:

- I *Coordination and dispatch support for specialists.*
- II *Intelligence support for specialists.*
- III *Intelligence support for on-the-ground staff.*
- IV *Coordination support for on-the-ground staff.*

Within every quadrant, we could describe many alternative scenarios. Even after deciding on the *who* and *how* cells, we still have several open questions. As an illustration, we can discuss some of the options for Quadrant I:

- *Who*: What does it mean to design a tool for a *safety specialist*? We could develop capabilities to map high-density areas, identify risk factors, design contingency planning templates, or provide tools for emergency response.
- *How*: What capabilities could help actively coordinate or control interventions?

We started with this initiating problem: *Safety might be compromised in densely populated venues*. Using the scenarios in Figure 7.1, we might rephrase this problem for every quadrant. For quadrant I, the problem could be stated as follows: *Can specialists use ClusterDetect to organize and dispatch help during safety-threatening events?*

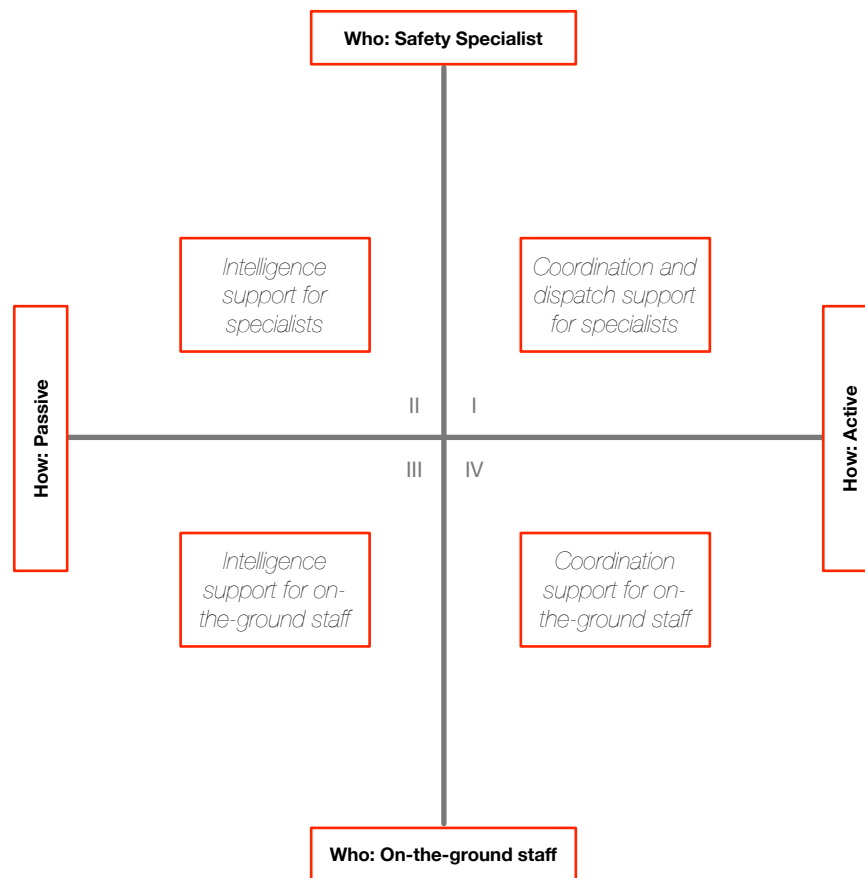
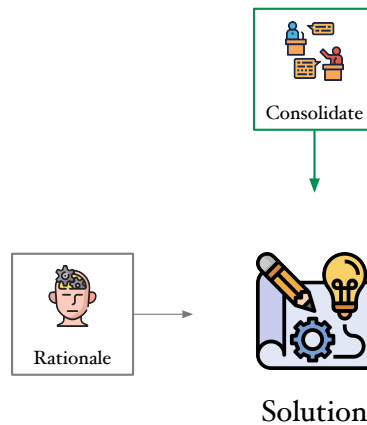


Figure 7.1: ClusterDetect Leverage Scenarios

Again, there could be many alternative formulations of the problem, and it is up to the team to formulate a meaningful statement for each of the four quadrants.

A problem is not just a problem but a perspective shaped by circumstances and beliefs. Working through scenarios triggers alternative viewpoints, allowing a reasoned conclusion to emerge. Leverage scenarios stimulate idea generation about the keystones of potential contributions before settling on a specific formulation. This is not a linear process in which we reach a single optimal solution step by step. Instead, it is often a nonlinear process that encourages exploration of different perspectives on the problem–solution relationship.



Chapter 8

Solution

Every great and deep difficulty bears in itself its own solution. It forces us to change our thinking in order to find it

*Niels Bohr
(1885–1962)*

Throughout this chapter, we will discuss the solution side of projects in more detail. A project is an ongoing conversation between problem and solution. Our understanding of the problem and potential solutions evolves through experimentation, research, and stakeholder discussions. In the *ClusterDetect* case, for example, we might develop a solution for various location types or user categories. We

might consider passive approaches, such as infrared or optical cameras, or active polling using Bluetooth or Wi-Fi protocols to estimate the density of people in an area.

Essence focuses on delivering value. Activities within a project unfold as ongoing interactions between the problem and our contributions. Therefore, we must consider whether we are addressing the right situation, solving it correctly, delivering value, and creating new strengths that will help us in future projects.

The solutions will rely on decision-makers' opinions, insights, and values. Our research, experiments, and evaluations will guide decisions while reflecting political priorities. Our choices could address questions like these:

- What is an effective solution to the problem?
- How will the contributions help resolve the issue?
- Are there limitations to the solution? Is it comprehensive? If not, can we accept the limitations?
- What value do our contributions provide?

8.1. Resolution

The resolution column in the Problem–Solution Canvas focuses on how and why we expect our contribution to resolve the problematic situation. Our understanding of the problem and how to solve it develops throughout a project. This growth might be iterative and continuous, but sometimes, it can be disruptive and cause us to revise—pivot—the project. Either way, this is a learning process.

Seeing a project as a learning process requires understanding the exploration that follows confronting a problem. Dewey calls this process *inquiry*.

A consequence of this exploration is that the way we solve the problem, and indeed the problem itself, might change in the course of the project. This calls for understanding what we aim for and how this goal might change. The goal we strive for is what Dewey calls an *end-in-view*, and we ultimately achieve what he calls an *end*. In this section, we will discuss these concepts.

8.1.1. Inquiry

Inquiry is a reflective process for understanding and resolving a problem. This process starts when we recognize an indeterminate situation as problematic (Section 6.1). The objective is to transform the indeterminate situation into a stable and determinate one.

Dewey defines inquiry as the controlled or directed transformation of an indeterminate situation into one that is determinate in its constituent distinctions

and relations, converting the elements of the original indeterminate situation into a unified whole (Dewey, 1938, pp. 104–105).

Through inquiry, we address problems by building on our existing knowledge and gaining new insights through both theoretical and practical approaches. Inquiry involves an interaction between individuals and their environment, in which new knowledge, concepts, meanings, and beliefs are created. *Transaction* (discussed in Section 7.1.2) is a critical element of this interaction.

Dewey aimed to connect theoretical thinking with practical application, scientific research with everyday knowledge, and scientific principles with common sense (Schön, 1992a, p. 121). At its core, inquiry is *transactional*, *open-ended*, and *inherently social*—it tries things out to see what works, embraces changes in conditions, scope, and focus, and considers the process to be social, where different insights and appreciations interact.

Following the discussion in Section 6.1, inquiry employs two kinds of operations working together (Dewey, 1938, p. 117):

- One deals with the *ideational* or *conceptual* subject matter—concepts to make the situation understandable and identify means and ends for resolving it. These operations suggest and require new observations, which in turn generate new facts. We reason about conditions and how to handle problematic situations, and develop concepts to denote and describe objects and events within them. Such reasoning can provide methods for effecting change in conditions, but alone, it cannot bring about any existential change.
- Another addresses *existential* subject matter: observations and experiments that interact with and alter objects and events in a given situation. We uncover previously hidden conditions that now appear essential, while downplaying other aspects that were once considered necessary. These activities challenge our reasoning and help us learn about objects, events, and, more broadly, the existential situation.

The only way to resolve an unsettled situation is to modify it and its elements. Experimental operations enable us to both alter and understand existing conditions. Reasoning about the problem can provide a means of effecting existential change, but cannot produce such change (Dewey, 1938, p. 118).

8.1.2. Ends and Ends-in-View

Having an aim for an activity is what Dewey calls an *end* or *end-in-view* (Dewey, 1957, p. 209). The ultimate *end* of inquiry is to resolve a problematic situation via existential changes. Such an end represents the termination of the project.

An *end-in-view* is an idea of an end to be reached (Dewey, 1938, p. 497). It represents the purpose of taking a step towards an overall resolving end. The end-in-view is the anticipated consequence of taking a step—the existential change to be effected by that step. The end-in-view represents why we think that particular step will move us closer to resolving the problematic situation. In this sense, the end-in-view becomes a procedural means (Dewey, 1938, p. 496).

Dewey offers this example:

The end-in-view of the man who sees an automobile approaching him is getting to a place of safety, not safety itself. The latter [...] is the end in the sense of close (Dewey, 1938, p. 166).

The difference between *end-in-view* and *end-as-final close* illustrates that termination of inquiry is stated as a procedural matter. Means and ends appear very different, but in fact they are not. They describe acts in the same reality but use different judgments. Dewey uses the term *end* to denote a series of collective acts. *Means* are separate acts in such a series. To think of the end is to extend and enlarge our view of the individual act. It means looking at the next act in perspective (Dewey, 1957, p. 35).

The next step in a project should be conceived and planned with a specific end-in-view. We must see the goal as a series of means to work toward a distant objective. Once we have visualized a far-off goal, we should consider strategy and tactics, turning *what* needs to be done into *how* we will achieve it. The end thus re-appears as a series of *what nexts* (Dewey, 1957, p. 36).

Converting an end into a series of means helps to conceive and define it clearly in light of how to reach it.

We do not know what we are really after until a course of action is mentally worked out (Dewey, 1957, p. 36).

To reach an end, we need to judge the scope and focus of the problem and set criteria for validating solutions. To test potential solutions, the team must establish relevant acceptance criteria. We also need to be able to represent possible solutions and evaluate the consequences of our actions to see if we are successful.

Table 8.1 presents examples of means and ends in software projects—the grey snippets in parentheses are examples from the *ClusterDetect* case. Items written in plain are tangible (existential). In contrast, items in italics symbolize items that may be ideational (not yet in existence) and represent ideas for actions, things to build, or hypothetical concepts, for example, regarding user characteristics. All items in the Ends column could also be listed in the Procedural means column since they may serve as intermediary or fleeting goals.

Material means	Procedural means	Ends, Ends-in-view
• Components built (modules for counting people) • Builds and releases (an experimental prototype) • Platforms and libraries chosen (drivers for sensors) • Objects and events (ticket-holders, concert venue) • Data available (contact data for ticket-holders and staff) • Infrastructure, digital ecosystem (relevant Android and IOS versions) • <i>Backlog items</i> (concert manager module) • <i>Presumed objects and events</i> (ticket-holders, staff, managers) • <i>Data presumed</i> (standardized venue information)	• Acceptance criteria (acceptable accuracy) • Test cases and results (Reject access for ticket-holder) • Prototypes (population density map for venue) • <i>Sprint (agreed procedure)</i> • <i>Planning (agreed procedure)</i> • <i>Test- and evaluation procedures (agreed procedure)</i> • <i>Daily stand-up meetings (agreed procedure)</i> • <i>Refactoring (agreed procedure)</i> • <i>Retrospectives (agreed procedure)</i> • <i>Demonstrations (agreed procedure)</i>	• Sprint goal (venue modeling with density thresholds) • Prospect (This app prevents crowding in indoor venues) • Plan (outline of expected iterations in the project) • Requirement (accuracy for population density) • User story (venue manager checks density map) (Ends-in-view may also serve as procedural means)

Table 8.1: Examples of means and ends

8.2. Solution

The main element in the solution cell is the *prospect*. We use this term to refer to the overall idea of our proposed solution. So, a prospect is the distant end-in-view for our design: what we believe would be a solution if the contributions developed in the project are used as intended. A prospect demonstrates an understanding of the problem and outlines what we believe to be a suitable solution. Therefore, the prospect reflects how we view the goal at the time of action.

The engineer, and more generally the designer, is concerned with how things ought to be—how they ought to be in order to attain goals, and to function (Simon, 1996, pp. 4-5).

It is called 'prospect' to acknowledge that our ideas can change over time. Therefore, the term reflects the adaptable nature of both the problem and the solution: While working on a solution, we might think we understand the problem, but new insights may emerge during the project, shifting our focus and opening new possibilities.

For the ClusterDetect case, we could formulate the following prospect: *ClusterDetect reduces congestion and disruption.*

Furthermore, we will use the term *warrant* to explain why solving the problem is essential. For example, in this case, the warrant could be: *Densely populated areas are common and dangerous.*

Finally, we will use the term *backing* to explain why the prospect would be an attractive and valuable solution to the problem. In this case, the backing could be formulated as follows: *The contribution is effective and inexpensive.*

8.3. Solution Scenarios

What combination of problem and solution generates the most value? Problems can be understood in different ways, and solutions can vary significantly. So, how can we combine various perspectives on a problem with potential solutions?

Solution scenarios develop and contrast alternative definitions of problems and solutions. They are designed to initiate projects centered around a value fulcrum. For this purpose, they define two axes and four unique quadrants, each representing mutually exclusive perspectives. These axes, in turn, are used to develop two understandings of the problem and two approaches to solving it.

Solution scenarios are therefore used to develop and contrast conceivable prospects. The objectives of a solution scenario are twofold: To meaningfully explore the problem and alternative ways to solve it. Accordingly, a solution scenario is essentially a combination of two sub-scenarios:

- *Problem scenarios* to explore the situation, and
- *Leverage scenarios* to explore the contribution.

	PROBLEM (VERTICAL)	LEVERAGE (VERTICAL)
<i>Who</i>	Who is affected? Who are the stakeholders?	Who will be part of the contribution? Who are the operators?
<i>Why</i>	Why is this a problem? Is it about money, health, convenience, safety, aesthetics, law, social responsibility, ecology, ...?	Why is this a contribution? Is it bespoke or generic? Is it contextual, universal, preventive, ameliorative, mitigating, compensating, active, passive, supportive, ...?

Table 8.2: Solution Scenarios—Candidate Questions for the Vertical Axis

Problem and leverage scenarios can be used independently as described in Section 6.3 and Section 7.3, but they are combined in solution scenarios.

Solution scenarios are based on the classic W5H2s (what, why, when, where, who, how, and how much). We will use the *who* or *why* questions for the vertical axis and the remaining five questions for the horizontal axis.

Table 8.2 covers the who and why questions for both core concerns simultaneously. We will use the *ClusterDetect* case as an illustration. This case concerns de-

tecting crowds in venues such as festivals and concert halls. The initiating problem was that *in crowded areas, safety could be compromised*. In this case, our dichotomies for the questions in Table 8.2 could be:

Problem (vertical)

- *Who* has the problem? One option could be an event planner. Another option could be on-the-ground staff handling crowd building.
- *Why* is this a problem? Why are we undertaking this project? The project might focus on a broad problem affecting a market or address the specific challenges of a particular venue.

Leverage (vertical)

- *Who* will operate the system? One option could be having safety assurance specialists work on-site. Another option might be to have on-the-ground staff manage crowd control during an event.
- *Why* is this considered a contribution? Will the system help prevent dangerous situations or assist in managing disasters? Will it focus on a specific scenario or address a broader range of issues? Will it be useful only for disasters, or will it have wider applications?

	PROBLEM (HORIZONTAL)	LEVERAGE (HORIZONTAL)
<i>What</i>	What type of problem? Is it simple, complicated, complex, or chaotic? (see Table 6.1 on page 84)	What are the inner characteristics of the contribution? Is it flexible, decomposed, integrated, platform-based, open, closed, standardized, ...?
<i>Where</i>	Where do we meet the problem? Where is it located or experienced? Where are those affected by it located?	Where will the contribution be used? Where will it have an effect? What part of the problem is targeted?
<i>When</i>	When does the problem occur? Is it constant, occasional, accidental, random, immediate, delayed, ...?	When will the contribution be deployed? When will the effects be seen (immediately, delayed, or gradually)? Will the contribution be used according to a plan, on demand, constantly, repeatedly, only once, ...?
<i>How</i>	How does the problem cause trouble? Does it affect directly, indirectly, as a network effect, as a side-effect, ...?	How will it work? Will the contribution work directly, indirectly, or via network effects? Will it be active or passive, intrusive or non-intrusive, proactive or reactive, switched or always-on?
<i>How much</i>	How much does the problem bother stakeholders? Can the problem be quantified by what is lost or what could be earned by solving it? Can it be quantified in terms of time, money, manpower, risk, ...? Is it about life, money, or comfort?	How much will the contribution cost? Can the expenses be quantified in terms of investment and operations? Can the contribution be quantified in terms of time, money, manpower, risk, ...?

Table 8.3: Solution Scenarios—Candidate Questions for the Horizontal Axis

Similarly, Table 8.3 covers the what, where, when, how, or how much questions for the situation and contribution concerns simultaneously.

Returning to the ClusterDetect case, our dichotomies for the questions in Table 8.3 could be:

Problem (horizontal)

- *What* type of problem? This question could concern the project's scope. One option could be *complicated* and limit the scope to addressing only known unknowns. Another option could be *complex* and accept that unknown unknowns might occur. This would allow for a broader scope, greater risk, and, hopefully, a stronger solution.
- *Where* do we meet the problem? Again, this question could affect the project's scope. One option is to focus on *specific venues* to reduce the scope, while another is to aim for *any venue* to increase utility.
- *When* do we meet the problem? The team could consider *predictable situations*, such as breaks or the end of an event. Alternatively, it could involve *accidental situations*, such as a fire breakout or a heart attack.
- *How* does the problem cause trouble? One alternative could be *direct dangers*, such as trampling or stampedes; another could be indirect dangers, such as crowding that prevents paramedics from reaching a visitor experiencing a heart attack.
- *How much* does the problem bother stakeholders? One effect could be *financial losses* if the venue is closed due to safety concerns; another could be *loss of life* in the event of an accident.

Leverage (horizontal)

- *What* characterizes the contribution? Is the system highly integrated for efficiency, or is it decomposed for flexibility and a longer lifespan?
- *Where* will the system have an effect? Will it work directly with people, or will it work with the location's infrastructure?
- *When* will the system be used? During planning and prevention work or in critical situations?
- *How* will the system work? Will it passively collect and present information, or will it actively coordinate or control interventions?
- *How much* will the system cost? Will cost considerations mainly focus on investments or operations? Will savings come primarily from reducing manpower costs or from minimizing risks?

To create a combined view of the situation and contribution, we use one question for the vertical axis and one for the horizontal axis, forming four quadrants. One question must come from a situation column, and the other must be from a contribution column.

We will select one question from each column that best stimulates our imagination and points to significant alternatives. There are no external criteria to help us make that choice.

For example, the team could choose the *who* cell—safety specialist vs. on-the-ground staff—as the vertical axis and the *when* cell—predictable vs. accidental—as the horizontal axis. If so, we will have the coordination system outlined in Figure 8.1.

Given this coordination system, the team can suggest many possible scenarios. The four outlined in Figure 8.1 are merely illustrations:

- I *Planning and Command & Control support for venue specialists.*
- II *Planning and intelligence support for venue specialists.*
- III *Planning and intelligence support for on-the-ground staff.*
- IV *Planning and coordination support for on-the-ground staff.*

We now have four quadrants, and while we could have used other W5H2 questions and defined the chosen axes differently, we have four meaningful scenarios.

We started with the initiating problem: *Safety might be compromised in densely populated venues.* Using the scenarios, we might rephrase this problem for every quadrant. For quadrant I, for example, the problem could be: *Can specialists utilize ClusterDetect to organize and send help during safety-threatening events?*

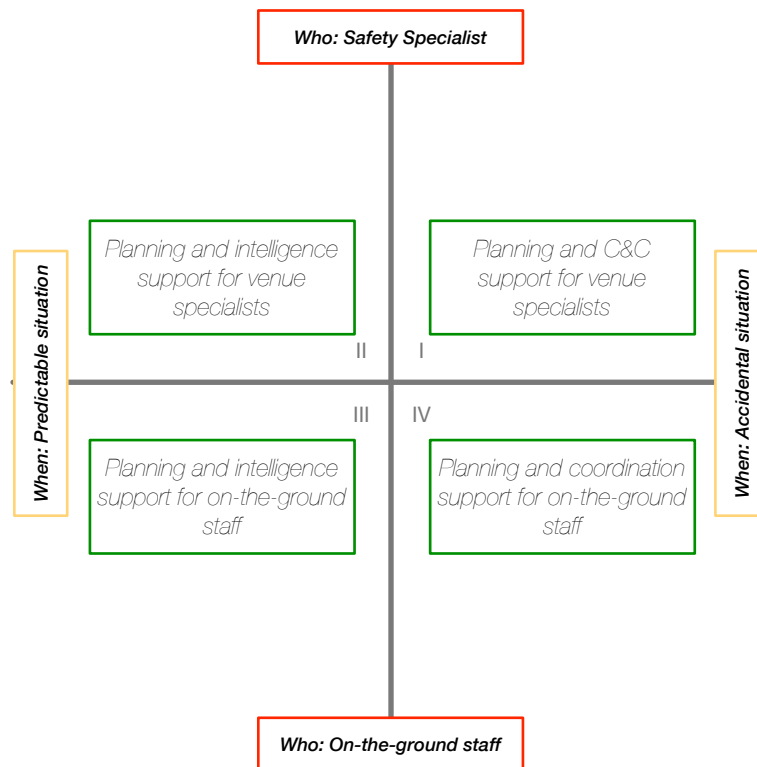
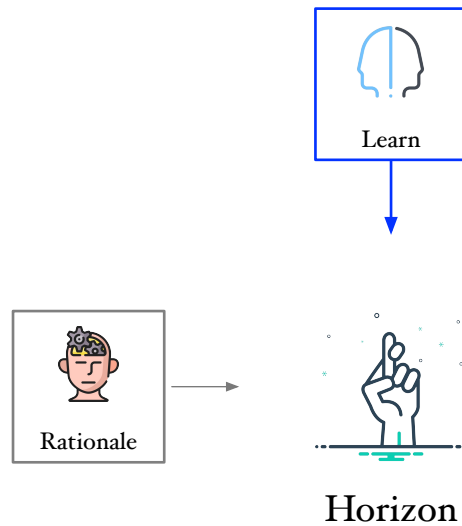


Figure 8.1: *ClusterDetect Solution Scenarios*

Once again, the solution scenarios are designed to generate various ideas regarding the problem and its potential solutions. However, there may be numerous al-

ternative formulations of the problem and its solution, leaving it to the team to create meaningful statements and choose among them.

A problem is not merely a problem, but a perspective shaped by the circumstances and beliefs surrounding it. Exploring different scenarios triggers various viewpoints before arriving at a reasoned conclusion. Solution scenarios stimulate idea generation concerning prospects before arriving at a particular formulation. This is not a linear process in which we step by step arrive at the single best formulation. In many ways, it is a random process that encourages inquiry into alternative views on the problem–solution pair. Still, hopefully, it is a process that ends with one meaningful formulation among competing ones.



Chapter 9

Horizon

We choose to go to the Moon [...] because that goal will serve to organize and measure the best of our energies and skills, because that challenge is one that we are willing to accept, one we are unwilling to postpone, and one we intend to win

John F. Kennedy
(1917–1963)

Learning at the rationale level is concerned with the current project and how it aligns with the long-term future we envision. The Kennedy quote opening this chapter is from his September 12, 1962, speech. In it, Kennedy expressed his support of the space program to the students at Rice University in Houston, Texas.

His speech occurred against several recent failures in the competition between the Soviet Union and the United States in critical areas of research and technology. The first artificial satellite to orbit Earth, Sputnik, was launched in 1957, demonstrating that the USA faced challenges in space research and rocket technology. This prompted the USA to engage in the space race, but by 1962, the United States was still lagging: Yuri Gagarin became the first man in space in 1961, while John Glenn followed as the first American in space the following year.

The Project Apollo followed Kennedy's national goal of *landing a man on the Moon and returning him safely to the Earth*. This mission was driven by a deeper goal: establishing technological leadership across multiple fields through space exploration. As a result, this mission, along with numerous smaller missions aimed at developing rockets, space capsules, lunar landing modules, space suits, and more, was driven by a broader objective: to mobilize resources, align initiatives, and enhance innovation and collaboration across sectors, with the overall goal of gaining an advantage in various disciplines considered essential for the future.

The three learning cells in the Problem-Solution Canvas are influenced by this same way of thinking, which is currently advocated by the EU and most notably by the Italian economist Mariana Mazzucato (Mazzucato, 2017; Mazzucato, 2018; Mazzucato, 2019).

Essence is aimed at design teams and projects; however, although the scale of such activities is much smaller than that of the Apollo Mission, we propose that a similar mindset is necessary today. We must view each project not only as a way to address the immediate problem but also as an opportunity to develop strengths that will help tackle future challenges.

A project is a goal-oriented activity with defined objectives and limited resources, but it allows us to create new leverage. Beyond solving the immediate issue it seeks to address, a project should also produce outcomes that equip us for the future. Therefore, we must keep our eyes on the horizon: What leverage will be helpful in the future?

In this chapter, we will address three aspects of this question:

- Pathways—how do we steer toward an uncertain and unprovable horizon? How can we learn from this and other projects to prepare for an elusive future?
- Horizon—how can we align what we do in our projects with a future we want to be part of? How can we predict changes in potential outer and inner environments that may assist us in steering toward a favorable future?
- Horizon hypotheses—what hypotheses might reflect this horizon and help steer towards it while simultaneously solving the problem we set out to solve?

9.1. Pathways

As a term, a pathway denotes a deliberate course of action—an approach to achieving desired outcomes.

However, the outcomes we aim for are, in our case, predictions or expectations for the future, which makes them hard to define. Our challenge is to create pathways for our ongoing activities that guide us toward an uncertain, unprovable horizon. At the same time, each project must fulfill its specific mission, and therefore every step toward this horizon must align with it.

This duality of pathways combines the purpose of immediate actions with the long-term goal that goes beyond the present. What steps should we take to accomplish the current mission? Can we coordinate those actions to prepare for the future we aim to be part of?

Projects that aim to solve problems involve actions intended to bring about change. These actions will be based on assumptions about how things work and how they respond to our actions.

Therefore, every intentional action must rely on explicitly or implicitly formulated hypotheses. Depending on how well the outcome aligns with expectations, each action either sustains or contests these hypotheses.

There is no inquiry that does not involve judgments of practice (Dewey, 1938, p. 174).

This practical judgment will seek evidence to determine whether objectives have been achieved. These objectives represent our understanding of what needs to change to solve the problem. Therefore, the evidence we look for reflects our understanding of the problem and the end we aim for. Deciding what evidence to look for is part of the inquiry, making ends-in-views operational and tangible.

Our goal is to turn the problematic situation into a determinate situation. So, what is a determinate situation, and how do we know if we have reached it? Understanding our goal helps us identify the evidence needed to determine if we are headed in the right direction. Dewey uses the term *end-in-view* to refer to the goal we want to reach (discussed further in Section 8.1.2).

In Essence, we have two types of end-in-view: *horizon end-in-view*, which focuses on preparing for a future beyond the current project, and *mission end-in-view*, which pertains to the current project. Both types may be subject to change, as they may be elusive and beyond our knowledge.

The mission end-in-view is the main topic of Chapter 19 whereas the horizon end-in-view is addressed in the following two sections.

9.2. Horizon

Three Horizons (Curry, 2015; Sharpe et al., 2016; Sharpe, 2020) is a futures method we will use to connect the long- and short-term steps, aligning what we do in our projects with the future we want to be part of.

As the name suggests, the Three Horizons method employs three different horizons:

- The first horizon (H1) represents *business as usual*—the way things are done now. H1 fits well with the present, but over time, it will become less fit for purpose as everything else changes. We will relate H1 to missions and discuss this further in Chapter 19.
- The second horizon (H2) represents the shift and clash of different ideas about the future. It is a domain where ideas are tested in response to apparent changes in markets, industries, macroeconomic forces, and major societal trends, including technological innovations. We will relate H2 to potential and discuss that further in Chapter 14.
- Horizon Three (H3) is long-term and represents an emerging future pattern, including technologies, application areas, norms, values, and economics based on ideas tested in H2. H3 can be plural, representing multiple competing patterns until one ultimately becomes dominant. We will explore H3 throughout the rest of this section.

What direction should we pursue for our innovation efforts? What long-term horizon should we set and prepare for? What leverage should we develop and test in future projects?

Looking for answers to these questions, we could find inspiration in three ideas from Chapter 1:

- *Technology push*: technology and science as drivers of change (Section 1.1.6). Should we base future leverage on technological and scientific advancements?
- *Market pull*: market features, users, economic developments, and more as drivers of change (Section 1.1.6). Should we base future leverage on major market, industry, and economic trends and shifts?
- *Innovation types*: product, process, position, and paradigm are four different variants of innovation (Table 1.1). Will we enhance our leverage by changing the kind of innovation?

We will consider these options one at a time.

9.2.1. Technology Push

Digital innovation relies on information technology, and advancements in science and technology will alter how we design and implement solutions. Information technology is evolving quickly. We have witnessed significant advancements in artificial intelligence, quantum computing, virtual reality, augmented reality, digital platforms, and more, and this trend will likely persist indefinitely.

Will these developments affect any of the leverage types we currently have? Will new leverage points develop, or will existing ones be threatened? Will we be able to handle new challenges better than others will?

For example:

- Will artificial intelligence be used to handle complex coding tasks, suggest more innovative code, or assist in refactoring intricate systems?
- Will AI aid in developing substantial abstractions and generalizations in our code?
- Will the increased use of distributed, microservice-based architectures provide significant advantages?
- Will we gain from the Internet of Things (IoT) and decentralized design paradigms?
- Will we be able to manage security considerations more effectively than others? Can we transform heightened security and privacy concerns into a competitive advantage?
- Can we create solutions with improved accessibility—new ways to interact without physical controls?
- Can we utilize direct neural interaction—e.g., through brain-computer interfaces (BCI)—to develop new leverage points?
- Will quantum computing become applicable for more practical applications? Will quantum computing introduce new leverage or challenge existing ones?

9.2.2. Market Pull

The term "market" is used broadly here to refer to the needs perceived by individuals or enterprises that digital solutions could address. This section will not focus only on trade or commercial contracts; any need that could prompt digital innovation is relevant here, regardless of whether it targets a commercial market.

There have been significant developments in business practices, industry organization, production locations, production methods, logistics, globalization, international and national regulations, income distribution, free trade, tariffs, a

growing middle class with greater purchasing power, crime, security, surveillance, warfare and new strategies for conducting it, as well as dependencies on critical infrastructure and innovative ways to protect and ensure resilience.

Will these developments create new opportunities or diminish the needs we have catered to thus far? Will new markets emerge, or will existing ones be at risk? Will we be able to target these markets more effectively than others?

For example:

- Will there be new markets or new needs? Will older ones disappear? Will competition change radically?
- Will our outer environment change? Will new service types emerge? New implements? New repositories? New customers or users?
- Will our market require adaptive and AI-driven interfaces? Personal interfaces? Predictive assistance?
- Will increased use of immersive technologies (augmented, virtual, or mixed reality) be required?
- Will the line between the physical and the digital blur? Will interaction become more context-aware?
- Will systems increasingly rely on modalities like voice and gestures? Will they need to interpret emotional cues and adjust to user moods or fatigue?
- Will increased focus on privacy and security affect our markets?
- Will bias mitigation and inclusivity cause changes in our markets?

9.2.3. Innovation Type

Will advances in technology or changing demands prompt changes to what we offer? If we provide products, will we continue to do so, or will we shift to a different type of offering? What contributions will help us succeed in the future? Will they be products, services, or processes?

Digital innovation affects products, services, processes, and business models that emerge within or between business units. Digital technology is a key factor in triggering or enabling value.

For example:

- *Product.* Will technological changes or changes in demand create opportunities for new or radically changed products or services?
- *Process.* Will such changes require new or radically improved production methods?
- *Position.* Will new products, services, processes, or business model opportunities emerge by fitting earlier solutions into new uses?

- *Paradigm*. Will new products, services, processes, or business models emerge from radical changes in the mental models of what a business is, who the users are, or what the market is?

9.3. Horizon Hypotheses

Following Herbert Simon (Simon, 1996), we seek an *appropriate* match between the situation and the solution. The situation includes the outer environment where we find the problem and where our contributions operate. In contrast, the solution consists of the substance and organization of our contributions, as well as their use.

We use *hypotheses* to ensure our design serves its intended purpose and meets our expected standards and end-in-view.

In Essence, hypotheses are tools for guiding us toward our goals. A hypothesis is a proposition that gives us a basis for reasoning about something we need to investigate further. It is a starting point for assessing whether the contribution aligns with the expected solution.

Hypotheses are used normatively while designing and as the basis for testing during implementation and validation. Therefore, they must be determinable. Conceptually, a hypothesis is an idea you test through research and experiments. It is a prediction that can be empirically tested. Hypotheses are verifiable or falsifiable and predict specific outcomes. A hypothesis is valuable even if it is proven false, because we learn from it: it enhances our understanding of the problem or solution.

The normative use of hypotheses is a key part of the design process. These hypotheses express what we expect to achieve if our contributions are used effectively, and later in the process, they are used to verify our expectations.

While we design, the hypotheses describe what we expect our contributions to achieve if used correctly. When we implement these contributions, we test our hypotheses through reviews or other methods to see if our expectations are confirmed.

Thus, hypotheses serve as a basis for reviewing a project's outcomes, but their formulation during the preceding design process helps clarify the project's goals. To explain these goals, we must specify what to look for to ensure our hypotheses are met, i.e., develop *criteria*.

The hypotheses for the Horizon cell have two foci: the present moment, where we examine whether the project provides a good solution to the defined problem, and the direction we wish the project to take to lead us toward our de-

sired horizon. The first focuses on the *mission end-in-view*, whereas the last concerns the *horizon end-in-view*.

At the level of *rationale*, an appropriate match between the situation and the solutions means a match between the problem and the resolution. We focus on the *problem*, *leverage*, and resolution cells at this level. The last cell is complex and has three parts: *prospect*, *warrant*, and *backing*.

- *Problem*: We hypothesize that the problem is well understood and defines the difficulty or challenge that our design must overcome.
- *Leverage*: We hypothesize that we employ technologies, components, information, or human resources that have the potential to deliver more substantial features, lower costs, greater flexibility, robustness, or other qualities that set our solution apart.
- *Prospect*: We hypothesize that the prospect solves the problem. This is inspired by what Ries calls a *leap of faith* (Ries, 2011, p. 81). This hypothesis posits our understanding of the problem and a meaningful way to solve it. It is the foundation for the design and is postulated without evidence. It must be tested and confirmed throughout the project. If it fails, the design is flawed.
- *Warrant*: We hypothesize that solving the problem is valued by stakeholders, customers, and users. This is inspired by what Ries calls the *growth hypothesis* (Ries, 2011, p. 61). This hypothesis posits that the solution will likely spread and become widely adopted among potential adopters.
- *Backing*: We hypothesize that the prospect is an attractive and valuable solution. This is inspired by what Ries calls the *value hypothesis* (Ries, 2011, p. 61). This hypothesis posits that the proposed contribution provides an attractive solution and that the benefits exceed the costs.

We want the project to address the present (mission end-in-view) and future (horizon end-in-view) rather than just solving the immediate problem that led to its creation. For that reason, we use two sets of hypotheses as shown in Table 9.1.

TIME	THE PRESENT	THE FUTURE
PROBLEM	The problem covers the issues faced by the stakeholders.	This problem will be significant in the future.
LEVERAGE	The leverage is vital for this project.	The leverage developed will help solve future challenges.
PROSPECT	The prospect solves the problem well.	This prospect is relevant to a category of future challenges.
WARRANT	The problem is worth solving.	This solution will help address future problems worth solving.
BACKING	The solution is effective and competitive.	This project will facilitate develop effective and competitive future solutions.

Table 9.1: Horizon hypotheses

In practice, we want to simplify the hypotheses and, if possible, combine them across the columns in Table 9.1. Ideally, every hypothesis should address both the present and the future.

Moreover, we want to consider the project as a whole, since the rationale at this level ensures it is sound. In addition to these cell-related hypotheses, we want to take an overall view of the project: Is the project beneficial for the team?

In the *ClusterDetect* example, these generic hypotheses could be formulated like this:

- The PROJECT creates strengths and opportunities.
- The PROBLEM is understood and generic.
- The LEVERAGE is future-proof.
- The PROSPECT solves the Problem.
- The BACKING suggests motivation to adopt.
- The WARRANT suggests growth.

The criteria should then interpret the hypotheses in a way that meaningfully covers the generic hypotheses in Table 9.1.



Part 3

Strategy: What We Develop



Chapter 10

Strategy: What We Develop

Chance favors the prepared mind

Louis Pasteur
(1822–1895)

The strategy part of this book examines how we can develop resources to improve our effectiveness in current and future projects. We focus on satisfying current stakeholders while identifying opportunities to use our insights in upcoming situations.

We will explore ways to create new resources or further develop existing ones. We will intentionally focus our efforts on what we believe will be crucial for upcoming projects, growth markets, or uses that seem to evolve or unfold in the future.

Such concerns are strategic. They relate to the duality between the immediate and the future. How can we tackle the current project while preparing for upcoming projects or challenges? Strategy involves formulating a comprehensive plan to address this overarching issue. This includes outlining the problem's scope, gathering essential elements for prospective solutions, and assessing the limitations of current solutions.

To paraphrase Simon, Essence emphasizes design by transforming existing conditions into preferred ones and creating artifacts that facilitate this change. Our strategic focus is to accomplish this by considering short-term and long-term development. This involves aligning the inner and outer environments, ensuring that what we build effectively addresses the current problem while preparing us for the future we envision.

Although we often assume that our problem is known and stable as we develop our solution, Essence is not designed for such fortunate situations. Instead, we anticipate scenarios where the internal or external environment will change significantly as we acquire more knowledge or as external factors influence our project. Gaining additional knowledge could provide new insights about our customers, users, or ongoing external projects relevant to us.

During our project, we might uncover previously overlooked leverage points and other factors that influence our situation. Since we cannot assume that the situation will remain constant throughout the project, we cannot depend on a viable master plan.

Based on these observations, projects need to prepare for major changes in the outer and inner environments. We should identify potential significant challenges and plan for possible scenarios in the design process.

10.1. VUCA

To help identify potential impactful challenges, we will use the VUCA framework based on leadership theories suggested by Warren Bennis and Burt Nanus (Bennis and Nanus, 1986). VUCA is an acronym for:

- *Volatility*—i.e., conditions that change rapidly or unpredictably.
- *Uncertainty*—i.e., needs and solutions that are unsettled.
- *Complexity*—i.e., external or internal parts that form intricate meshes.
- *Ambiguity*—i.e., causes and effects that are mixed up.

These four terms are not meant to suggest everything is in flux and impossible to prepare for. Instead, VUCA points to four challenge types that should be addressed differently (Bennett and Lemoine, 2014).

In Essence, we use VUCA to outline four categories of challenges that need four different responses in the design. At the strategy level, Essence aims to bridge between the inner and outer environments through the contribution of artifacts. Therefore, VUCA challenges may originate in either environment.

The outer environment refers to the surroundings in which our problem resides and where the contributions we design will be implemented. External services, implements, repositories, and users are examples of elements in the outer

environment. Such an environment can be challenging to work with, and we might use the VUCA framework to suggest ways to handle or even leverage these difficulties.

Similarly, we will use the framework to discuss challenges within the inner environment, as well as the substance and organization of our designs and implementations. By characterizing existing or upcoming challenges, we might find alternative ways to ameliorate or exploit challenges that will likely affect us.

In a small note in Harvard Business Review, Bennett and Lemoine organize VUCA along two axes (Bennett and Lemoine, 2014). These axes describe our *ability to predict the results of our actions* relative to our *knowledge about the situation*. The axes define a two-by-two matrix with cells for volatility, uncertainty, complexity, and ambiguity.

Traditionally, the VUCA framework is mainly used for leadership and organizational strategy, whereas Essence focuses on design challenges. For that reason, we will recast the VUCA challenges as shown in Table 10.1.

		KNOWLEDGE ABOUT THE SITUATION	
ABILITY TO PREDICT THE RESULTS OF ACTIONS	HIGH	COMPLEXITY	VOLATILITY
		External or internal parts that form intricate meshes. The situation or contribution has many interconnected parts and variables. Some information may be available, but the volume can be overwhelming.	Conditions that change rapidly or unpredictably. The situation or contribution is unstable or may change unexpectedly, but is not necessarily hard to understand. Knowledge about the change is often available.
		AMBIGUITY	UNCERTAINTY
	LOW	Causes and effects that are mixed up. Causal relationships are completely unclear. The logic of the situation changes with our actions, and stakeholders might be unsure of their preferences.	Needs and solutions that are unsettled. Basic causes and effects are known, but there is a lack of knowledge about possible changes. The future is, therefore, unpredictable despite available information.
		Low	High

Table 10.1: VUCA Challenges (inspired by (Bennett and Lemoine, 2014))

The table outlines four categories of design project challenges. The following four subsections will propose possible causes or indicators for these categories.

10.1.1. Volatility

Volatility refers to conditions that change rapidly or unpredictably. Knightian uncertainties belong to this category. Large-scale examples include wars, pandemics,

and logistics disruptions caused by accidents in the Suez Canal, missile attacks in the Red Sea, or political earthquakes in the US. Unexpected societal, economic, or technological changes could significantly impact a project's foundation:

- *Outer environment:* Sudden disruptive changes, such as shifts in legislation, technological advancements, market conditions, revenues, the economy, and competition.
- *Inner environment:* Sudden disruptive changes in leverage points, such as technologies, components, information, or people available.

10.1.2. Uncertainty

Uncertainty refers to unsettled needs and solutions. Breakups of international institutions and alliances, pending legislation, trade agreement negotiations, proposed joint ventures, mergers, and pending product launches exemplify such uncertainties. Current uncertainties surrounding generative AI, social media regulations, privacy concerns, and more may significantly impact design projects and be managed differently across different parts of the world.

Uncertainties could significantly impact the reasoning about both the Outer and inner environments of a design project:

- *Outer environment:* Unsettled essential legislative changes, technological breakthroughs, or changes in market conditions, revenues, the economy, and competition, among other things.
- *Inner environment:* Unsettled significant changes in leverage points, such as technologies, components, information, or people available for building contributions.

10.1.3. Complexity

Complexity occurs when external or internal parts of a system form intricate meshes. According to Simon, in such systems, the whole is more than the sum of its parts, and given the properties of the parts, it is not trivial *to infer the properties of the whole* (Simon, 1962, p. 468).

Complexity may arise during a design project, initially or gradually, as the project progresses and the environment becomes better understood.

- *Outer environment:* The situation involves many elements that interact in non-simple ways, and the properties of the whole cannot be easily worked out.
- *Inner environment:* The contribution similarly involves many elements that interact in non-simple ways, and the properties of the whole cannot easily be understood.

10.1.4. Ambiguity

Ambiguity describes situations in which causes and effects are mixed. Decisions or predictions are difficult to make when actions change the system's logic. Likewise, honoring a customer's requirements is problematic if that customer is unclear about what is wanted. Such situations involve unclear or conflicting expectations from customers, users, and other stakeholders.

Ambiguity might be visible from the start of a project, but often it only becomes discernible or manifest as part of the inquiry.

- *Outer environment:* The logic, needs, and constraints of the outer environment are unknown and unavailable, or causal relationships are unclear.
- *Inner environment:* The contribution alters the logic, needs, and constraints of the outer environment, or leads to unforeseen changes during development and implementation.

10.2. Complex Systems

This section will discuss complex systems based on Herbert Simon's work. Complexity is a significant challenge in design, whether we consider the outer or inner environment. A common challenge in design is managing complexity within time and resource constraints.

According to Simon, a complex system *is made up of a large number of parts that interact in a nonsimple way*, and it is not a trivial matter to deduce the properties of the whole from the properties of the parts (Simon, 1962, p. 468).

Sometimes, this challenge can be mitigated by a widespread property of complex systems: near decomposability. Simon argues that many complex systems have *a nearly decomposable, hierarchic structure*, and this quality enables us to *understand, to describe, and even to "see" such systems and their parts* (Simon, 1962, p. 477). Simon continues:

Most of the complex structures found in the world are enormously redundant, and we can use this redundancy to simplify their description (Simon, 1962, p. 481).

To Simon, designers can exploit this near-decomposability property to simplify the behavior within and among system parts (Simon, 1962, p. 482).

Simon argues that complex systems will develop from hierarchies over time, since such *systems will evolve from simple systems much more rapidly if there are stable intermediate forms than if there are not* (Simon, 1962, p. 473). Simon hypothesizes that

such complex forms will be hierarchical, since they have the time to evolve (Simon, 1962, p. 473).

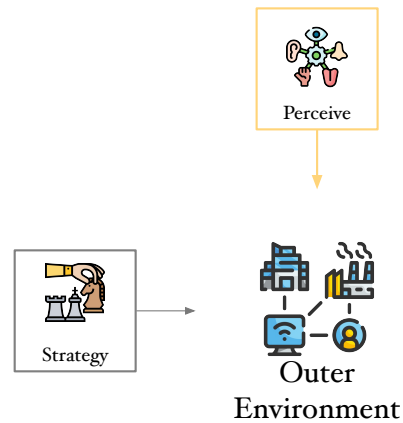
Near decomposability will be a recurring theme in the remaining chapters in the strategy part of this book.

10.3. Chapters in this Part

In this chapter, we discussed the strategy for the current project and what lies ahead; how we can build resources to improve our effectiveness in both current and future projects.

The following four chapters in this part are:

- Chapter 11 (*Outer Environment*) explores the surroundings in which our problem resides and where our design contributions will operate.
- Chapter 12 (*Inner Environment*) discusses the substance and organization of what we design and implement in the project.
- Chapter 13 (*Evolvability*) is concerned with the qualification of our solution; are there limitations to our solution, and are these limitations acceptable?
- Chapter 14 (*Potential*) discusses the relationship between the mission and the horizon end-in-view—i.e., the relationship between immediate goals and the long-term developments.



Chapter 11

Outer Environment

*Reality is merely an illusion,
albeit a very persistent one*

*Albert Einstein
(1879–1955)*

In Essence, we focus on how problems are perceived within a context we are somewhat familiar with, and how our perceptions evolve as we work to solve them.

At the strategy level, we will describe the situation in terms of what Herbert Simon (Simon, 1996) called the *outer environment*: the surroundings where our problem resides, and where our contributions will operate. To Simon, this outer

environment determines the conditions for goal attainment that we must meet (Simon, 1996, p. 11).

In Essence, we assume the world to be hypercomplex, ambiguous, and uncertain, and in such a world, we can not presume the outer environment to be given or known. Our world is constantly and dynamically shaped by the assumptions and interactions we encounter while designing our contributions: *Designing is learning*. A settled outer environment meaningfully reflects our understanding of the problem at a given time. Therefore, we assume that the conditions for achieving goals may change as we gain experience from what we design and experiment with.

11.1. The Outer Environment

The outer environment reflects the problematic situation in which we have a notion of a problem. How this problem is conceived serves as a basis for identifying its *constituents*.

The constituents are the parts of a problem that enable inquiry to move forward. Identifying the constituents is essential for establishing a link between the problem and the solution. Constituents are understood through our appreciative system—our insights and inclinations. We may, therefore, overlook constituents because we fail to recognize their existence or importance.

One clear issue is defining boundaries: How well do we understand the outer environment we need to engage with? There may have been a time when a project's boundaries were almost taken for granted. Determining what was inside or outside these boundaries would follow logically from a clearly defined problem. Once these boundaries were established, we would also recognize the interfaces between the outer and inner environments. Perhaps we once believed ecologies to be well-defined and mostly unchanging throughout a project's duration.

Such conditions still exist, but we can no longer assume they are typical. Our environments will not simply be *handed* to us; they will be *created* and increasingly built independently by others through new technologies, systems, components, software libraries, services, sources of information, national or global standards, and so on. As a result, our environments are often shaped or constructed by others. In this case, we don't know our environment from the beginning, but we must start the project with assumptions about it.

Of course, some parts of the outer environment will seem given and trivial. Such parts could be stakeholders, users, equipment, etc., which are usually central to the situation we are inquiring about.

For the ClusterDetect case, this outer environment might include access points to enclosed spaces, boundaries of these areas, locations of restrooms and food outlets, stages, and more. We might, for example, want to prevent people from entering crowded areas at a concert. To do this, we could focus on individuals and their locations, or we could differentiate between small groups and larger crowds. Therefore, we could analyze population density from either an atomistic or an aggregate perspective. The approach we select could depend on conscious decisions, conventions, habitual behaviors we are unaware of, or accidental choices. Often, we do not even consider our options when perceiving the elements of a situation.

In the *atomistic* example, these constituents might occur to us:

- Individual persons and their location.
- The layout of the music venue—seats, apron, front of the stage, doors, aisles, etc.

Based on this information, the solution could be to calculate and monitor population density for each segment of the venue, and we might consider using wearable transmitters, such as mobile phones, to track the location of every individual in each segment.

In the *aggregate* example, we could track group formations and develop strategies to intervene before groups turn into crowds. Our focus could then shift to group sizes. In this example, we might not need to monitor the venue layout or individual locations; instead, we could use heat maps from infrared cameras, Wi-Fi network analysis, Bluetooth Low Energy beacons, or similar technologies.

In our efforts to understand a situation, we create concepts: What elements do we observe? What qualities do they have? Which categories do those elements belong to? *Elements* are what we perceive as part of the situation, and the *whole* is the sum of the elements we identify as involved in this situation—the scope we consider relevant.

Some elements are *existential*—they exist in the physical world—and others are *ideational*. For example, the aisles and exits in a concert hall are existential. In contrast, if a fire breaks out, our idea of an escape route is ideational and depends on how we perceive the fire's location and how we think the audience will move to escape. Both existential and ideational elements are part of our later investigations and may change over time.

Old objects may undergo modification through the tests which are put upon them in new problems (Dewey, 1938, pp. 520-521).

The difference between existential and ideational elements can be subtle. The *Higgs Boson* was an ideational concept—a theoretical idea proposed in 1964—until experiments at CERN's Large Hadron Collider in July 2012 confirmed its exis-

tence. *Gravitational waves* were ideational predictions made by Einstein in 1916 based on his general theory of relativity; they were observed empirically in February 2016.

In the ClusterDetect case, the team could base the solution on technology that generates heat maps to show group sizes. This would assume the availability of affordable sensors with enough accuracy. Such sensors would be ideational until candidates are found and validated.

Dewey identifies two broad categories of elements: objects and events. Both objects and events are parts of a situation, which is not just a simple collection of objects and events but a context perceived as meaningful (Dewey, 1938, p. 66).

The words meaningful and perceived highlight the duality between the objective and the subjective. A situation encompasses real objects, events, relations, and more; however, what they are part of is essentially a mental construct. Objects and events are always situated within and observed in an experienced surrounding world, and such observations help to make sense of a field:

There is always a field in which observation of this or that object or event occurs. Observation of the latter is made for the sake of finding out what that field is with reference to some active adaptive response to be made in carrying forward a course of behavior (Dewey, 1938, p. 67).

Subjects observe objects and events, and when they observe, they also perceive; isolating the two would not make sense. Animals and humans developed sense organs to better survive in a challenging world, so observation and interpretation are inseparable. In common-sense inquiry, an object or an event is understood only in terms of what it signifies and how the situation should be dealt with.

We live and act in connection with the existing environment, not in connection with isolated objects, even though a singular thing may be crucially significant in deciding how to respond to total environment (Dewey, 1938, p. 68).

Dewey uses the term *object* to refer to a thing with properties and effects. Additionally, objects are part of our language, which mirrors our experience.

A tool or machine, for example, is not simply a simple or complex physical object having its own physical properties and effects, but is also a mode of language. For it says something, to those who understand it, about operations of use and their consequences (Dewey, 1938, p. 46).

[...] things exist as objects for us only as they have been previously determined as outcomes of inquiries. When used in carrying on new inquiries in new problematic situations, they are known as objects in virtue of prior inquiries which warrant their assertibility (Dewey, 1938, p. 119).

This understanding of objects is similar to how object-oriented programming understands objects as entities with identity, state, and behavior (Mathiassen et al., 2000, p. 51). When we know an object, we also gain knowledge about it through experience: what it is, its attributes, possible conditions and states, and how it might behave. If an object is part of a shared experience, we also have a shared understanding and language.

During periods of rapid change—driven by technological, scientific, or environmental revolutions—we observe the emergence of knowledge gaps or corridors in which the same object is understood differently by different actors. Users and designers might speak different languages, have different interpretations, and see a variety of opportunities and threats in the same object.

Moreover, the object we know from experience might take on new meanings—a new language, different understandings of usage, and other effects—when used in new situations. Objects have a changeable nature, even when they are not directly altered.

In the *ClusterDetect* case, individuals and family groups could be considered objects, since close family members will likely organize themselves and stay together at the concert. The project could register knowledge about individuals and family groups in attributes, conditions, and behaviors.

Dewey uses the term event to describe transitory objects that can be represented—like any object—by nouns or substantives. To Dewey, events possess substance. Events are characterized by *a delimiting beginning, an interval, and a termination* (Dewey, 1938, p. 223). This allows us to understand events as temporal or moving through stages, and to distinguish, for example, between the beginning and end of a concert.

In the *ClusterDetect* case, an intermission in a concert could be an event during which audience groups might dissolve and new ones arise. How the *ClusterDetect* system should intervene to reduce the risk of crowd building during intermissions may depend on available data, intervention options, and other factors.

11.2. Outer Environment Taxonomy

The outer environment links the problem to its manifestations by definition. Manifestations are usually perceivable; they tend to be tangible or visible and have objective qualities. However, it is not always clear if we perceive them directly or indirectly, through data or by sensing or observing them. Additionally, we can limit our scope of observation either broadly or narrowly. We might include or exclude parts of the manifestations' context.

Determining what constitutes the outer environment is complex, and it is unlikely that we will discover a single, undeniable configuration. Our deliberations

might revolve around the strengths and weaknesses arising from including or excluding a particular part, and how our choices might open up external opportunities or leave us vulnerable to external threats.

To help understand and evaluate the many possibilities, we need a *taxonomy* to create an overview and generate ideas. Taxonomies facilitate divergent thinking by prompting thoughts about various categories of items. Such prompts can help provoke ideas about the outer environment.

CATEGORIES	CHARACTERISTICS
<i>External services</i>	Services that are used without modification and reduce what is left for us to deal with.
<i>External implements</i>	Equipment, sensors, or actuators available in the outer environment and accessed via trivial interfaces.
<i>External repositories</i>	Repositories for collecting, storing, disseminating, and/or aggregating information via trivial interfaces.
<i>External people</i>	Generally available people, professionals, stakeholders, authorities, networks, organizations, etc.

Table 11.1: Four Outer Environment Categories

We suggest a taxonomy with the following categories (summarized in Table 11.1):

External services. As the Internet expands and connects us to everything, what external services might be available to actively assist in the job and help solve the problems we still need to address? By external services, we mean those used without modification that reduce what we have to manage. For example, providers of marketing information, equipment tracking, maintenance, streaming services, digital infrastructure, and more.

External implements. New equipment, sensors, actuators, and more are constantly emerging. How much should a solution connect with current or future external devices? External implements refer to equipment, sensors, actuators, and other components that can be accessed and used through trivial interfaces.

External repositories. As the amount and quality of information relevant to our problem increase, how should a solution interact with external repositories, and to what extent? By external repositories, we mean those that collect, store, disseminate, and/or aggregate information for access and use through trivial interfaces.

External people. As digital technology increasingly permeates our professional and personal lives, our interactions with external people and practices will expand as well. How deeply and extensively will we connect with these external individuals? The term “external people” includes a variety of specific persons, groups, stakeholders, authorities, social or professional networks, organizations, and others who may engage with, or be engaged by, our system.

11.3. VUCA Strategies for Outer Environment

Having decided on the elements of the outer environment, our next step is to explore possible VUCA issues and determine how to handle them. As mentioned in Section 10.1, VUCA is an acronym for four types of challenges to a design project:

- *Volatility*—i.e., conditions that change rapidly or unpredictably.
- *Uncertainty*—i.e., needs and solutions that are unsettled.
- *Complexity*—i.e., external or internal parts that form intricate meshes.
- *Ambiguity*—i.e., causes and effects that are mixed up.

If any VUCA challenge characterizes the outer environment for our project, we should consider strategies to ameliorate or avoid them.

11.3.1. Volatility

If sudden disruptive changes occur in legislation, technology, markets, revenues, the economy, or competition, among other factors, we may prepare for the consequences and develop strategies to manage them.

Bennet and Lemoine recommend creating slack and accumulating resources such as inventory or talent (Bennett and Lemoine, 2014). However, in projects funded by outside parties, such a strategy might not be feasible because we may not know what to stockpile or which talent to attract. Alternatively, we can include force majeure clauses in the contract. These clauses could allow for delays and the addition of resources to the project.

Strategically, we might prepare to *act, sense, and respond*. Snowden and Boone describe this situation as chaotic and argue that searching for the right answers would be pointless since the relationships between cause and effect shift constantly. Such circumstances call for us to *act to establish order, then sense where stability is present and from where it is absent, and then respond by working to transform the situation from chaos to complexity, where the identification of emerging patterns can both help prevent future crises and discern new opportunities* (Snowden and Boone, 2007, p. 74).

One example of a volatile situation is the sudden shutdown of venues during COVID-19. If we assume the ClusterDetect project targets concert venues and that an immediate, indefinite shutdown occurs, all venues will close quickly, so no customers will need a ClusterDetect system. What options do you have?

One option is to modify your customer segment to include law enforcement or other authorities, and to adjust the system to report situations involving large crowds.

Another option is to change who will use the system, focusing on individuals and alerting them if they are about to enter an area with too many people. The customers paying for the service could be malls, and the users could be visitors who use the service.

Both options require the project to redefine the outer environment and aim at a new product for a different market. The team would *act, sense, and respond*.

Alternatively, the project could use a *force majeure* clause to re-negotiate with the original customers, assuming the shutdown will end and normal conditions will be restored relatively soon.

11.3.2. Uncertainty

Suppose we face disruptive changes in legislation, technological breakthroughs, market conditions, revenues, the economy, and competition, among other things. Still, we do not yet know when or what they will entail. In that case, we might prepare contingency plans for the options we can identify and try to determine dependencies so that the unsettled changes affect as few parts of a project or design as possible.

For example, we can evaluate pending legislation considering terror attacks like the Crocus City Hall attack in Krasnogorsk, the Bataclan Theatre attack in Paris, or the Dubrovka Theater Hostage crisis in Moscow. Such laws might include new requirements for the ClusterDetect system.

Terror-related requirements for capabilities to assist in rescue or anti-terror operations might arise. These capabilities could include, for example:

- A verified list of attendees based on ticket control.
- Photos of all attendants for identification purposes.
- Real-time mapping of people's locations.
- Analyzing movement patterns to help locate the attackers.
- Switching to infrared lighting in parts of the venue to give responders tactical advantages.

Until that legislation is finalized, the project must face significant uncertainty. Likewise, the fire at Le Constellation bar in Crans-Montana, Switzerland, will probably lead to stricter safety regulations for public entertainment venues. This may also create uncertainties for the project.

One possible approach is to create contingency plans. This will demand extra effort and may result in some outcomes going unused. Another approach could be to isolate the dependencies of these unresolved issues so they affect only a few parts of the design.

11.3.3. Complexity

Complexity occurs in situations involving many elements that interact in complex ways, making it difficult to infer the properties of the whole.

The outer environment might be complex at the start of a project, or it may become clearer as the project progresses and the environment is better understood.

For example, the ClusterDetect project could become more complicated if its focus is unclear or too broad. This could happen if the requirements include prediction, planning, monitoring, interventions, centralized and decentralized operations, rescue support, and terror prevention. In this case, complexity increases due to many different features.

Complexity can also increase if the project targets multiple markets, including different countries, local regulations, customer types, user categories, and venue types. Having too many user categories also increases complexity.

Furthermore, complexity could increase if the system interacts with other systems, such as local IT systems at each venue. Here, complexity rises due to external dependencies.

In some situations, complexity can be reduced by limitations such as:

- Fewer manifestations
- Fewer user categories
- Fewer dependencies

In other situations, complexity can not be reduced below a certain level if the project is to remain relevant.

11.3.4. Ambiguity

Ambiguity involves challenges where causes and effects are confused. Making decisions or predictions is hard when actions alter the system's logic. We often see causation as a one-way relationship. However, the link between actions and effects is often two-way.

Meeting requirements can be difficult when users or customers are unsure about their needs or when multiple customers and users are involved. These situations can lead to vague or conflicting expectations from customers, users, and other stakeholders.

In the outer environment, ambiguity means that the logic, needs, and constraints are unknown or unavailable, or causal relationships are entirely unclear. Ambiguity might be visible from the start of a project, but often it becomes discernible or emerges as part of the inquiry.

Strategically, we might prepare to *probe, sense, and respond*. Snowden and Boone describe this situation as unpredictable and in flux. Why things happen can only be understood in retrospect. In this situation, Snowden and Boone suggest that instructive patterns can emerge from safe-to-fail experiments. *That is why, instead of attempting to impose a course of action, leaders must patiently allow the path to reveal itself. They need to probe first, then sense, and then respond* (Snowden and Boone, 2007, p. 74).

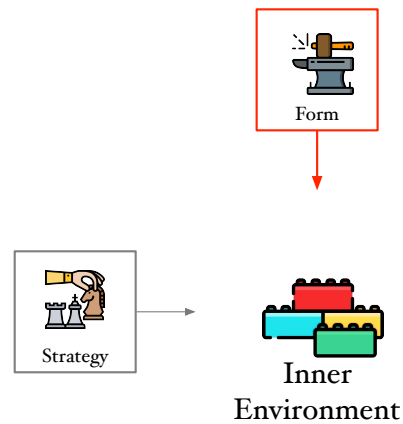
In such situations, the customer's requirements and needs might emerge through experiments with prototypes, metaphors, innovation games (Hohmann, 2006), and similar approaches to identify patterns, suggest trends, and make sense of what could be system requirements.

For the ClusterDetect case, prototypes and similar approaches could be used to gather requirements for a specific, typical venue or purpose, and then use those requirements to begin creating value and growth opportunities for a general market.

The strategies are summarized in Table 11.2.

KNOWLEDGE ABOUT THE SITUATION			
ABILITY TO PRE-DICT THE RESULTS OF ACTIONS	HIGH	COMPLEXITY External or internal parts that form intricate meshes. STRATEGIES • Fewer manifestations • Fewer user categories • Fewer dependencies	VOLATILITY Conditions that change rapidly or unpredictably. STRATEGIES • Stockpile resources. Include force majeure clauses • Act, sense, and respond
		AMBIGUITY Causes and effects that are mixed up. STRATEGIES • Probe, sense, and respond • Look for patterns to suggest trends and sense-making	UNCERTAINTY Needs and solutions that are unsettled. STRATEGIES • Prepare contingency plans • Isolate dependencies
	LOW	LOW	HIGH

Table 11.2: VUCA: Strategies for Outer Environment



Chapter 12

Inner Environment

Intelligence is the ability to adapt to change

*Stephen Hawking
(1942–2018)*

This chapter discusses the challenges our contribution must address in response to the anticipated changes. Will we need to adapt to volatility, uncertainty, complexity, or ambiguity? What strategies could help us create a design that adjusts to these changes?

The need to adapt can arise from dynamics in the outer environment or changes in the means available for our design—such as components, technologies,

data, staff, and more. This interaction between the internal and external environments necessitates reciprocal adaptation. This adaptation is inherently dynamic and iterative, as we gain a deeper understanding of both environments through our experiences.

In design projects, our experiences tend to grow through interactions—conversations with the materials of the situation—as Donald Schön describes it (Schön, 1992a, pp. 125-127). Designers or practitioners take action, observe how the materials respond, and adjust accordingly. This pattern is common among professionals such as engineers, architects, and even artists.

Conceptually, materials can be ambiguous. We usually think of materials as physical things or matter from which we can build or perform actions. However, materials can also include facts, information, or visualizations used to create or act in specific ways. As a result, models, ideas, frameworks, and similar items are sometimes considered as materials.

Examples of such uses could be:

- Computational models are used to run simulations and interpret the results.
- Strategic thinking is used to develop and adapt policies or roadmaps for enterprise innovation. Chapter 9 covers the development of horizons and properties, while Chapter 14 illustrates strategic thinking.
- Scenario development offers different options for an activity or project. Problem scenarios (Section 6.3), leverage scenarios (Section 7.3), and solution scenarios (Section 8.3) in *Essence* serve as models for generating alternatives, selecting from them, or even combining various options into a new direction.
- Conceptual models are used to develop, evaluate, and critique a design or its implementation outcomes. Visual inquiry tools like the Business Model Canvas and the Problem-Solution Canvas exemplify such models.

Specialized conceptual models are utilized in various professions related to digital innovation.

- In *building and urban design*, for example, Christopher Alexander's pattern language is used to create and characterize livable spaces (Alexander, Christopher et al., 1977). Patterns outline a problem and provide a solution supporting both the creation of solutions and the identification of their characteristics. This method is also applied in software design (Gamma, 1995) and application development (Rising, 1998).
- There is a strong tradition of model-based engineering in *systems and software design*. In this approach, models support requirements elicitation, design, analysis, verification, and validation. The Unified Modeling Language (UML) and Systems Modeling Language (SysML) provide many tools for model-driven development. More modern approaches include

microservice architectures, in which applications are composed of a collection of loosely coupled services.

- In *interaction design*, conceptual models are widely utilized in user- or activity-centered design approaches. Additionally, wireframes and storyboards are commonly employed.

Using models, concepts, frameworks, and similar tools challenges our reasoning. When we apply specific abstractions to analyze our designs, these abstractions require us to express our reasoning through them, simplifying ideas to what they are meant to illustrate. In this manner, our reasoning is shaped and refined into something with a clear focus and scope. Working within the inner environment, utilizing models, concepts, frameworks, and similar tools enables us to shape and refine our architecture for enhanced analysis, improvement, and sharing.

12.1. Inner Environment

Chapter 11 describes the *outer environment* as the surroundings in which our problem resides and where our design contributions will operate. This chapter describes our contribution as an *inner environment*—the substance and organization of our design and implementation. This definition is based on Simon’s definition: *The substance and organization of the artifact itself* (Simon, 1996, p. 6), where the word ‘artifact’ is Simon’s term for what we design.

Simon describes the artifact as a meeting point between the outer and inner environments. The inner environment interfaces with the outer environment, having substance and organization that support this connection. Simon says this about the relation:

If the inner environment is appropriate to the outer environment, or vice versa, the artifact will serve its intended purpose. (Simon, 1996, p. 6)

In Essence, the inner environment is the architecture and the primary components used to exploit the leverage and implement the desired capabilities.

In the case of *ClusterDetect*, we would likely prioritize flexibility and effectiveness as key factors for maintaining an appropriate relationship between the inner and outer environments: Since we can choose from various technologies, components, data, or personnel available for crowd measurement, we may want to work experimentally and therefore value a design that adapts to changes in configuration and remains practical in its implementation and use.

12.2. Near-Decomposability as a General Strategy

Essence highlights near-composability as a general design strategy for the inner environment. Herbert Simon introduced the concept of near-decomposability in his work on hierarchical systems (Simon, 1962; Simon, 1996). Near-decomposability refers to systems that can be separated into semi-independent (not fully independent) subsystems, which interact weakly with each other while maintaining strong internal cohesion.

This principle is often articulated in software engineering as low coupling and high cohesion. Endres and Rombach state this as Constantine's law (Endres and Rombach, 2003, p. 43):

A structure is stable if cohesion is strong and coupling is weak.

The intuition underlying this principle is that changes to the structure (system) can be implemented with *minimal consideration of the effect on other pieces of the system*.

Sarasvathy—a student of Herbert Simon—notes that near-decomposability boosts the system's likelihood of survival (Sarasvathy, 2003, p. 213):

artifacts that incorporate the property of near-decomposability in their structure endure better since they allow parts of the structure to be modified, or even destroyed and rebuilt, while retaining the rest of the entity relatively unchanged.

This approach allows individual components to be modified whenever their inherent limitations pose a challenge or when significant changes in the external environment require the system to adapt. In fact, due to their near-decomposability, individual components can also be repurposed in new systems.

A further consequence of near-decomposability is that it fosters simplicity in design. It enables a system to be designed as simply as possible at any given moment and to add more functionality as needed (Beck, 2000).

Using near-decomposability as a general design strategy has several implications for the resulting inner environment:

- Modularity and scalability arise from the ability to modify or scale systems more quickly by working on subsystems independently.
- Robustness and resilience result because local failures are less likely to cascade to other parts of the system.
- Optimization and specialization are possible because subsystems can be modified locally.

- Incremental design through stepwise refinement and iterative improvements is possible without a complete overhaul of the entire system.

Near-decomposability is essential for creating scalable, maintainable, and resilient information systems architectures. This principle helps manage complexity by maintaining relative independence among subsystems while ensuring that databases, APIs, microservices, and other components communicate in an organized way.

In *interaction design*, near-decomposability influences navigation and interaction with systems. This principle enables progressive disclosure, meaning complexity is revealed only when necessary; interfaces can be context-aware and adapt to the user, and they can be modular, sharing a common framework.

12.3. VUCA Strategies for Inner Environment

By definition, the inner environment shapes the structure and main components we use to leverage our strengths and develop valuable capabilities. The boundary between the inner and outer environments is somewhat interconnected. What is viewed as part of the outer environment influences which components we build and how we arrange them within the inner environment. Likewise, we need to design essential components to achieve desired capabilities while making the best use of our leverage points.

We will typically employ principles of near-decomposability to enhance robustness and flexibility, thereby improving the long-term survivability of our components. Furthermore, we aim to prepare the inner environment and its key components for future challenges by exploring and addressing potential VUCA challenges. As mentioned in Section 10.1, VUCA is an acronym for four types of challenges to a design project:

- *Volatility*—i.e., conditions that change rapidly or unpredictably.
- *Uncertainty*—i.e., needs and solutions that are unsettled.
- *Complexity*—i.e., external or internal parts that form intricate meshes.
- *Ambiguity*—i.e., causes and effects that are mixed up.

If any challenges jeopardize our project's inner environment, we should seek strategies to mitigate or avoid them.

Overall strategies:

- *Volatility*: Components designed like *Lego* blocks, with a clear interface and function, will always be useful in new combinations and will help meet emerging needs. If we abandon our original design ideas, we will still retain valuable components for new designs.

- *Uncertainty*: It is essential to be ready for the potential outcomes of already known uncertainties. By doing so, we can decrease their impact.
- *Complexity*: To maintain flexibility and keep the system's complexity low, it should be designed using near-decomposable parts.
- *Ambiguity*: Experiment and apply natural selection to identify which components endure.

12.3.1. Volatility

Knighitian uncertainties belong to this category. Large-scale examples include wars, pandemics, and logistics disruptions caused by events at the Suez Canal or the Red Sea. Instabilities and unexpected societal, economic, or technological changes could significantly impact a project's foundation.

Volatility in the inner environment could arise from sudden, disruptive changes in leverage points, such as technologies, components, information, or people available to build a solution.

To tackle volatility, we should design for rapidly changing or unpredictable conditions. Possible strategies could include building *generic capabilities*.

- General or non-specific abilities that can be easily adapted to changes.
- Design components with multiple lines of tearing to withstand changing environments. Such systems should be decomposable in various ways and thus recomposable into new wholes (inspired by (Augier and Sarasvathy, 2004, p. 187)).

A further strategy is to use *microservice architectures* or *platforms*.

- A collection of loosely coupled, fine-grained services.
- Services that can be developed and deployed independently of others.

12.3.2. Uncertainty

Ongoing election campaigns, pending legislation, trade agreement negotiations, proposed joint ventures, mergers, and upcoming products illustrate uncertainties.

Uncertainty concerning the inner environment could be:

- The impact of generative AI technologies on system design, requirements elicitation, hardware specifications, energy consumption, and more remains largely unknown.
- Social media regulations.
- Privacy concerns and more.
- Disruptive shifts in leverage points, such as technologies, components, information, or people that could be part of our contributions.

Such uncertainties may significantly impact design projects and might even be treated differently in different parts of the world.

Designing in uncertainty involves addressing needs and solutions that are still unresolved. Strategies could include:

Using *information hiding* ((Parnas, 1972))

- Hide difficult design decisions and decisions that are likely to change in separate modules.

Isolating uncertainties in modules.

- Reduce the impact of uncertainty to influence as few modules as possible. Prevent or limit the spread of errors and failures.

Information hiding is founded on what Endres and Rombach refer to as Parnas' Law: *Only what is hidden can be changed without risks* (Endres and Rombach, 2003, p. 45). By hiding unnecessary details, the principle of information hiding aims to lower complexity and improve modularity, security, and maintainability. This approach allows changes to a subsystem's internal components without affecting other subsystems.

12.3.3. Complexity

The inner environment sometimes includes many elements that interact in non-simple ways, making it difficult to understand the properties of the entire system. According to Simon, in such systems, the whole is more than just the sum of its parts, and determining the properties of the whole from those of the individual parts is not straightforward (Simon, 1962, p. 468).

Complexity can arise when external or internal parts become intricate meshes. To reduce complexity, we might use these strategies:

Modularize into *nearly decomposable* systems (Simon, 1962):

- Systems in which the interactions among the subsystems are weak but not negligible.

Use *divide-and-conquer* patterns.

- Try splitting complex tasks into smaller subtasks and addressing them individually.

As discussed earlier in this chapter, modularizing into near-decomposable systems enhances the system's survival chances by simplifying adjustments and boosting reusability.

There are fundamental limits to the divide-and-conquer strategy. Some complex problems—such as chaotic systems—cannot be broken down into smaller parts (Snowden and Boone, 2007). For a system to be near-decomposable, the interactions among subsystems should be weaker than those within subsystems.

12.3.4. Ambiguity

Ambiguity can arise when our contribution alters the logic, needs, or constraints of the outer environment, or when it causes unpredictable changes. It can also arise from customers or users who are unclear about their needs or have conflicting expectations. Ambiguity might be visible from the start of a project, but often it only becomes discernible or manifest as part of the inquiry.

Ambiguity occurs when causes and effects are mixed up. To handle this, we must understand the environmental responses to determine what to focus on and what to ignore. Simon argues that *selectivity can always be equated with some kind of feedback of information from the environment* (Simon, 1962, p. 473).

To address this, we might use the following strategies:

- *Experimentation*—use prototypes, minimum viable products, and similar approaches to maximize user feedback.
- Use problem-solving as a *natural selection* process. Simon asserts that *problem solving [...] involves nothing more than varying mixtures of trial and error and selectivity* (Simon, 1962, p. 472).

The process of natural selection relies on trial and error, where successful components survive and lay the foundation for further development. As a result, these components evolve into stable configurations through rapid adaptation.

		KNOWLEDGE ABOUT THE SITUATION	
ABILITY TO PREDICT THE RESULTS OF ACTIONS	HIGH	COMPLEXITY	VOLATILITY
		External or internal parts that form intricate meshes. STRATEGIES • Modularize • Divide & conquer	Conditions that change rapidly or unpredictably. STRATEGIES • Build generic capabilities • Use microservice architecture
		AMBIGUITY	UNCERTAINTY
	LOW	Causes and effects that are mixed up. STRATEGIES • Experimentation (MVP) • Problem-solving as natural selection	Needs and solutions that are unsettled. STRATEGIES • Information hiding (Parnas) • Isolate uncertainties in modules
		LOW	HIGH

Table 12.1: VUCA: Strategies for Inner Environment

The VUCA strategies for the inner environment are summarized in Table 12.1.